

Circle

Question1

$A(4, 3)$, $B(2, 5)$ are two points. If P is a variable point on the same side as that of the origin with respect to the line AB and is at most at a distance of 5 units from the mid-point of AB , then the locus of P is

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

$$x^2 + y^2 - 6x - 8y = 0$$

B.

$$x^2 + y^2 - 6x - 8y \leq 0, x + y - 7 < 0$$

C.

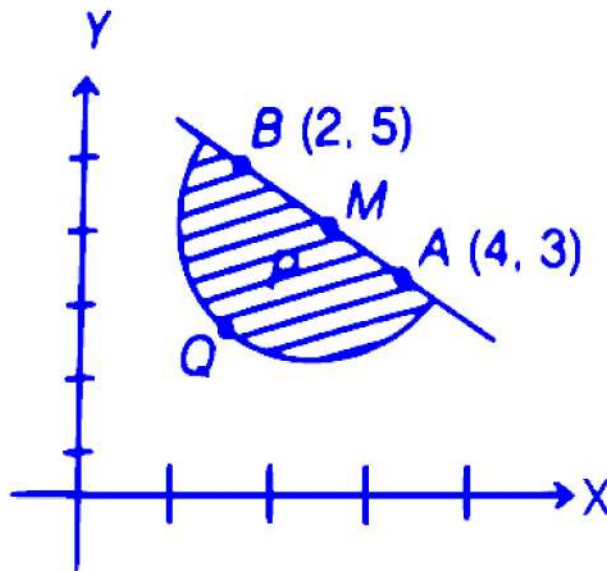
$$x^2 + y^2 + 6x + 8y - 25 = 0, x + y - 7 \leq 0$$

D.

$$x^2 + y^2 - 6x + 8y \geq 0, x + y - 7 < 0$$

Answer: B

Solution:



$$m = \left(\frac{2+4}{2}, \frac{5+3}{2} \right) = (3, 4)$$

$$\text{and } mQ \leq 5$$

$$\text{So, } P \text{ should be in the shaded region thus, } AB : (y - 3) = \frac{3-5}{4-2}(x - 4)$$

$$\Rightarrow y = -x + 4 + 3$$

$$\Rightarrow x + y - 7 = 0$$

and equation of the semi-circle

$$(x - 3)^2 + (y - 4)^2 = (5)^2$$

$$\Rightarrow x^2 - 6x + 9 + y^2 - 8y + 16 = 25$$

$$\Rightarrow x^2 + y^2 - 6x - 8y = 0$$

Hence, shaded region is

$$x^2 + y^2 - 6x - 8y \leq 0, x + y - 7 < 0$$

Question2

The circles $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 + 2x + 4y - 11 = 0$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

cut each other orthogonally

B.

do not meet

C.

intersect at the points lying on the line $4x + 8y - 7 = 0$

D.

touch each other at the point lying on the line $4x + 8y - 7 = 0$

Answer: C

Solution:

$$x^2 + y^2 - 2x - 4y - 4 = 0 \quad \dots (i)$$

$$c_1 = (1, 2)$$

$$\text{and } r_1 = \sqrt{(1)^2 + (2)^2 - (-4)} = 3$$

$$x^2 + y^2 + 2x + 4y - 11 = 0 \quad \dots (ii)$$

$$c_2 = (-1, -2) \text{ and}$$

$$r_2 = \sqrt{(-1)^2 + (-2)^2 - (-11)} = 4$$

$$\begin{aligned} c_1 c_2 &= \sqrt{(-1 - 1)^2 + (-2 - 2)^2} \\ &= \sqrt{18} < r_1 + r_2 = 7 \end{aligned}$$

So, both circles intersect, each others.

Subtracting Eq. (i) from Eq. (ii), we get $4x + 8y - 7 = 0$

Hence, intersects at the points lying on the line $4x + 8y - 7 = 0$

Question3

If the line $4x - 3y + 7 = 0$ touches the circle

$x^2 + y^2 - 6x + 4y - 12 = 0$ at (α, β) , then $\alpha + 2\beta =$

Options:

A.

3

B.

-1

C.

1

D.

-3

Answer: C**Solution:**

$$\begin{aligned}x^2 + y^2 - 6x + 4y - 12 &= 0 \\ \Rightarrow (x^2 - 6x + 9) + (y^2 + 4y + 4) &= 12 + 9 + 4 \\ \Rightarrow (x - 3)^2 + (y + 2)^2 &= (5)^2\end{aligned}$$

Centre = $(3, -2)$ and radius = 5

The line $4x - 3y + 7 = 0$ touches the circle at the point (α, β)

Foot of perpendicular from $(3, -2)$ to the line

$$\begin{aligned}4x - 3y + 7 = 0 \text{ is} \\ \frac{x - x_0}{a} = \frac{y - y_0}{b} = -\frac{ax_0 + by_0 + c}{a^2 + b^2} \\ \Rightarrow \frac{x - 3}{4} = \frac{y + 2}{-3} = -\frac{4(3) - 3(-2) + 7}{3^2 + 4^2} = -1\end{aligned}$$

$$\text{Thus, } \frac{x-3}{4} = -1 \Rightarrow x = -1$$

$$\text{and } \frac{y+2}{-3} = -1 \Rightarrow y = +1$$

$$\text{so, } (\alpha, \beta) = (-1, 1)$$

$$\text{Hence, } \alpha + 2\beta = -1 + 2 = 1$$

Question4

The slope of the common tangent drawn to the circles

$$x^2 + y^2 - 4x + 12y - 216 = 0 \text{ and } x^2 + y^2 + 6x - 12y + 36 = 0 \text{ is}$$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

1

B.

-1

C.

5/12

D.

12/7

Answer: C

Solution:

$$\begin{aligned}x^2 - 4x + y^2 + 12y &= 216 \\ \Rightarrow (x - 2)^2 + (y + 6)^2 &= 256 \\ c_1 &= (2, -6), r_1 = 16 \\ \text{and } x^2 + 6x + y^2 - 12y &= -36 \\ \Rightarrow (x + 3)^2 + (y - 6)^2 &= 9 \\ \therefore c_2 &= (-3, 6), r_2 = 3 \\ c_1 c_2 &= \sqrt{(2 + 3)^2 + (-6 - 6)^2} \\ &= \sqrt{169} = 13\end{aligned}$$

$$\text{and } r_1 - r_2 = 16 - 3 = 13$$

$\therefore c_1 c_2 = |r_1 - r_2|$ the circle touches internally.

then, the slope of line joining centers

$$= \frac{6 - (-6)}{3 - 2} = -12/5$$

$\therefore$ the slope of tangent

$$= \frac{-1}{(-12/5)} = 5/12$$

Question5

If r_1 and r_2 are radii of two circles touching all the four circles $(x \pm r)^2 + (y \pm r)^2 = r^2$, then $\frac{r_1+r_2}{r} =$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

$$\frac{\sqrt{2}+1}{2}$$

B.

$$3\sqrt{2}$$

C.

$$2\sqrt{2}$$

D.

$$\frac{3+\sqrt{2}}{4}$$

Answer: C

Solution:

The four given circles are

$$(x \pm r)^2 + (y \pm r)^2 = r^2$$

centers are $(r, r), (r, -r), (-r, -r), (-r, r)$ the circle touches all four given centered at the origin $(0, 0)$, due to the symmetry.

One circle is internally tangent to the region by the four circles, with radius $(r_1) = r(\sqrt{2} - 1)$

Other circles is externally tangent to the region, with radius $(r_2) = r(\sqrt{2} + 1)$

$$\therefore r_1 + r_2 = r(\sqrt{2} - 1 + \sqrt{2} + 1) = 2\sqrt{2}r$$

$$\text{Hence, } \frac{r_1+r_2}{r} = \frac{2\sqrt{2}r}{r} = 2\sqrt{2}$$

Question6

If the equation of the circle having the common chord to the circles $x^2 + y^2 + x - 3y - 10 = 0$ and $x^2 + y^2 + 2x - y - 20 = 0$ as its diameter is $x^2 + y^2 + \alpha x + \beta y + \gamma = 0$, then $\alpha + 2\beta + \gamma =$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

0

B.

1

C.

-1

D.

2

Answer: A

Solution:

$$\because x^2 + y^2 + x - 3y - 10 = 0 \text{ and } x^2 + y^2 + 2x - y - 20 = 0$$

So, common chord will be

$$\begin{aligned} (x^2 + y^2 + x - 3y - 10) - (x^2 + y^2 + 2x - y - 20) &= 0 \\ \Rightarrow x + 2y - 10 &= 0 \Rightarrow x = 10 - 2y \end{aligned}$$

put the ' x ' in first circle

$$\begin{aligned} (10 - 2y)^2 + y^2 + (10 - 2y) - 3y - 10 &= 0 \\ \Rightarrow 100 + 4y^2 - 40y + y^2 - 2y + 10 - 3y - 10 &= 0 \end{aligned}$$

$$\Rightarrow y^2 - 9y + 20 = 0$$

$$\Rightarrow (y - 5)(y - 4) = 0$$

$$\text{when } y = 4 \Rightarrow x = 2$$

$$\text{when } y = 5 \Rightarrow x = 0$$

so, intersection points are $(2, 4)$ and $(0, 3)$ which are end-points of diameters.

So, equation of the circle

$$\begin{aligned}(x - 0)(x - 2) + (y - 4)(y - 5) &= 0 \\ \Rightarrow x(x - 2) + (y - 4)(y - 5) &= 0 \\ \Rightarrow x^2 + y^2 - 2x - 9y + 20 &= 0\end{aligned}$$

Comparing with

$$\begin{aligned}x^2 + y^2 + \alpha x + \beta y + \gamma &= 0 \\ \alpha = -2, \beta = -9 \text{ and } \gamma &= 20\end{aligned}$$

Therefore, $\alpha + 2\beta + \gamma$

$$= -2 - 18 + 20 = 0$$

Question7

If $A(\cos \alpha, \sin \alpha)$, $B(\sin \alpha, -\cos \alpha)$, $C(1, 2)$ are the vertices of a $\triangle ABC$, then the locus of its centroid is

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$3(x^2 + y^2) - 2x - 4y + 1 = 0$$

B.

$$x^2 + y^2 - 2x - 4y + 1 = 0$$

C.

$$x^2 + y^2 - 2x - 4y + 3 = 0$$

D.

$$2(x^2 + y^2) - 2x - 4y + 5 = 0$$

Answer: A

Solution:

Let the centroid of the triangle be at (h, k) .

Step 1: Finding the Centroid Coordinates

The centroid's x -coordinate (h) is the average of all three x -coordinates: $h = \frac{\cos \alpha + \sin \alpha + 1}{3}$ So,
 $3h - 1 = \cos \alpha + \sin \alpha$

The centroid's y -coordinate (k) is the average of all three y -coordinates: $k = \frac{\sin \alpha - \cos \alpha + 2}{3}$ So,
 $3k - 2 = \sin \alpha - \cos \alpha$

Step 2: Squaring Both Sides

Square both sides of each equation to remove the trigonometric functions:

$$(3h - 1)^2 = (\cos \alpha + \sin \alpha)^2 \quad \dots (i)$$

$$(3k - 2)^2 = (\sin \alpha - \cos \alpha)^2 \quad \dots (ii)$$

Step 3: Add the Two Equations

Add equations (i) and (ii): $(3h - 1)^2 + (3k - 2)^2 = (\cos \alpha + \sin \alpha)^2 + (\sin \alpha - \cos \alpha)^2$

Notice that $(\cos \alpha + \sin \alpha)^2 + (\sin \alpha - \cos \alpha)^2 = 2 \sin^2 \alpha + 2 \cos^2 \alpha = 2$

$$\text{So, } (3h - 1)^2 + (3k - 2)^2 = 2$$

Step 4: Locus Equation

Replace h by x and k by y because the locus is about every possible point (x, y) :

$$(3x - 1)^2 + (3y - 2)^2 - 2 = 0$$

Step 5: Expand to Standard Form

Expanding and simplifying:

$$(3x - 1)^2 + (3y - 2)^2 - 2 = 0$$

$$9x^2 - 6x + 1 + 9y^2 - 12y + 4 - 2 = 0$$

$$9x^2 + 9y^2 - 6x - 12y + 3 = 0$$

$$3(x^2 + y^2) - 2x - 4y + 1 = 0$$

Question 8

A circle passing through origin cuts the coordinate axes at A and B . If the straight line AB passes through a fixed point (x_1, y_1) , then the locus of the centre of the circle is

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$\frac{x_1}{x} + \frac{y_1}{y} = 1$$

B.

$$x_1y = xy_1$$

C.

$$xy_1 + yx_1 = 2$$

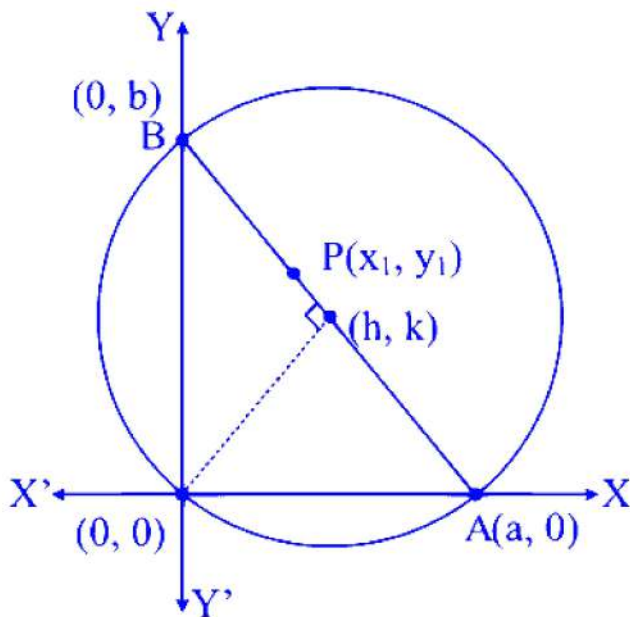
D.

$$\frac{x_1}{x} + \frac{y_1}{y} = 2$$

Answer: D

Solution:

$\therefore AB$ will be the diameter



$\therefore$ Let equation of circle

$$x^2 + y^2 + 2gx + 2fy = 0$$

$(a, 0)$ lies on the circle

$$\therefore a^2 + 2ga = 0 \Rightarrow a = -2g$$

$(0, b)$ lies on the circle

$$\therefore b^2 + 2fb = 0 \Rightarrow b = -2f$$

∴ Equation of line AB

$$\frac{x}{-2g} + \frac{y}{-2f} = 1 \quad [\because (x_1, y_1) \text{ lies on } L]$$

$$\Rightarrow \frac{x}{-g} + \frac{y}{-f} = 2 \Rightarrow \frac{x_1}{-g} + \frac{y_1}{-f} = 2$$

∴ Locus is centre $(-g, -f)$ is $\frac{x_1}{x} + \frac{y_1}{y} = 2$

Question9

If (α, β) is the external centre of similitude of the circles $x^2 + y^2 = 3$ and $x^2 + y^2 - 2x + 4y + 4 = 0$, then $\frac{\beta}{\alpha} =$

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

-3

B.

-2

C.

2

D.

3

Answer: B

Solution:

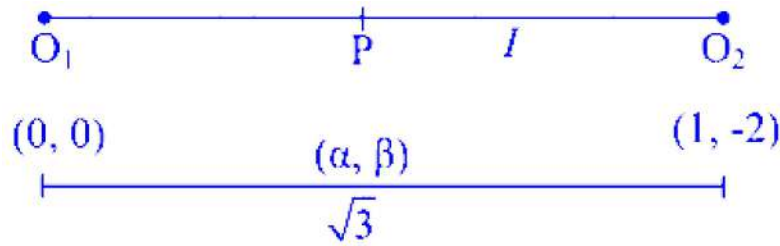
Given, $C_1 : x^2 + y^2 = 3$,

$O_1(0, 0), r_1 = \sqrt{3}$

and $C_2 : x^2 + y^2 - 2x + 4y + 4 = 0$

$O_2(1, -2), r_2 = 1$

Point P divides O_1O_2 externally in $r_1 : r_2$



$$\therefore \alpha = \frac{0 \times \sqrt{3} - 1(1)}{\sqrt{3} + 1} = \frac{-1}{\sqrt{3} + 1} \text{ and}$$

$$\beta = \frac{0 \times \sqrt{3} - (-2)}{\sqrt{3} + 1} = \frac{2}{\sqrt{3} + 1}$$

$$\Rightarrow \frac{\beta}{\alpha} = -2$$

Question10

The equation of the circle touching the lines $|x - 2| + |y - 3| = 4$ is

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$x^2 + y^2 - 6x - 4y + 5 = 0$$

B.

$$x^2 + y^2 - 4x - 6y + 5 = 0$$

C.

$$x^2 + y^2 - x - 2y - 5 = 0$$

D.

$$x^2 + y^2 - 2x - y - 5 = 0$$

Answer: B

Solution:

Given, $|x - 2| + |y - 3| = 4$

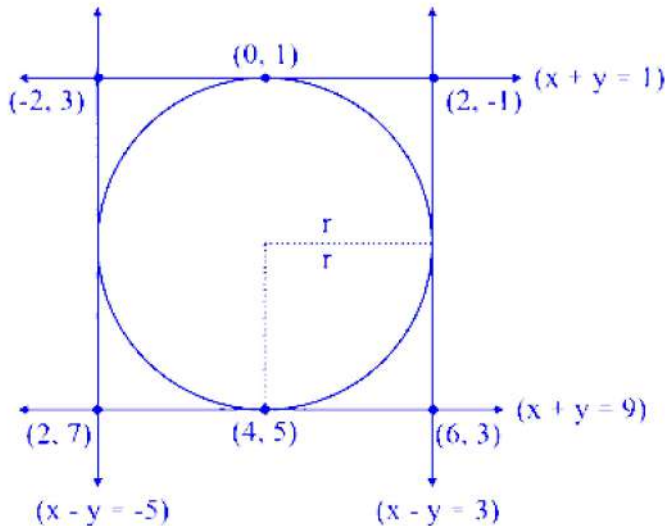
$\therefore x - 2 + y - 3 = 4$

$\Rightarrow x + y = 9$ and $x - 2 - y + 3 = 4$

$\Rightarrow x - y = 3$ and $-x + 2 + y - 3 = 4$

$\Rightarrow x - y = -5$ and $-x + 2 - y + 3 = 4$

$\Rightarrow x + y = 1$



$\therefore$ Centre $\left(\frac{4 + 0}{2}, \frac{5 + 1}{2} \right) = (2, 3)$

and $r = \frac{\sqrt{4^2 + 4^2}}{2} = 2\sqrt{2}$

$\therefore$ Equation of circle :

$$(x - 2)^2 + (y - 3)^2 = (2\sqrt{2})^2$$

$\Rightarrow x^2 + y^2 - 4x - 6y + 5 = 0$

Question11

If the chord joining the points $(1, 2)$ and $(2, -1)$ on a circle subtends an angle of $\frac{\pi}{4}$ at any point on its circumference, then the equation of such a circle is

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$x^2 + y^2 + 6x - 2y + 5 = 0$$

B.

$$x^2 + y^2 - 6x - 2y + 5 = 0$$

C.

$$x^2 + y^2 - 6x + 2y + 5 = 0$$

D.

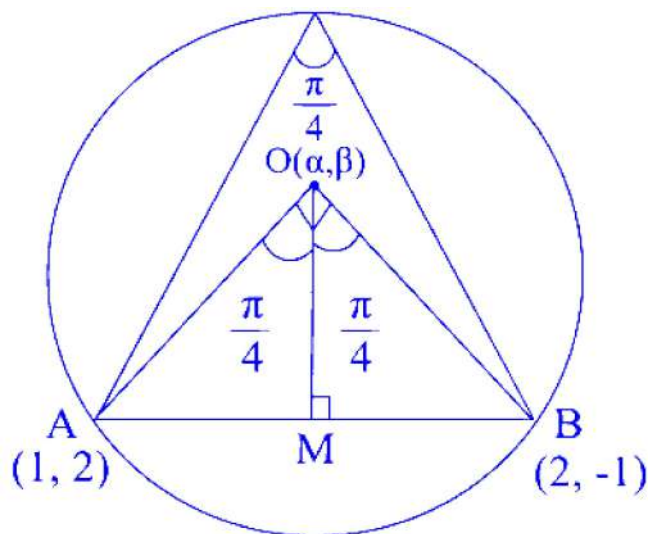
$$x^2 + y^2 + 6x + 2y + 5 = 0$$

Answer: B

Solution:

$$AB = \sqrt{1^2 + 9} = \sqrt{10}$$

$$AM = \frac{\sqrt{10}}{2}$$



$$\therefore OA = r = \sqrt{\frac{10}{4} + \frac{10}{4}} = \sqrt{\frac{20}{4}} = \sqrt{5}$$

Let $O(\alpha, \beta)$

$$OA = OB$$

$$(\alpha - 1)^2 + (\beta - 2)^2 = (\alpha - 2)^2 + (\beta + 1)^2$$

$$\Rightarrow -2\alpha + 1 - 4\beta + 4 = -4\alpha + 4 + 2\beta - 1$$

$$\Rightarrow 2\alpha - 6\beta = 0$$

$$\Rightarrow \alpha = 3\beta \quad \dots (i)$$

$$\therefore m_{OA} \cdot m_{OB} = -1$$

$$\Rightarrow \frac{\beta-2}{\alpha-1} \cdot \frac{\beta+1}{\alpha-2} = -1$$

$$\Rightarrow (\beta - 2)(\beta + 1) + (\alpha - 1)(\alpha - 2) = 0$$

$$\Rightarrow \beta^2 - \beta - 2 + \alpha^2 - 3\alpha + 2 = 0$$

$$\Rightarrow \beta^2 - \beta + 9\beta^2 - 9\beta = 0$$

$$\Rightarrow 10\beta^2 = 10\beta$$

$$\Rightarrow \beta = 0, 1$$

$$\alpha = 0, 3$$

$\therefore$ Centre $(0, 0)$ or $(3, 1)$

$\therefore$ Equation of circle

$$x^2 + y^2 = 5 \text{ or } (x - 3)^2 + (y - 1)^2 = 5$$

$$\Rightarrow x^2 + y^2 - 6x - 2y + 5 = 0$$

Question12

The equation of the circle which cuts all the three circles

$$4(x - 1)^2 + 4(y - 1)^2 = 1, 4(x + 1)^2 + 4(y - 1)^2 \text{ and}$$

$$4(x + 1)^2 + 4(y + 1)^2 = 1 \text{ orthogonally is}$$

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$4x^2 + 4y^2 = 49$$

B.

$$4(x - 1)^2 + 4(y + 1)^2 = 1$$

C.

$$(x - 1)^2 + (y + 1)^2 = 4$$

D.

$$4x^2 + 4y^2 = 7$$

Answer: D

Solution:

Given,

$$S_1 : 4(x-1)^2 + 4(y-1)^2 = 1$$
$$\Rightarrow (x-1)^2 + (y-1)^2 = \frac{1}{4}$$

$$O_1(1, 1), r_1 = \frac{1}{2}$$

$$S_2 : 4(x+1)^2 + 4(y-1)^2 = 1$$
$$\Rightarrow (x+1)^2 + (y-1)^2 = \frac{1}{4}$$

$$O_2(-1, 1), r_2 = 1/2$$

$$\text{and } S_3 : 4(x+1)^2 + 4(y+1)^2 = 1$$
$$\Rightarrow (x+1)^2 + (y+1)^2 = \frac{1}{4}$$

$$O_3(-1, -1), r_3 = \frac{1}{2}$$

$$\therefore s_1 - s_2 = 0$$

$$\Rightarrow -2x - 2y + 2x - 2y = 0$$

$$x = 0$$

$$S_2 - s_3 = 0$$

$$\Rightarrow 2y = 0$$

$$\Rightarrow y = 0$$

$\therefore$ Centre of required circle = (0, 0)

$$\text{radius} = L = \sqrt{s_1}$$

$$= \sqrt{1 + 1 - \frac{1}{4}} = \sqrt{\frac{7}{4}}$$

$\therefore$ Equation of circle

$$x^2 + y^2 = \frac{7}{4}$$

$$\Rightarrow 4x^2 + 4y^2 = 7$$

Question13

$A(a, 0)$ is a fixed point and θ is a parameter such that $0 < \theta < 2\pi$. If $P(a \cos \theta, a \sin \theta)$ is a point on the circle $x^2 + y^2 = a^2$ and $Q(b \sin \theta, -b \cos \theta)$ is a point on the circle $x^2 + y^2 = b^2$, then the locus of the centroid of the $\triangle APQ$ is

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

a circle with centre at $\left(\frac{a}{3}, 0\right)$ and radius $\left(\frac{\sqrt{a^2+b^2}}{3}\right)$

B.

a circle with centre at $(a, 0)$ and radius $\left(\frac{\sqrt{a^2+b^2}}{3}\right)$

C.

a parabola with focus at $\left(\frac{a}{3}, 0\right)$

D.

a parabola with focus at $(a, 0)$

Answer: A

Solution:

Let the coordinates of the centroid be $G(x, y)$ of the $\triangle APQ$ is

$$x = \frac{x_A + x_P + x_Q}{3}, y = \frac{y_A + y_P + y_Q}{3}$$

Given the points $A(a, 0)$, $P(a \cos \theta, a \sin \theta)$ and $Q(b \sin \theta, -b \cos \theta)$

The coordinates of the centroid are

$$x = \frac{a + a \cos \theta + b \sin \theta}{3}$$
$$y = \frac{0 + a \sin \theta - b \cos \theta}{3}$$

Let $x = h$ and $y = k$, then,

$$h = \frac{a + a \cos \theta + b \sin \theta}{3}, k = \frac{a \sin \theta - b \cos \theta}{3}$$
$$\Rightarrow 3h - a = a \cos \theta + b \sin \theta$$
$$3k = a \sin \theta - b \cos \theta$$

Squaring and adding these equations

$$(3h - a)^2 + (3k)^2 = (a \cos \theta + b \sin \theta)^2 + (a \sin \theta - b \cos \theta)^2$$

$$\begin{aligned}
&= a^2 \cos^2 \theta + b^2 \sin^2 \theta + 2ab \cos \theta \sin \theta + a^2 \sin^2 \theta + b^2 \cos^2 \theta - 2ab \cos \theta \sin \theta \\
&= a^2 + b^2
\end{aligned}$$

Replacing h with x and k with y , we get the locus of the centroid as

$$\begin{aligned}
(3x - a)^2 + (3y)^2 &= a^2 + b^2 \\
\Rightarrow \left(x - \frac{a}{3}\right)^2 + y^2 &= \frac{a^2 + b^2}{9} \\
\Rightarrow \left(x - \frac{a}{3}\right)^2 + y^2 &= \left(\frac{\sqrt{a^2 + b^2}}{3}\right)^2
\end{aligned}$$

So, the locus of the centroid is

$$\left(x - \frac{a}{3}\right)^2 + y^2 = \left(\frac{\sqrt{a^2 + b^2}}{3}\right)^2$$

It is a circle with centre at $\left(\frac{a}{3}, 0\right)$ and radius $\frac{\sqrt{a^2 + b^2}}{3}$

Question14

If the equation of the circle passing through the point $(8, 8)$ and having the lines $x + 2y - 2 = 0$ and $2x + 3y - 1 = 0$ as its diameters is $x^2 + y^2 + px + qy + r = 0$, then $p^2 + q^2 + r =$

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

244

B.

100

C.

-44

D.

44

Answer: C

Solution:

Given, lines are $x + 2y - 2 = 0$ and

$$2x + 3y - 1 = 0$$

Solving these two lines, we get

$$x = -4 \text{ and } y = 3$$

Thus, the center of circle is $(-4, 3)$.

Since, the general eq. of circle is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius.

$$\begin{aligned} \text{So, } (x - (-4))^2 + (y - 3)^2 &= r^2 \\ \Rightarrow (x + 4)^2 + (y - 3)^2 &= r^2 \quad \dots (i) \end{aligned}$$

But, the circle passes through the point $(8, 8)$,

$$\begin{aligned} \text{So, } (8 + 4)^2 + (8 - 3)^2 &= r^2 \\ 12^2 + 5^2 &= r^2 \\ 169 &= r^2 \\ \Rightarrow r &= 13 \end{aligned}$$

Now, from Eq. (i), we get

$$\begin{aligned} (x + 4)^2 + (y - 3)^2 &= r^2 = 169 \\ \Rightarrow x^2 + 8x + 16 + y^2 - 6y + 9 &= 169 \\ \Rightarrow x^2 + y^2 + 8x - 6y - 144 &= 0 \end{aligned}$$

Comparing with the given equation

$$x^2 + y^2 + px + qx + r = 0$$

We get $p = 8, q = -6, r = -144$

$$\begin{aligned} \therefore p^2 + q^2 + r &= 8^2 + (-6)^2 + (-144) \\ &= 64 + 36 - 144 \\ &= -44 \end{aligned}$$

Question15

If $2x - 3y + 1 = 0$ is the equation of the polar of a point $P(x_1, y_1)$ with respect to the circle $x^2 + y^2 - 2x + 4y + 3 = 0$, then $3x_1 - y_1 =$

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

$$\frac{1}{3}$$

B.

$$-3$$

C.

$$3$$

D.

$$-\frac{1}{3}$$

Answer: C

Solution:

Given circle is

$$x^2 + y^2 - 2x + 4y + 3 = 0$$

$$\text{So, center} = (-g, -f) = \left(-\frac{-2}{2}, \frac{4}{2}\right)$$

$$= (-1, 2) \text{ and } c = 3$$

Also, the equation of the polar of a point (x_1, y_1) w.r.t the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$\Rightarrow xx_1 + yy_1 - (x + x_1) + 2(y + y_1) + 3 = 0$$

$$\Rightarrow xx_1 + yy_1 - x - x_1 + 2y + 2y_1 + 3 = 0$$

$$\Rightarrow (x_1 - 1)x + (y_1 + 2)y - x_1 + 2y_1 + 3 = 0 \quad \dots (i)$$

Given, polar equation is

$$2x - 3y + 1 = 0 \quad \dots (ii)$$

By comparing the coefficients of two equations, we get

$$\begin{aligned} \frac{x_1 - 1}{2} &= \frac{y_1 + 2}{-3} \\ &= \frac{-x_1 + 2y_1 + 3}{1} \end{aligned}$$

$$\Rightarrow \frac{x_1 - 1}{2} = \frac{y_1 + 2}{-3} \text{ and}$$

$$\frac{x_1 - 1}{2} = \frac{-x_1 + 2y_1 + 3}{1}$$

$$\begin{aligned} \Rightarrow -3x_1 + 3 &= 2y_1 + 4 \text{ and } x_1 - 1 \\ &= -2x_1 + 4y_1 + 6 \end{aligned}$$

$$\Rightarrow 3x_1 + 2y_1 = -1 \text{ and } 3x_1 - 4y_1 - 7 = 0$$

By solving these two equations, we get $x_1 = \frac{5}{9}$ and $y_1 = \frac{-4}{3}$

$$\begin{aligned}
 \text{Now, } 3x_1 - y_1 &= 3\left(\frac{5}{9}\right) - \left(\frac{-4}{3}\right) \\
 &= \frac{5}{3} + \frac{4}{3} = \frac{9}{3} = 3 \\
 \therefore 3x_1 - y_1 &= 3
 \end{aligned}$$

Question16

If a unit circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ touches the circle $S' \equiv x^2 + y^2 - 6x + 6y + 2 = 0$ externally at the point $(-1, -3)$, then $g + f + c =$

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

0

B.

1

C.

15

D.

17

Answer: D

Solution:

Given,

$$s \equiv x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\text{And } s' \equiv x^2 + y^2 - 6x + 6y + 2 = 0$$

$$\Rightarrow (x^2 - 6x) + (y^2 + 6y) = -2$$

$$\begin{aligned}
 \Rightarrow (x^2 - 6x + 9) + (y^2 + 6y + 9) \\
 = -2 + 9 + 9
 \end{aligned}$$

$$\Rightarrow (x - 3)^2 + (y + 3)^2 = 16$$

Thus, the center of circle S' is $(3, -3)$ and radius, $r' = 4$

$$\text{Now, } S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\Rightarrow (x + g)^2 + (y + f)^2 = g^2 + f^2 - c$$

So, the center of circle S is $(-g, -f)$ and radius,

$$r = \sqrt{g^2 + f^2 - c}$$

Since, radius of circle S is 1 .

$$\therefore \sqrt{g^2 + f^2 - c} = 1$$

$$\Rightarrow g^2 + f^2 - c = 1$$

Also, two circle touch externally,

So, distance between $(-g, -f)$ and

$$(3, -3) = r + r'$$

$$\Rightarrow \sqrt{(-g - 3)^2 + (-f + 3)^2} = 4 + 1$$

$$\Rightarrow (g + 3)^2 + (f - 3)^2 = 25$$

$$\Rightarrow g^2 + 6g + 9 + f^2 - 6f + 9 = 25$$

$$\Rightarrow g^2 + f^2 + 6g - 6f + 18 - 25 = 0$$

$$\Rightarrow g^2 + f^2 + 6g - 6f = 7 \quad \dots (i)$$

Since, point $(-1, -3)$ lies on both circles, it satisfies both circle.

$\therefore$ For circles

$$(-1)^2 + (-3)^2 + 2g(-1) + 2f(-3) + c = 0$$

$$\Rightarrow 1 + 9 - 2g - 6f + c = 0$$

$$\Rightarrow -2g - 6f + c = -10 \quad \dots (ii)$$

For circle S' ,

$$(-1)^2 + (-3)^2 - 6(-1) + 6(-3) + 2 = 0$$

$$\Rightarrow 1 + 9 + 6 - 18 + 2 = 0$$

$$\Rightarrow 0 = 0$$

Now, the slope of line joining $(-g, -f)$ and $(-1, -3)$ is equal to slope of line joining $(-1, -3)$ and $(3, -3)$

$$(-f + 3) = \left(\frac{-3 + 3}{3 + 1} \right) (-g + 1)$$

$$\Rightarrow (-f + 3) = 0$$

$$\Rightarrow f = 3$$

Put $f = 3$ into Eq. (i), we get

$$g^2 + 9 + 6g - 6(3)$$

$$\Rightarrow g^2 + 9 + 6g - 18 = 7$$

$$\Rightarrow g^2 + 6g - 16 = 0$$

$$\Rightarrow (g + 8)(g - 2) = 0$$

$$\Rightarrow g = -8 \text{ or } g = 2 (\text{not possible})$$

Now, put $g = 2, f = 3$ into Eq. (ii), we get

$$\begin{aligned} -2(2) - 6(3) + c &= -10 \\ \Rightarrow c &= -10 + 4 + 18 = 12 \end{aligned}$$

$$\text{Now, } g + f + c = 2 + 3 + 12 = 17$$

Question17

$3x + 4y - 43 = 0$ is a tangent to the circle $S \equiv x^2 + y^2 - 6x + 8y + k = 0$ at a point P . If C is the centre of the circle and Q is a point which divides CP in the ratio $-1 : 2$, then the power of the point Q with respect to the circle $S = 0$ is

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

50

B.

21

C.

0

D.

5

Answer: C

Solution:

$$\begin{aligned} S &\equiv x^2 + y^2 - 6x + 8y + k = 0 \\ \Rightarrow (x - 3)^2 + (y + 4)^2 &= 25 - k \end{aligned}$$

Thus, the center of the circle C is at $(3, -4) = (x_1, y_1)$.

Let point P as (x_2, y_2) and point Q is the point that divides the line segment CP in the ratio $-1 : 2$

Also, $3x + 4y - 43 = 0$

$$\Rightarrow y = \frac{-3}{4}x + \frac{43}{4}$$

So, slope of tangent line = $\frac{-3}{4}$

And slope of line $CP = -\left(\frac{-4}{3}\right) = \frac{4}{3}$

Thus, equation of line CP is

$$\Rightarrow y - (-4) = \frac{4}{3}(x - 3)$$

$$\Rightarrow y + 4 = \frac{4}{3}(x - 3)$$

$$\Rightarrow y = \frac{4}{3}x - 8$$

Since, P lies on both the lines $3x + 4y - 43 = 0$ and the line CP ,

$$\therefore 3x + 4\left(\frac{4}{3}x - 8\right) - 43 = 0$$

$$\Rightarrow 3x + \frac{16}{3}x - 32 - 43 = 0$$

$$\Rightarrow \frac{25}{3}x = 75$$

$$\Rightarrow x = 9$$

$$\text{Now, } y = \frac{4}{3}x - 8 = \frac{4}{3}(9) - 8$$

$$= 12 - 8 = 4$$

So, $P = (9, 4)$

Using section formula, we find the coordinates of $Q(x, y)$.

$$x = \frac{2x_1 - x_2}{2 - 1} = \frac{2(3) - 9}{1} = -3$$

$$y = \frac{2y_1 - y_2}{2 - 1} = \frac{2(-4) - 4}{1} = -12$$

So, $Q = (x, y) = (-3, -12)$

The power of point Q w.r.t to circle, $x^2 + y^2 - 6x + 8y + k = 0$ is

$$\begin{aligned} &(-3)^2 + (-12)^2 - 6(-3) + 8(-12) + k \\ &= 9 + 144 + 18 - 96 + k \\ &= 75 + k \quad \dots (i) \end{aligned}$$

Now, the line $3x + 4y - 43 = 0$ is tangent to the circle, the radius of the circle is the distance from the center $C(3, -4)$ to the line $3x + 4y - 43 = 0$

$$\begin{aligned} \text{So, } r &= \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}} \\ &= \frac{|3(3) + 4(-4) - 43|}{\sqrt{3^2 + 4^2}} \\ &= \frac{|9 - 16 - 43|}{5} = \frac{50}{5} = 10 \end{aligned}$$

Now, $r = \sqrt{25 - k}$

$$\Rightarrow 10 = \sqrt{25 - k} \Rightarrow 25 - k = 100$$

$$\Rightarrow k = -75$$

Put $k = -75$ in Eq. (i), we get The power of Q w.r.t the circle

$$= 75 + k = 75 - 75 = 0$$

Question18

If the radical axis of the circles $x^2 + y^2 + 2gx + 2fy + c = 0$ and $2x^2 + 2y^2 + 3x + 8y + 2c = 0$ touches the circle $x^2 + y^2 + 2x + 2y + 1 = 0$, then

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

$$\text{either } g = \frac{3}{2} \text{ or } f \neq 2$$

B.

$$\text{either } g \neq \frac{3}{4} \text{ or } f = \frac{1}{2}$$

C.

$$\text{either } g = \frac{3}{4} \text{ or } f = 2$$

D.

$$\text{either } g = \frac{1}{2} \text{ or } f = \frac{3}{4}$$

Answer: C

Solution:

$$s_1 : x^2 + y^2 + 2gx + 2fy + c = 0$$

$$s_2 : 2x^2 + 2y^2 + 3x + 8y + 2c = 0$$

$$\Rightarrow x^2 + y^2 + \frac{3}{2}x + 4y + c = 0$$

$$s_3 : x^2 + y^2 + 2x + 2y + 1 = 0$$

$$\text{Radical axis } s_1 - s_2 = 0$$

$$\begin{aligned}
&\Rightarrow (x^2 + y^2 + 2gx + 2fy + c) \\
&\quad - (x^2 + y^2 + \frac{3}{2}x + 4y + c) + 0 \\
&\Rightarrow 2gx + 2fy - \frac{3x}{2} - 4y = 0 \\
&\Rightarrow (2g - \frac{3}{2})x + (2f - 4)y = 0 \\
&\Rightarrow (4g - 3)x + (4f - 8)y = 0
\end{aligned}$$

The radical axis touches the circle s_3 , where radius of s_3 .

$$= \sqrt{(-1)^2 + (-1)^2 - 1} = 1 \text{ and center is } (-1, -1)$$

The perpendicular distance from the center of s_3 to the radical axis = radius of s_3

$$\begin{aligned}
&\frac{|(4g - 3) \cdot (-1) + (4f - 8)(-1)|}{\sqrt{(4g - 3)^2 + (4f - 8)^2}} = 1 \\
&\Rightarrow \frac{|-4g + 3 - 4f + 8|}{\sqrt{(4g - 3)^2 + (4f - 8)^2}} = 1 \\
&\Rightarrow \frac{|-4g - 4f + 11|}{\sqrt{(4g - 3)^2 + (4f - 8)^2}} \\
&\Rightarrow \frac{(-4g - 4f + 11)^2}{(4g - 3)^2 + (4f - 8)^2} \\
&\Rightarrow \frac{16g^2 + 16f^2 + 121 + 32gf - 88g - 88f}{16g^2 - 24g + 9 + 16f^2 - 64f + 64} \\
&\Rightarrow \frac{32gf - 64g - 24f + 48}{8gf - 16g - 6f + 12} = 0 \\
&\Rightarrow \frac{8g(f - 2) - 6(f - 2)}{(8g - 6)(f - 2)} = 0 \\
&\Rightarrow \frac{(4g - 3)(f - 2)}{(4g - 3)(f - 2)} = 0 \\
&\Rightarrow 4g - 3 = 0 \text{ or } f - 2 = 0 \\
&\Rightarrow g = \frac{3}{4} \text{ or } f = 2
\end{aligned}$$

Either $g = \frac{3}{4}$ or $f = 2$ is the required condition.

Question19

After the coordinate axes are rotated through an angle $\frac{\pi}{4}$ in the anti-clockwise direction without shifting the origin, if the equation $x^2 + y^2 - 2x - 4y - 20 = 0$ transforms to $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ in the new coordinate system, then

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} =$$

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

-20

B.

-25

C.

-30

D.

-35

Answer: B

Solution:

$$\because x^2 + y^2 - 2x - 4y - 20 = 0$$

After rotation of axes through $\theta = \pi/4$, it transforms to $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ in the given equation

$$a = 1, h = 0, b = 1, g = -1, f = -2, c = -20$$

After rotation of $\theta = \pi/4$

$$\begin{aligned} a' &= a \cos^2 \theta + b \sin^2 \theta + 2h \cos \theta \sin \theta \\ &= 1 \left(\frac{1}{\sqrt{2}} \right)^2 + 1 \left(\frac{1}{\sqrt{2}} \right)^2 + 0 = 1 \end{aligned}$$

$$\begin{aligned} b' &= a \sin^2 \theta + b \cos^2 \theta - 2h \sin \theta \cos \theta \\ &= 1 \left(\frac{1}{\sqrt{2}} \right)^2 + 1 \left(\frac{1}{\sqrt{2}} \right)^2 - 0 = 1 \end{aligned}$$

also,

$$h' = (b - a) \sin \theta \cos \theta + h (\cos^2 \theta - \sin^2 \theta)$$

$$= 0 + 0 = 0$$

$$\text{Similarly, } x' = x \cos \theta - y \sin \theta = \frac{1}{\sqrt{2}}(x - y)$$

$$\text{and } y' = x \sin \theta + y \cos \theta = \frac{1}{\sqrt{2}}(x + y)$$

$$\text{then } -2x' - 4y' = \frac{-2x + 2y - 4x - 4y}{\sqrt{2}}$$

$$= \frac{-6x - 2y}{\sqrt{2}}$$

$$= -3\sqrt{2}x - \sqrt{2}y = 2gx + 2fy$$

$$\text{then } g = -3/\sqrt{2}, f = -1/\sqrt{2}, c = -20$$

So,

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = \begin{vmatrix} 1 & 0 & -3/\sqrt{2} \\ 0 & 1 & -1/\sqrt{2} \\ -3/\sqrt{2} & -1/\sqrt{2} & -20 \end{vmatrix}$$

$$= 1 \left(-20 - \frac{1}{2} \right) - \frac{3}{\sqrt{2}} \left(+ \frac{3}{\sqrt{2}} \right)$$

$$= -41/2 - 9/2 = -25$$

Question20

If the circles $x^2 + y^2 + 5kx + 2y + k = 0$ and $2x^2 + 2y^2 + 2kx + 3y - 1 = 0, k \in R$ intersect at points P and Q then the line $4x + 5y - k = 0$ passes through P and Q for

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

exactly one value of k

B.

exactly two values of k

C.

no value of k

D.

infinitely many value of k

Answer: C

Solution:

Given equation of circle

$$x^2 + y^2 + 5kx + 2y + k = 0$$

$$\text{and } 2x^2 + 2y^2 + 2kx + 3y - 1 = 0$$

Equation of common chord

$$8kx + y + 2k + 1 = 0$$

Since, this chord is coincide with

$$4x + 5y - k = 0$$

$$\frac{8k}{4} = \frac{1}{5} = \frac{2k+1}{-k}$$

$$\text{Solving } \frac{8k}{4} = \frac{1}{5}$$

$$\text{we get } k = \frac{1}{10}$$

$$\text{Solving } 8\frac{k}{4} = \frac{2k+1}{-k} \Rightarrow 2k^2 + 2k + 1 = 0$$

No solution

$\therefore$ No values of k possible

Question21

The slope of one of the direct common tangents drawn to the circles $x^2 + y^2 - 2x + 4y + 1 = 0$ and $x^2 + y^2 - 4x - 2y + 4 = 0$ is

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

0

B.

$$\frac{4}{3}$$

C.

$$\frac{3}{4}$$

D.

1

Answer: B

Solution:

$$c_1 : x^2 + y^2 - 2x + 4y + 1 = 0$$

$$\Rightarrow (x-1)^2 + (y+2)^2 = 4$$

$$\text{Centre } (c_1) = (1, -2), r_1 = 2$$

$$\text{and } c_2 : x^2 + y^2 - 4x - 2y + 4 = 0$$

$$\Rightarrow (x-2)^2 + (y-1)^2 = 1$$

$$\text{Centre } (c_2) = (2, 1), r_2 = 1$$

$$c_1 c_2 = d = \sqrt{(2-1)^2 + (1+2)^2} = \sqrt{10}$$

Let the slope of direct common tangent be m the general line is

$$y = mx + c$$

this line has perpendicular distance r_1 from c_1 and r_2 from c_2

$$\therefore \left| \frac{m(1) - (-2) + c}{\sqrt{1+m^2}} \right| = 2$$

$$\Rightarrow |m + 2 + c| = 2\sqrt{1+m^2} \quad \dots (i)$$

$$\text{and } \left| \frac{m(2) - 1 + c}{\sqrt{1+m^2}} \right| = 1$$

$$\Rightarrow |2m + c - 1| = \sqrt{1+m^2} \quad \dots (ii)$$

Put the value of 'c' from Eq. (i) to Eq. (ii)

$$2m - 1 + (-m - 2 + 2\sqrt{m^2 + 1}) = \sqrt{m^2 + 1}$$

$$\Rightarrow m = 3 - \sqrt{m^2 + 1}$$

Squaring both sides, we get

$$m^2 = m^2 + 10 - 6\sqrt{m^2 + 1}$$

$$\Rightarrow \sqrt{m^2 + 1} = 5/3$$

$$\Rightarrow m^2 + 1 = \frac{25}{9}$$

$$\Rightarrow m^2 = 16/9$$

$$\Rightarrow m = 4/3$$

Question22

If $(1, a), (b, 2)$ are conjugate points with respect to the circle $x^2 + y^2 = 25$, then $4a + 2b =$

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

25

B.

50

C.

100

D.

150

Answer: B

Solution:

We are told that $(1, a)$ and $(b, 2)$ are *conjugate points* with respect to the circle

$$x^2 + y^2 = 25.$$

Step 1. Condition for conjugate points

If (x_1, y_1) and (x_2, y_2) are conjugate points with respect to the circle $x^2 + y^2 = r^2$,

then the polar of one point passes through the other.

Equation of the *polar* of (x_1, y_1) w.r.t. $x^2 + y^2 = r^2$ is:

$$x_1x + y_1y = r^2$$

Step 2. The point $(1, a)$

For the point $(1, a)$, the polar is:

$$1 \cdot x + a \cdot y = 25$$

That is:

$$x + ay = 25.$$

Step 3. The point $(b, 2)$ lies on this polar

Since the points $(1, a)$ and $(b, 2)$ are conjugate, the point $(b, 2)$ **lies on the polar** of $(1, a)$.

So:

$$b + a(2) = 25$$

$$b + 2a = 25 \quad (1)$$

Step 4. Symmetry of conjugate points

Similarly, the polar of $(b, 2)$ is:

$$bx + 2y = 25.$$

And $(1, a)$ must lie on this polar:

$$b(1) + 2(a) = 25$$

Wait—that would give same, yes so far matches (1). Let's check the other way:

Substitute $(x, y) = (1, a)$:

$$b \cdot 1 + 2a = 25$$

This is the *same* as (1). So it's consistent.

Step 5. We need $4a + 2b = ?$

From (1): $b + 2a = 25$.

Multiply both sides by 2:

$$2b + 4a = 50$$

So,

$4a + 2b = 50.$

 **Final Answer: 50**

Option B

Question23

If the pole of the line $x + 2by - 5 = 0$ with respect to the circle $S \equiv x^2 + y^2 - 4x - 6y + 4 = 0$ lies on the line $x + by + 1 = 0$, then the polar of the point $(b, -b)$ with respect to the circle $S = 0$ is

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

$$5y - 6 = 0$$

B.

$$y - 6 = 0$$

C.

$$x + 5y - 6 = 0$$

D.

$$5x + y - 6 = 0$$

Answer: A

Solution:

Let (h, k) be the pole of the line

$$x + 2by - 5 = 0$$

with respect to the circle

$$s \equiv x^2 + y^2 - 4x - 6y + 4 = 0$$

$$\therefore hx + ky - 2(x + h) - 3(y + k) + 4 = 0$$

$$(h - 2)x + (k - 3)y - 2h - 3k + 4 = 0$$

is coincide with $x + 2by - 5 = 0$

$$\frac{h-2}{1} = \frac{k-3}{2b} = \frac{2h+3k-4}{5}$$
$$\Rightarrow \frac{h-2}{1} = \frac{k-3}{2b}$$

$$2bh - k = 4b - 3 \quad \dots (i)$$

$$\text{and } \frac{h-2}{1} = \frac{2h+3k-4}{5}$$

$$h - k = 2 \quad \dots (ii)$$

Solving Eqs. (i) and (ii), we get

$$h = \frac{4b-5}{2b-1}, k = \frac{-3}{2b-1}$$

(h, k) lie on line $x + by + 1 = 0$

$$\frac{4b-5}{2b-1} - \frac{3b}{2b-1} + 1 = 0$$

$$4b-5-3b+2b-1=0$$

$$=2$$

∴ Pole of the line from pole $(2, -2)$ is

$$(2-2)x + (-2-3)y - 2(2) - 3(-2) + 4 = 0$$

$$\Rightarrow -5y + 6 = 0$$

$$\Rightarrow 5y - 6 = 0$$

Question24

If $P(\alpha, \beta)$ is the radical centre of the circles

$$S \equiv x^2 + y^2 + 4x + 7 = 0, S' \equiv 2x^2 + 2y^2 + 3x + 5y + 9 = 0 \text{ and}$$

$S'' \equiv x^2 + y^2 + y = 0$, then the length of the tangent drawn from P to $S' = 0$ is

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

5

B.

8

C.

4

D.

2

Answer: D

Solution:

The given equations are

$$s : x^2 + y^2 + 4x + 7 = 0$$

$$s' : 2x^2 + 2y^2 + 3x + 5y + 9 = 0$$

$$\Rightarrow x^2 + y^2 + \frac{3}{2}x + \frac{5}{2}y + \frac{9}{2} = 0$$

$$s'' : x^2 + y^2 + y = 0$$

Radical axis of s and s' : $s - s' = 0$

$$\Rightarrow 4x - \frac{3}{2}x - \frac{5}{2}y + 7 - \frac{9}{2} = 0$$

$$\Rightarrow \frac{5}{2}x - \frac{5}{2}y + \frac{5}{2} = 0$$

$$\Rightarrow x - y + 1 = 0 \quad \dots (i)$$

Radical axis of s and s'' : $s - s'' = 0$

$$\Rightarrow 4x - y + 7 = 0 \quad \dots (ii)$$

From Eqs. (i) and (ii), we have

$$x = -2, y = -1$$

$\therefore$ Radical centre is $P(-2, -1)$

Length of tangent from P to s'

$$\begin{aligned} &= \sqrt{(-2)^2 + (-1)^2 + \frac{3}{2}(-2) + \frac{5}{2}(-1) + \frac{9}{2}} \\ &= \sqrt{4 + 1 - 3 + 2} = \sqrt{4} = 2 \end{aligned}$$

Question 25

When the axes are rotated through an angle θ about origin in anti-clockwise direction and then translated to the new origin $(2, -2)$, if the transformed equation the equation of $x^2 + y^2 = 4$ is

$X^2 + Y^2 + aX + bY + c = 0$ then $a + b + c =$

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

4

B.

8

C.

0

D.

12

Answer: A

Solution:

Substitute $x = X + 2$ and $y = Y - 2$ into the original equation, $x^2 + y^2 = 4$ we get;

$$\begin{aligned}(X + 2)^2 + (Y - 2)^2 &= 4 \\ \Rightarrow X^2 + 4 + 4X + Y^2 + 4 - 4Y &= 4 \\ \Rightarrow X^2 + Y^2 + 4X - 4Y + 4 &= 0 \\ \therefore a + b + c &= 4 - 4 + 4 = 4\end{aligned}$$

Question26

From a point $P(-4, 0)$, two tangents are drawn to the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ touching the circle at A and B . If the equation of the circle passing through P, A and B is $x^2 + y^2 + 2gx + 2fy + c = 0$, then $(g, f) =$

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

$$\left(-1, \frac{3}{2}\right)$$

B.

$$\left(\frac{3}{2}, -1\right)$$

C.

$$\left(\frac{1}{2}, \frac{-3}{2}\right)$$

D.

$$\left(1, \frac{-3}{2}\right)$$

Answer: D

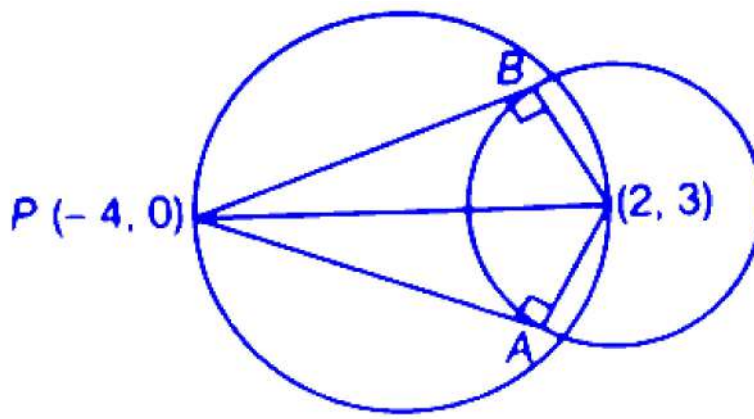
Solution:

Given, point $P(-4, 0)$

Equation of circle

$$x^2 + y^2 - 4x - 6y - 12 = 0$$

Centre $(2, 3)$



Centre of circle passing through P, A, B is mid point of OP

$$\therefore \left(\frac{-4+2}{2}, \frac{0+3}{2}\right) = \left(-1, \frac{3}{2}\right)$$

$$\therefore (g, f) = \left(1, \frac{-3}{2}\right)$$

Question27

If the equation of the polar of the point $(\alpha, -1)$ with respect to the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ is $y = \beta$, then $4(\alpha + \beta) =$

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

B.

7

C.

-6

D.

0

Answer: A

Solution:

Equation of polar is $T = 0$

$$\text{i.e., } \alpha x + y(-1) - 4 \cdot \frac{1}{2}(x + \alpha)$$

$$-6 \cdot \frac{1}{2}(y + (-1)) - 12 = 0$$

$$\Rightarrow \alpha x - y - 2x - 2\alpha - 3y + 3 - 12 = 0$$

$$\Rightarrow (\alpha - 2)x - 4y - 2\alpha - 9 = 0 \quad \dots (i)$$

But equation of polar is $y = \beta$ given

So, coefficient of x must be zero.

$$\text{i.e., } \alpha - 2 = 0 \Rightarrow \alpha = 2$$

$$\text{Thus, } 0 - 4y - 2 \times (2) - 9 = 0 \quad (\text{from Eq. (i)})$$

$$\Rightarrow -4y = 13$$

$$\Rightarrow y = \frac{-13}{4} = \beta \quad (\text{given})$$

$$\therefore 4(\alpha + \beta) = 4 \left(2 - \frac{13}{4} \right) = -5$$

Question 28

If θ is the angle between the tangents drawn from the point $(-1, -1)$ to the circle $x^2 + y^2 - 4x - 6y + c = 0$ and $\cos \theta = -\frac{7}{25}$, then the radius of the circle is

Options:

A.

4

B.

1

C.

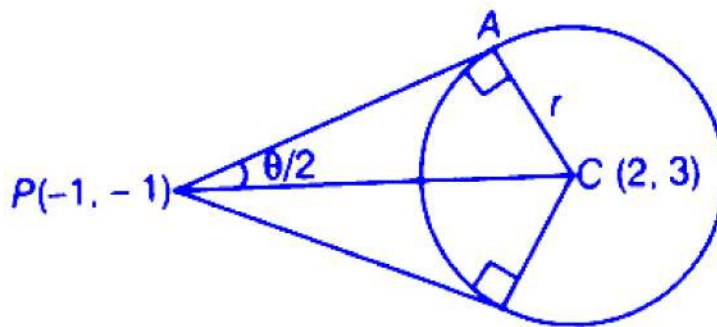
2

D.

3

Answer: A**Solution:**

Here centre of circle = $C = (2, 3)$ let $P \equiv (-1, -1)$ and $CP = 5$



Let radius = r

$$\therefore \sin \frac{\theta}{2} = \frac{r}{CP}$$

$$\Rightarrow r^2 = (5)^2 \sin^2 \frac{\theta}{2}$$

$$\therefore \cos \theta = \frac{-7}{25} \quad (\text{given})$$

$$1 - 2 \sin^2 \frac{\theta}{2} = \frac{-7}{25} \Rightarrow 2 \sin^2 \frac{\theta}{2} = 1 + \frac{7}{25}$$

$$\Rightarrow \sin^2 \frac{\theta}{2} = \frac{16}{25} = 25 \cdot \frac{16}{25} = 16$$

$$\therefore r = 4 \quad (\because r > 0)$$

Question29

If the power of the point $(1, 6)$ with respect to the circle $x^2 + y^2 + 4x - 6y - a = 0$ is **-16**, then $a =$

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

7

B.

11

C.

13

D.

21

Answer: D

Solution:

As we know that, power of a point

$$\text{W.r.t a circle} = \left(\sqrt{S_1}\right)^2 = S_1$$

$$\therefore 1 + 36 + 4 - 36 - a = -16$$

$$\Rightarrow a = 5 + 16 = 21$$

Question30

The radius of the circle passing through the points of intersection of the circles $x^2 + y^2 + 2x + 4y + 1 = 0$, $x^2 + y^2 - 2x - 4y - 4 = 0$ and intersecting the circle $x^2 + y^2 = 6$ orthogonally is

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

$$\sqrt{19}$$

B.

5

C.

$$\sqrt{39}$$

D.

4

Answer: C

Solution:

The equation of a circle passing through the intersection of two circles

$$x^2 + y^2 + 2x + 4y + 1 = 0 \text{ and}$$

$$x^2 + y^2 - 2x - 4y - 4 = 0 \text{ is}$$

$$(x^2 + y^2 + 2x + 4y + 1)$$

$$+ \lambda (x^2 + y^2 - 2x - 4y - 4) = 0, \text{ where } \lambda$$

is a parameter.

$$\Rightarrow (1 + \lambda)x^2 + (1 + \lambda)y^2 + (2 - 2\lambda)x + (4 - 4\lambda)y + 1 - 4\lambda = 0$$

$$\Rightarrow x^2 + y^2 + \frac{2(1 - \lambda)}{(1 + \lambda)}x + \frac{4(1 - \lambda)}{(1 + \lambda)}y + \frac{1 - 4\lambda}{1 + \lambda} = 0 \dots (i)$$

Applying the orthogonal condition,

$$2 \left(\frac{1 - \lambda}{1 + \lambda} \right) \times 0 + 2 \times \frac{2(1 - \lambda)}{(1 + \lambda)} \times 0$$

$$= \frac{1 - 4\lambda}{1 + \lambda} - 6$$

$$\Rightarrow \frac{1 - 4\lambda}{1 + \lambda} = 6$$

$$\Rightarrow 6 + 6\lambda = 1 - 4\lambda \Rightarrow \lambda = \frac{-1}{2}$$

Substituting $\lambda = \frac{-1}{2}$ in Eq. (i) and simplifying we get

$$x^2 + y^2 + 6x + 12y + 6 = 0$$

$\therefore$ Radius of this circle

$$= \sqrt{3^2 + 6^2 - 6} = \sqrt{39}$$

Question31

A circle passing through the point $(1, 0)$ makes an intercept of length 4 units on X -axis and an intercept of length $2\sqrt{11}$ units on Y -axis. If the centre of the circle lies in the fourth quadrant, then the radius of the circle is

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

$$4\sqrt{5}$$

B.

$$3$$

C.

$$2\sqrt{5}$$

D.

$$5$$

Answer: C

Solution:

Let the x -intercepts be $(a, 0)$ and

$$(a + 4, 0)$$

Let the y -intercepts be $(0, b)$ and

$$(0, b + 2\sqrt{11})$$

$$\text{Center of circle} = (a + 2, b + \sqrt{11})$$

Since, the center is in the fourth quadrant $h > 0$ and $k < 0$

$$\Rightarrow a + 2 > 0 \text{ and } b + \sqrt{11} < 0$$

$$\text{Equation of circle } (x - h)^2 + (y - k)^2 = r^2$$

$$\Rightarrow (1 - (a + 2))^2 + (0 - (b + \sqrt{11}))^2 = r^2$$

$$\Rightarrow (-a - 1)^2 + (-b - \sqrt{11})^2 = r^2$$

Substituting $(a, 0)$ gives

$$(a - (a + 2))^2 + (0 - (b + \sqrt{11}))^2 = r^2$$

$$\Rightarrow (-2)^2 + (-b - \sqrt{11})^2 = r^2$$

substituting $(a + 4, 0)$ gives

$$\Rightarrow (a + 4 - (a + 2))^2 + (0 - (b + \sqrt{11}))^2 = r^2$$

$$\Rightarrow 2^2 + (-b - \sqrt{11})^2 = r^2$$

Similarly, $(0, b)$ gives

$$(0 - (a + 2))^2 + (b - (b + \sqrt{11}))^2 = r^2$$

substituting $(0, b + 2\sqrt{11})$

$$(0 - (a + 2))^2 + (b + 2\sqrt{11} - (b + \sqrt{11}))^2 = r^2$$

$$(-a - 2)^2 + (\sqrt{11})^2 = r^2$$

Solving above equation, we get

$$r^2 = 20$$

$$r = 2\sqrt{5}$$

Question32

If $(\frac{1}{10}, \frac{-1}{5})$ is the inverse point of a point $(-1, 2)$ with respect to the circle $x^2 + y^2 - 2x + 4y + c = 0$ then $c =$

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

4

B.

-4

C.

2

D.

-2

Answer: B

Solution:

The inverse point is $\left(\frac{1}{10}, \frac{-1}{5}\right)$

The original point is $(-1, 2)$

The circle equation is

$$x^2 + y^2 - 2x + 4y + c = 0$$

Centre of the circle is $(1, -2)$

The original point is $(-1, 2)$

$$\begin{aligned}\text{The distance} &= \sqrt{(-1 - 1)^2 + (2 - (-2))^2} \\ &= \sqrt{4 + 16} = \sqrt{20}\end{aligned}$$

The centre of the circle is $(1, -2)$

The inverse point is $\left(\frac{1}{10}, \frac{-1}{5}\right)$

$$\begin{aligned}\text{The distance} &= \sqrt{\left(\frac{1}{10} - 1\right)^2 + \left(\frac{-1}{5} + 2\right)^2} \\ &= \frac{9\sqrt{5}}{10}\end{aligned}$$

$$\Rightarrow r^2 = d_1 d_2 = \sqrt{20} \times \frac{9\sqrt{5}}{10} = 9$$

$$\Rightarrow 5 - c = 9 \Rightarrow c = -4$$

Question33

If the equation of the circle lying in the first quadrant, touching both the coordinate axes and the line $\frac{x}{3} + \frac{y}{4} = 1$ is

$$(x - c)^2 + (y - c)^2 = c^2, \text{ then } c =$$

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

1 or 4

B.

2 or 3

C.

1 or 6

D.

2 or 5

Answer: C

Solution:

$$\frac{x}{3} + \frac{y}{4} = 1$$

$$\Rightarrow 4x + 3y = 12$$

$$\Rightarrow 4x + 3y - 12 = 0$$

distance from the centre (c, c) to the line

$$4x + 3y - 12 = 0$$

$$d = \frac{|4c + 3c - 12|}{\sqrt{25}}$$

$$= \frac{|4c + 3c - 12|}{5}$$

$$\Rightarrow \frac{|7c - 12|}{5} = c$$

$$\Rightarrow |7c - 12| = 5c$$

$$\Rightarrow 7c - 12 = 5c \text{ and } 7c - 12 = -5c$$

$$c = 6, c = 1$$

Question 34

If the point of contact of the circles $x^2 + y^2 - 6x - 4y + 9 = 0$ and $x^2 + y^2 + 2x + 2y - 7 = 0$ is (α, β) , then $7\beta =$

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

$$5\alpha$$

B.

$$2\alpha$$

C.

$$3\alpha$$

D.

$$4\alpha$$

Answer: D

Solution:

$$\text{Circle I : } x^2 + y^2 - 6x - 4y + 9 = 0$$

$$\text{Circle II : } x^2 + y^2 + 2x + 2y - 7 = 0$$

$$\text{Point of contact : } (\alpha, \beta)$$

$$\Rightarrow (x - 3)^2 + (y - 2)^2 = 4$$

$$\text{Center} = (3, 2)$$

$$(x + 1)^2 + (y + 1)^2 = 9$$

$$\text{centre } (-1, -1)$$

$$\text{Slope of line } (m) = \frac{3}{4}$$

$$\text{Equation of line } y - 2 = \frac{3}{4}(x - 3)$$

$$\Rightarrow 3x - 4y - 1 = 0$$

$$\text{The point of contact } (\alpha, \beta) \text{ lies on the line } 3x - 4y - 1 = 0$$

$\Rightarrow$ Radii are $r_1 = 2$ and $r_2 = 3$

Ratio = 2 : 3

Using section formula

$$\alpha = \frac{2(-1) + 3(3)}{2 + 3} = \frac{7}{5}, \beta = \frac{2(-1) + 3(2)}{2 + 3}$$

$$\Rightarrow \beta = \frac{4}{5} \Rightarrow 7\beta = 7 \times \frac{4}{5} = 4 \cdot \frac{7}{5}$$

$$\therefore 7\beta = 4\alpha$$

Question 35

If the circles $x^2 + y^2 - 2\lambda x - 2y - 7 = 0$ and $3(x^2 + y^2) - 8x + 29y = 0$ are orthogonal, then $\lambda =$

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

4

B.

3

C.

2

D.

1

Answer: D

Solution:

$$x^2 + y^2 - \frac{8}{3}x + \frac{29}{3}y = 0$$

For the first circle, $g_1 = -\lambda$, $f_1 = -1$, $c_1 = -7$ for the second circle $g_2 = \frac{-4}{3}$,

$$f_2 = \frac{29}{6}, c_2 = 0$$

Orthogonality condition.

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$\Rightarrow 2(-\lambda) \left(\frac{-4}{3} \right) + 2(-1) \left(\frac{29}{6} \right) = -7 + 0$$

$$\Rightarrow \frac{8}{3}\lambda - \frac{29}{3} = -7$$

$$\Rightarrow \frac{8}{3}\lambda = \frac{29}{3} - 7$$

$$\Rightarrow \frac{8}{3}\lambda = \frac{8}{3} \Rightarrow \lambda = 1$$

Question36

If Q is the inverse point of $P(-1, 1)$ with respect to the circle $x^2 + y^2 - 2x + 2y = 0$, then the line containing Q is

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

$$x - 3y - 2 = 0$$

B.

$$x - y + 1 = 0$$

C.

$$x + y - 2 = 0$$

D.

$$2x - 3y + 5 = 0$$

Answer: A

Solution:

Here Q is the inverse point of $P(-1, 1)$ with respect to the circle

$$x^2 + y^2 - 2x + 2y = 0$$

$$\Rightarrow (x - 1)^2 + (y + 1)^2 = 2$$

Centre of circle $O \equiv (1, -1)$

$$\therefore OP \cdot OQ = r^2$$

$$\Rightarrow OQ = \frac{r^2}{OP} = \frac{2}{\sqrt{(1+1)^2 + (-1-1)^2}}$$
$$= \frac{2}{\sqrt{8}} = \frac{1}{\sqrt{2}}$$

OP, OQ are collinear

$$\therefore \text{Slope of } OP = -1$$

Q point lie on OPQ

$$\therefore \frac{x-1}{\cos \theta} = \frac{y+1}{\sin \theta} = OQ$$

$$\Rightarrow \frac{x-1}{\frac{-1}{\sqrt{2}}} = \frac{y+1}{\frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}}$$

$$x-1 = \frac{-1}{2}, \quad y+1 = \frac{1}{2}$$

$$x = \frac{1}{2}, \quad y = \frac{-1}{2}$$

$$\therefore Q \equiv \left(\frac{1}{2}, \frac{-1}{2}\right) \text{ lie on line}$$

$$x - 3y - 2 = 0$$

Question37

If the circle passing through $(3, 5)$, $(5, 5)$ and $(3, -3)$ cuts the circle $x^2 + y^2 + 2x + 2fy = 0$ orthogonally, then $f =$

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

-12

B.

-3

C.

-15

D.

Answer: D**Solution:**

Equation of circle passes through $(3, 5)$, $(5, 5)$ and $(3, -3)$ is

Let equation be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \dots (i)$$

This is passes through $(3, 5)$, $(5, 5)$ and $(3, -3)$

$$6g + 10f + c = -34 \quad \dots (ii)$$

$$10g + 10f + c = 50 \quad \dots (iii)$$

$$6g - 6f + c = -18 \quad \dots (iv)$$

From Eqs. (i), (ii) and (iii) $g = -4$,

$$f = -1 \text{ and } c = 0$$

Required equation of circle

$$x^2 + y^2 - 8x - 2y = 0 \quad \dots (v)$$

and given circle is

$$x^2 + y^2 + 2x + 2fy = 0$$

are orthogonal

$$2(-4)(1) + 2(-1)f = 0 + 0$$

$$f = -4$$

Question38

Length of the common chord of two circles of same radius is $2\sqrt{17}$. If one of the two circles is $x^2 + y^2 + 6x + 4y - 12 = 0$, then acute angle between the two circles is

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

$$\frac{\pi}{2}$$

B.

$$\sin^{-1}\left(\frac{3}{5}\right)$$

C.

$$\cos^{-1}\left(\frac{9}{25}\right)$$

D.

$$\tan^{-1}\left(\frac{9}{17}\right)$$

Answer: C

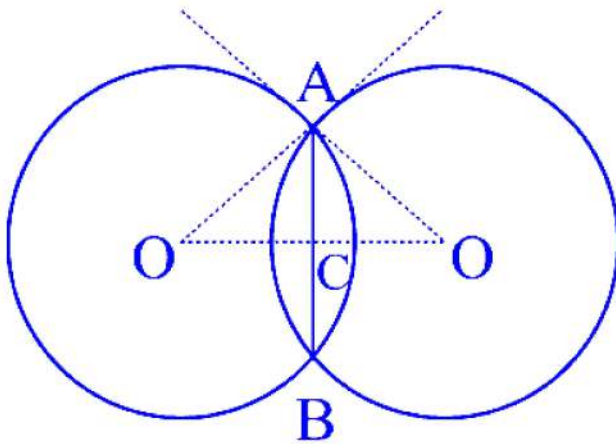
Solution:

Given that length of common chord of two circles = $2\sqrt{17}$, with same radii one circle is

$$\Rightarrow x^2 + y^2 + 6x + 4y - 12 = 0$$

$$r = \sqrt{3^2 + 2^2 + 12} = 5$$

both circles have same radii = 5



$$AB = 2\sqrt{17}$$

$$AC = \sqrt{17}$$

In $\triangle AO'C$

$$O'C^2 = O'A^2 - AC^2$$

$$= 25 - 17 \Rightarrow O'C = 2\sqrt{2}$$

$$OO' = 4\sqrt{2}$$

$$\Rightarrow \cos \theta = \frac{O'A^2 + OA^2 - O'O^2}{2(OA')(OA)}$$

$$\Rightarrow \cos \theta = \frac{25 + 25 - 32}{50} \Rightarrow \cos \theta = \frac{9}{25}$$

$$\theta = \cos^{-1}\left(\frac{9}{25}\right)$$

Question39

A circle $S \equiv x^2 + y^2 - 16 = 0$ intersects another circle $S' = 0$ of radius 5 units such that their common chord is of maximum length. If the slope of that chord is $\frac{3}{4}$, then the centre of such a circle $S' = 0$ is

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

$$\left(\frac{9}{5}, \frac{12}{5}\right)$$

B.

$$\left(\frac{5}{9}, -\frac{12}{5}\right)$$

C.

$$\left(-\frac{9}{5}, \frac{12}{5}\right)$$

D.

$$\left(\frac{3}{5}, \frac{4}{5}\right)$$

Answer: C

Solution:

Given that

$$S \equiv x^2 + y^2 - 16 = 0$$

Let equation of

$$S' \equiv (x - h)^2 + (y - k)^2 = 5^2$$

Common chord $S_2 - S_1 = 0$

$$2xh + 2ky + 9 = k^2 + h^2$$

Given slope = $\frac{3}{4}$

$$\frac{-2h}{2k} = \frac{3}{4} \Rightarrow h = -\frac{3}{4}k$$

The chord of maximum length is diameter

$\therefore (0, 0)$ lies on common chord

$$h^2 + k^2 = 9$$

$$\frac{9}{16}k^2 + k^2 = 9 \Rightarrow k = \pm \frac{12}{5}$$

$$h = \mp \frac{9}{5}$$

$$\text{re} \equiv \left(\frac{-9}{5}, \frac{12}{5} \right), \left(\frac{9}{5}, \frac{-12}{5} \right)$$

Question40

Let θ be the angle between the circles $S \equiv x^2 + y^2 + 2x - 2y + c = 0$ and $S' \equiv x^2 + y^2 - 6x - 8y + 9 = 0$. If c is an integer and $\cos \theta = \frac{5}{16}$, then the radius of the circle $S = 0$ is

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

2

B.

4

C.

3

D.

1

Answer: A

Solution:

Given equation of circles,

$$S \equiv x^2 + y^2 + 2x - 2y + c = 0$$

$$S' \equiv x^2 + y^2 - 6x - 8y + 9 = 0$$

$$(g_1, f_1) \equiv (-1, 1), c_1 = c$$

$$(g_2, f_2) \equiv (3, 4), c_2 = 9$$

Angle between circle S and S' be θ

$$\cos \theta = \frac{2(g_1g_2 + f_1f_2) - (c_1 + c_2)}{2\sqrt{g_1^2 + f_1^2 - c_1}\sqrt{g_2^2 + f_2^2 - c_2}}$$

$$\Rightarrow \cos \theta = \frac{2(-3 + 4) - c - 9}{2\sqrt{2 - c}\sqrt{25 - 9}}$$

$$\Rightarrow \frac{5}{16} = \frac{-c - 7}{2\sqrt{2 - c}(4)}$$

$$\Rightarrow 5 \times \sqrt{2 - c} = -2(c + 7)$$

$$\Rightarrow 25(2 - c) = 4(c^2 + 14c + 49)$$

$$\Rightarrow 4c^2 + 81c + 146 = 0$$

$$\Rightarrow c = -2\frac{-146}{8}$$

$$\therefore c = -2$$

$$\text{Radius} = \sqrt{1 + 1 + 2} = 2$$

Question41

If a circle S passes through the origin and makes an intercept of length 4 units on the line $x = 2$, then the equation of the curve on which the centre of S lies is

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

$$y^2 - 4x = 8$$

B.

$$y^2 + 4x = 8$$

C.

$$x^2 + 4y = 8$$

D.

$$x^2 - 4y = 8$$

Answer: B

Solution:

Let centre (h, k) and radius be r of circle S

$$\therefore S \equiv (x - h)^2 + (y - k)^2 = r^2$$

Circle passes through origin

$$\therefore r^2 = h^2 + k^2$$

Given, the line $x = 2$ intersects the circle at two points let these points be $(2, y_1)$ and $(2, y_2)$ mid-points of these points is $(2, k)$

Now, the distance of $(2, k)$ to $(2, y_1)$ is $2 y_1 = k + 2$ and $y_2 = k - 2$

$(2, k + 2)$ and $(2, k - 2)$ are intersection point radius of the circle.

$$h^2 + k^2 = (h - 2)^2 + (k - (k + 2))^2$$

$$h^2 + k^2 = h^2 + 4 - 4h + 4$$

$$k^2 = -4h + 8 \Rightarrow k^2 + 4h = 8$$

Taking locus, we get

$$y^2 + 4x = 8$$

Question42

A circle touches the line $2x + y - 10 = 0$ at $(3, 4)$ and passes through the point $(1, -2)$. Then, a point that lies on the circle is

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

$(5, 4)$

B.

$(4, 5)$

C.

$(-5, 4)$

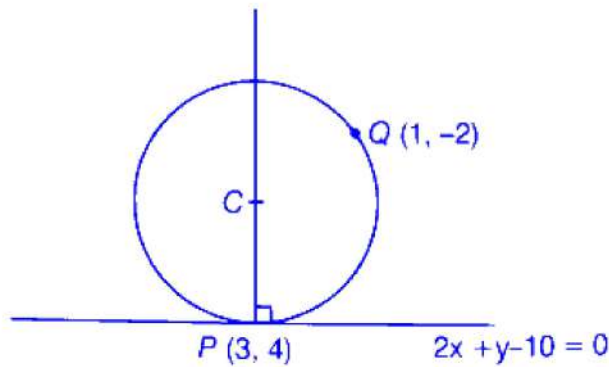
D.

$(4, -5)$

Answer: C

Solution:

Given, line $2x + y - 10 = 0$ touches the circle at the point $(3, 4)$.



$\therefore$ Perpendicular to this line passes through centre of circle.

Equation of line perpendicular to $2x + y - 10 = 0$ is

$$x - 2y + \lambda = 0$$

$$\Rightarrow 3 - 8 + \lambda = 0 \Rightarrow \lambda = 5$$

$$\therefore x - 2y + 5 = 0$$

Centre of the circle be $C(2k - 5, k)$

Now radius $r^2 = CP^2 = CQ^2$

$$r^2 = (2k - 8)^2 + (k - 4)^2 = (2k - 6)^2 + (k + 2)^2$$

On solving this, we get $k = 2$

$\therefore$ Centre $(-1, 2)$ and radius $= \sqrt{20}$

Equation of circle be

$$(x + 1)^2 + (y - 2)^2 = 20$$

Clearly $(-5, 4)$ lies on the circle.

Question43

If (a, b) is the common point for the circles $x^2 + y^2 - 4x + 4y - 1 = 0$ and $x^2 + y^2 + 2x - 4y + 1 = 0$, then $a^2 + b^2 =$

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

$$\frac{1}{5}$$

B.

$$5$$

C.

$$25$$

D.

$$\frac{1}{25}$$

Answer: A

Solution:

(a, b) is the common point for the two given circle

$$a^2 + b^2 - 4a + 4b - 1 = 0 \quad \dots (i)$$

$$a^2 + b^2 + 2a - 4b + 1 = 0 \quad \dots (ii)$$

$$\begin{array}{ccccccc} - & - & - & + & - & & \\ \hline \end{array}$$

Subtracting Eqs. (ii) from (i), we get

$$-6a + 8b - 2 = 0$$

$$-3a + 4b - 1 = 0$$

$$4b = 3a + 1 \quad \dots (iii)$$

From Eqs. (i) and (iii), we get

$$a^2 + \left(\frac{3a+1}{4}\right)^2 - 4a + 3a + 1 - 1 = 0$$

$$\Rightarrow 25a^2 - 10a + 1 = 0$$

$$(5a - 1)^2 = 0$$

$$a = \frac{1}{5} \text{ and } b = \frac{2}{5}$$

$$\text{Now, } a^2 + b^2 = \frac{1}{25} + \frac{4}{25} = \frac{5}{25} = \frac{1}{5}$$

Question44

The angle between the tangents drawn from the point $(2, 2)$ to the circle $x^2 + y^2 + 4x + 4y + c = 0$ is $\cos^{-1} \left(\frac{7}{16} \right)$. If two such circles exist, then sum of the values of c is

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

16

B.

20

C.

-20

D.

-16

Answer: D

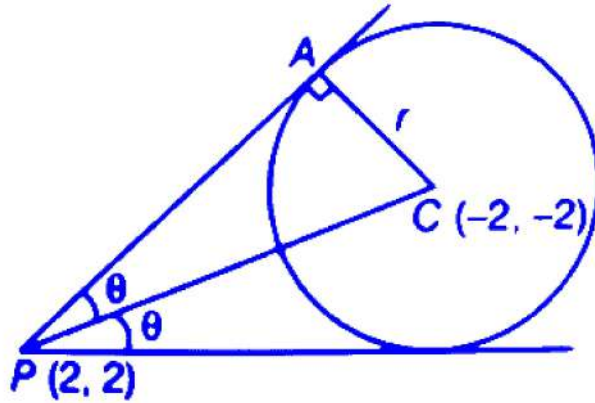
Solution:

We have the circle

$$x^2 + y^2 + 4x + 4y + c = 0$$

$$C = \text{Centre} = (-2, -2)$$

$$AC = \text{radius} = \sqrt{8 - C}$$



Angle between tangents

$$\cos 2\theta = \frac{7}{16}, CP = \sqrt{16 + 16} = 4\sqrt{2}$$

$$\therefore \sin \theta = \frac{3}{4\sqrt{2}}$$

$$\text{Using } \sin \theta = \frac{r}{CP} = \frac{\sqrt{8-C}}{4\sqrt{2}} = \frac{3}{4\sqrt{2}}$$

$$\therefore 8 - C = 9$$

$$C = -1$$

Now, another circle is possible when angles be supplementary

$\therefore \theta$ is changed by $\pi - 2\theta$

$$\therefore \sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

$$\therefore \cos \theta = \sqrt{\frac{1 + \cos 2\theta}{2}} = \frac{\sqrt{23}}{4\sqrt{2}}$$

$$\therefore \cos \theta = \frac{\sqrt{8-C}}{4\sqrt{2}} = \frac{\sqrt{23}}{4\sqrt{2}}$$

$$\therefore 8 - c = 23$$

$$-c = -15$$

$\therefore$ Now, sum of all possible values of C .

$$= -1 - 15 = -16$$

Question 45

If the circle $S = x^2 + y^2 + 2gx + 4y + 1 = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x - 3 = 0$, then the radius of circle $S = 0$ is

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

5

B.

$\sqrt{12}$

C.

25

D.

12

Answer: B

Solution:

We have the circle S

$$S = x^2 + y^2 + 2gx + 4y + 1 = 0$$

and the second circle be S_1 ,

$$S_1 = x^2 + y^2 - 2x - 3 = 0$$

$\therefore$ Centre $(1, 0)$ and radius $= 2$

Now, common chord of circles S and S_1 is

$$S - S_1 = 0$$

$$(2g + 2)x + 4y + 4 = 0$$

If a circle bisects the circumference of another circle, then common chord is a diameter of the second circle.

$$2g + 2 + 4 = 0 \Rightarrow g = -3$$

$$\therefore S = x^2 + y^2 - 6x + 4y + 1 = 0$$

Centre $(3, -2)$

$$\text{Radius} = \sqrt{9 + 4 - 1} = \sqrt{12}$$

Question46

From a point P on the circle $x^2 + y^2 = 4$, two tangents are drawn to the circle $x^2 + y^2 - 6x - 6y + 14 = 0$. If A and B are the points of contact of those lines, then the locus of the centre of the circle passing through the points P , A and B is

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

$$x^2 + y^2 - 3x - 3y + 4 = 0$$

B.

$$2x^2 + 2y^2 + 6x + 6y - 7 = 0$$

C.

$$x^2 + y^2 + 3x + 3y - 4 = 0$$

D.

$$2x^2 + 2y^2 - 6x - 6y + 7 = 0$$

Answer: D

Solution:

Circle 1 given $x^2 + y^2 = 4$

Circle 2 is given,

$$x^2 + y^2 - 6x - 6y + 14 = 0$$

Centre $(3, 3)$ radius $= 2$

Let $P(x_1, y_1)$ be a point on circle 1 .

$\therefore$ Equation of chord of contact AB is

$$xx_1 + yy_1 - 3(x + x_1) - 3(y + y_1) + 14 = 0$$

$$(x_1 - 3)x + (y_1 - 3)y - 3x_1 - 3y_1 + 14 = 0$$

Now, equation of circle passing through P , A and B is

$$x^2 + y^2 - 6x - 6y + 14 + \lambda [(x_1 - 3)x + (y_1 - 3)y - 3x_1 - 3y_1 + 14] = 0$$

We have this circle passes through $P(x_1, y_1)$

$$\Rightarrow x_1^2 + y_1^2 - 6x_1 - 6y_1 + 14 + \lambda [(x_1 - 3)x_1 + (y_1 - 3)y_1 - 3x_1 - 3y_1 + 14] = 0$$

$$\Rightarrow x_1^2 + y_1^2 - 6x_1 - 6y_1 + 14 + \lambda [x_1^2 + y_1^2 - 3x_1 - 3y_1 - 3x_1 - 3y_1 + 14] = 0$$

Put $x_1^2 + y_1^2 = 4$

$$\Rightarrow 4 - 6x_1 - 6y_1 + 14 + \lambda (4 - 6x_1 - 6y_1 + 14) = 0$$

$$\Rightarrow 18 - 6x_1 - 6y_1 + 14 + \lambda (18 - 6x_1 - 6y_1) = 0$$

$$\therefore 1 + \lambda = 0$$

$$\lambda = -1$$

$\Rightarrow$ Equation of circle be

$$(x^2 + y^2 - 6x - 6y + 14) - [(x_1 - 3)x + (y_1 - 3)y - 3x_1 - 3y_1 + 14] = 0$$

$$\Rightarrow x^2 + y^2 - (3 + x_1)x - (3 + y_1)y + 3x_1 + 3y_1 = 0$$

Centre $\left(\frac{3+x_1}{2}, \frac{3+y_1}{2} \right)$

$h = \frac{3+x_1}{2}$ and $k = \frac{3+y_1}{2}$

$$x_1 = 2h - 3 \quad y_1 = 2k - 3$$

$$x_1^2 + y_1^2 = 4$$

$$(2h - 3)^2 + (2k - 3)^2 = 4$$

$$4h^2 + 4k^2 - 12h - 12k + 14 = 0$$

$$h^2 + k^2 - 3h - 3k + \frac{7}{2} = 0$$

$\therefore$ Taking locus of (h, k) , we get

$$2x^2 + 2y^2 - 6x - 6y + 7 = 0$$

Question 47

If the product of the lengths of the perpendicular drawn from the ends of a diameter of the circle $x^2 + y^2 = 4$ on the line $x + y + 1 = 0$ is maximum, then the two ends of that diameter are

Options:

A.

$$(-2, 0), (2, 0)$$

B.

$$(\sqrt{3}, 1), (-\sqrt{3}, -1)$$

C.

$$(\sqrt{2}, \sqrt{2}), (-\sqrt{2}, -\sqrt{2})$$

D.

$$(0, 2), (0, -2)$$

Answer: C**Solution:**

Let the ends of diameter of the given circle $x^2 + y^2 = 4$ is

$$(2 \cos \theta, 2 \sin \theta), \text{ then other must be } (-2 \cos \theta, -2 \sin \theta)$$

Then, length of perpendicular from the extremities of diameter to the line $x + y + 1 = 0$ is

$$\begin{aligned} & \frac{2(\cos \theta + \sin \theta) + 1}{\sqrt{2}} \\ & \times \frac{-2(\cos \theta + \sin \theta) + 1}{\sqrt{2}} = P(\text{ say }) \\ \Rightarrow P &= \frac{1 - 4(\cos \theta + \sin \theta)^2}{2} \\ &= \frac{1 - 4(1 + \sin 2\theta)}{2} \\ \Rightarrow P &= \frac{-3 - 4 \sin 2\theta}{2} \\ \Rightarrow \frac{dP}{d\theta} &= \frac{1}{2}(-4) \cos 2\theta \cdot 2 = 0 \\ \Rightarrow \cos 2\theta &= 0 \Rightarrow 2\theta = \frac{\pi}{2} \\ \theta &= \frac{\pi}{4} \end{aligned}$$

Hence, the points are $(\sqrt{2}, \sqrt{2})$ and $(-\sqrt{2}, -\sqrt{2})$

Question 48

If the intercept made by a variable circle on the X -axis and Y -axis are 8 and 6 units respectively, then the locus of the centre of the circle is

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

$$x^2 - y^2 + 28 = 0$$

B.

$$y^2 - x^2 - 7 = 0$$

C.

$$x^2 - y^2 - 28 = 0$$

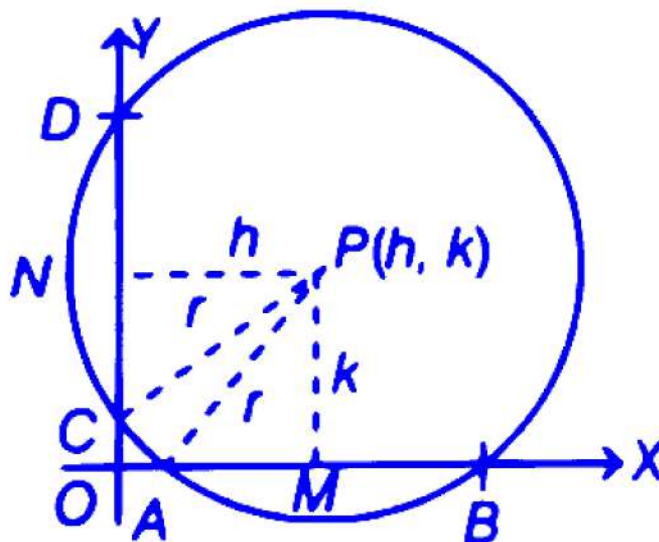
D.

$$x^2 - y^2 - 7 = 0$$

Answer: D

Solution:

Let the centre of circle be (h, k) and radius be r .



Clearly, $PM = k$

$$PN = h$$

$$AM = 4, CN = 3$$

$$r^2 = 4^2 + k^2$$

$$r^2 = 3^2 + h^2$$

$$\therefore 16 + k^2 = 9 + h^2$$

$$h^2 - k^2 - 7 = 0$$

Taking locus of $P(h, k)$, we get

$$x^2 - y^2 - 7 = 0$$

Question49

The slope of the non-vertical tangent drawn from the point $(3, 4)$ to the circle $x^2 + y^2 = 9$ is

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

$$\frac{2}{3}$$

B.

$$\frac{3}{2}$$

C.

$$\frac{7}{24}$$

D.

$$\frac{24}{7}$$

Answer: C

Solution:

Equation of tangent to the circle,

$$x^2 + y^2 = 9 \text{ is } y = mx + 3\sqrt{1+m^2}$$

Tangent passes through the point (3, 4),

$$(4 - 3m)^2 = 9(1 + m^2)$$

$$\Rightarrow 16 + 9m^2 - 24m = 9 + 9m^2$$

$$\Rightarrow 24m = 7$$

$$m = \frac{7}{24}$$

Question50

If the acute angle between the circles

$$S \equiv x^2 + y^2 + 2kx + 4y - 3 = 0 \text{ and}$$

$$S' \equiv x^2 + y^2 - 4x + 2ky + 9 = 0 \text{ is } \cos^{-1} \left(\frac{3}{8} \right) \text{ and the centre of}$$

$S' = 0$ lies in the first quadrant, then the radical axis of $S = 0$ and $S' = 0$ is

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

$$x - 5y + 6 = 0$$

B.

$$x - 5y - 4 = 0$$

C.

$$5x - y - 6 = 0$$

D.

$$5x - y - 4 = 0$$

Answer: A

Solution:

$$S = x^2 + y^2 + 2kx + 4y - 3 = 0$$

$$C_1 = (-k, -2) \text{ radius } r_1 = \sqrt{k^2 + 4 + 3}$$

$$S' = x^2 + y^2 - 4x + 2ky + 9 = 0$$

$$C_2 = (2, -k) \text{ radius } = \sqrt{4 + k^2 - 9}$$

$$= \sqrt{k^2 - 5}$$

Now, angle between two circles $S = 0$ and $S' = 0$ is

$$\cos(180^\circ - \theta) = \frac{(r_1^2 + r_2^2 - d^2)}{2r_1 r_2}$$

where r_1, r_2 are the radii and d is the distance between their centres.

$$= \frac{k^2 + 7 + k^2 - 5 - (k + 2)^2 - (k - 2)^2}{2\sqrt{k^2 + 7}\sqrt{k^2 - 5}}$$

$$\Rightarrow -\cos \theta = \frac{2k^2 + 2 - (2k^2 + 8)}{2\sqrt{k^2 + 7}\sqrt{k^2 - 5}}$$

$$= \frac{-6}{2\sqrt{k^2 + 7}\sqrt{k^2 - 5}}$$

$$\Rightarrow \cos \theta = \frac{3}{\sqrt{k^2 + 7}\sqrt{k^2 - 5}} = \frac{3}{8}$$

$$\Rightarrow (k^2 + 7)(k^2 - 9) = 64$$

$$\Rightarrow k^4 + 2k^2 - 99 = 0$$

$$\Rightarrow (k^2 + 11)(k^2 - 9) = 0 \Rightarrow k^2 = 9$$

$$\Rightarrow k = \pm 3$$

Centre of $S' = 0$ lies in I quadrant

$$[\therefore k = -3]$$

Now, radical axis of $S = 0$ and $S' = 0$ is

$$x(2k + 4) + y(4 - 2k) - 12 = 0$$

$$\Rightarrow -2x + 10y - 12 = 0$$

$$x - 5y + 6 = 0$$

Question 51

The circumference of a circle passing through the point $(4, 6)$ with two normals represented by $2x - 3y + 4 = 0$ and $x + y - 3 = 0$ is

AP EAPCET 2024 - 23th May Morning Shift

Options:

A. 5π

B. 10π

C. 25π

D. 8π

Answer: B

Solution:

The two given normal equations pass through the center of the circle. By solving the equations $2x - 3y + 4 = 0$ and $x + y - 3 = 0$, we find:

$$x = 1, \quad y = 2$$

Thus, the coordinates of the center of the circle are $(1, 2)$. Given that the point $(4, 6)$ lies on the circumference of the circle, we can calculate the radius as follows:

$$\text{Radius} = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{9 + 16} = 5$$

The circumference of the circle is then given by:

$$2\pi r = 10\pi$$

Question52

If the line through the point $P(5, 3)$ meets the circle $x^2 + y^2 - 2x - 4y + \alpha = 0$ at $A(4, 2)$ and $B(x_1, y_1)$, then $PA \cdot PB$ is equal to

AP EAPCET 2024 - 23th May Morning Shift

Options:

A. 6

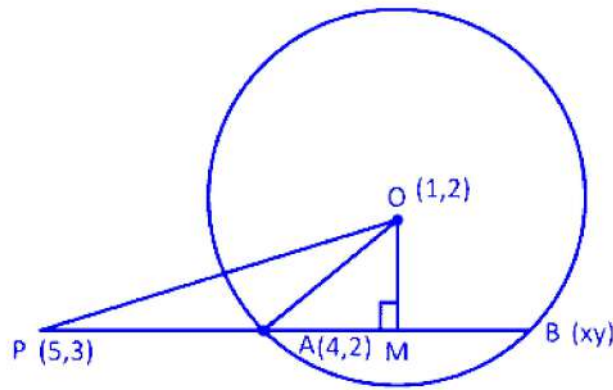
B. 12

C. 9

D. 8

Answer: D

Solution:



Coordinate of center of circle $x^2 + y^2 - 2x - 4y = 0$ is $(1, 2)$

$OM \perp AB$ and M , also M is mid-point of AB .

Radius of circle $= \sqrt{(4-1)^2 + (2-2)^2} = 3$

Equation of line PB is

$$y - 3 = \left(\frac{2-3}{4-5} \right) (x - 5)$$

$$x - y = 2$$

$$OM = \frac{|1 - 2 - 2|}{\sqrt{1^2 + 1^2}} = \frac{3}{\sqrt{2}}$$

$$\begin{aligned} AM &= \sqrt{OA^2 - OM^2} \\ &= \sqrt{9 - \frac{9}{2}} = \frac{3}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} PB &= PA + 2AM \\ &= \sqrt{(5-4)^2 + (3-2)^2} + 2 \times \frac{3}{\sqrt{2}} \\ &= \sqrt{2} + \frac{6}{\sqrt{2}} = \frac{8}{\sqrt{2}} \end{aligned}$$

$$\text{Now, } PA \times PB = \sqrt{2} \times \frac{8}{\sqrt{2}} = 8.$$

Question53

Consider the point $P(\alpha, \beta)$ on the line $2x + y = 1$. If the P and $(3, 2)$ are conjugate points with respect to the circle $x^2 + y^2 = 4$, then $\alpha + \beta$ is equal to

AP EAPCET 2024 - 23th May Morning Shift

Options:

- A. 3
- B. -1
- C. -5
- D. 7

Answer: A

Solution:

P and $A(3, 2)$ are conjugates points of the circle $x^2 + y^2 = 4$.

Polar of A is $3x + 2y = 4$

Polar of A passes through P

$$3\alpha + 2\beta = 4 \quad \dots (i)$$

Also, point P is on the line $2x + y = 1$

$$\Rightarrow 2\alpha + \beta = 1 \quad \dots (ii)$$

On solving Eqs. (i) and (ii), we get

$$\alpha = -2 \text{ and } \beta = 5$$

$$\text{Hence, } \alpha + \beta = -2 + 5 = 3.$$

Question 54

If $(1, 3)$ is the mid-point of a chord of the circle $x^2 + y^2 - 4x - 8y + 16 = 0$, then the area of the triangle formed by that chord with the coordinate axes is

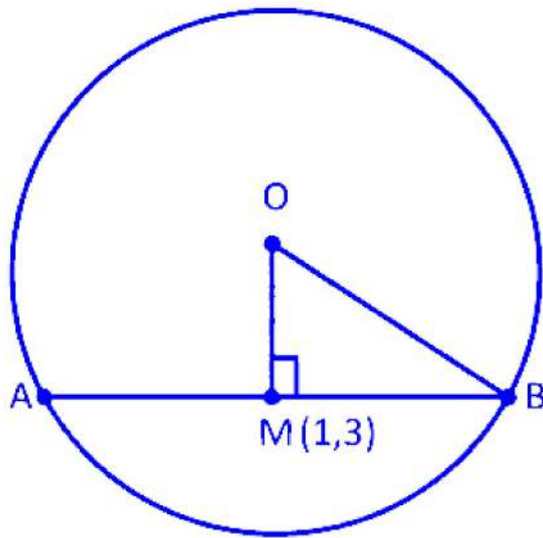
AP EAPCET 2024 - 23th May Morning Shift

Options:

- A. 16
- B. 8
- C. 4
- D. $8\sqrt{2}$

Answer: B

Solution:



Center of the given circle is $(2, 4)$ and radius of the given circle

$$\begin{aligned} &= \sqrt{4 + 16 - 16} \\ &= 2 \end{aligned}$$

As the perpendicular line from center bisects the chord,

$$AM = MB$$

$$\text{Slope of line } OM = \frac{4-3}{2-1} = 1$$

$$\text{Slope of chord } AB = -1 [\because AM \perp OM]$$

Equation of chord AB is

$$y - 3 = -1(x - 1)$$

$$x + y = 4$$

$$\frac{x}{4} + \frac{y}{4} = 1$$

$$\therefore x \text{ intercept} = y \text{ intercept} = 4$$

$\therefore$ Required area of triangle

$$= \frac{1}{2} \times 4 \times 4 = 8 \text{ sq unit}$$

Question55

If the circles $x^2 + y^2 + 2\alpha x + 2y - 8 = 0$ and $x^2 + y^2 - 2x + ay - 14 = 0$ intersect orthogonally, then the distance between their centres is

AP EAPCET 2024 - 23th May Morning Shift

Options:

A. $\sqrt{242}$

B. $\sqrt{970}$

C. $\sqrt{629}$

D. $\sqrt{541}$

Answer: C

Solution:

If the two circle meets orthogonal, then

$$\begin{aligned} 2g_1g_2 + 2f_1f_2 &= c_1 + c_2 \\ 2a(-1) + a &= -8 - 14 \\ a &= 22 \end{aligned}$$

Center of two given circle are $(-22, -1)$ and $(1, -11)$

$\therefore$ Required distance

$$\begin{aligned} &= \sqrt{(1 + 22)^2 + (-11 + 1)^2} \\ &= \sqrt{529 + 100} \\ &= \sqrt{629} \end{aligned}$$

Question56

If the axes are rotated through angle ' α ', then the number of values of α such that the transformed equation of $x^2 + y^2 + 2x + 2y - 5 = 0$ contains no linear terms is

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. 0

B. 1

C. 2

D. Infinite

Answer: A

Solution:

The given equation is:

$$x^2 + y^2 + 2x + 2y - 5 = 0$$

We need to consider the transformation when the axes are rotated by an angle α . The equations for the transformed coordinates are:

$$x = X \cos \alpha - Y \sin \alpha$$

$$y = X \sin \alpha + Y \cos \alpha$$

Substituting these into the given equation, we get:

$$(X \cos \alpha - Y \sin \alpha)^2 + (X \sin \alpha + Y \cos \alpha)^2 + 2(X \cos \alpha - Y \sin \alpha) + 2(X \sin \alpha + Y \cos \alpha) - 5 = 0$$

Expanding and simplifying, the equation becomes:

$$X^2 + Y^2 + 2X \cos \alpha - 2Y \sin \alpha + 2X \sin \alpha + 2Y \cos \alpha - 5 = 0$$

Further simplifying, we have:

$$X^2 + Y^2 - 2X(\cos \alpha + \sin \alpha) + 2Y(\cos \alpha - \sin \alpha) - 5 = 0$$

For the transformed equation to have no linear terms, the coefficients of X and Y must be zero. Therefore, we set:

$$\cos \alpha + \sin \alpha = 0$$

$$\cos \alpha - \sin \alpha = 0$$

However, both equations cannot be simultaneously satisfied. Since these conditions are not possible together, there is no value of α that can make the transformed equation free of linear terms.

Question57

The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If $Q = (3, 4)$ and $R = (-4, 3)$, then $\angle QPR$ is equal to

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. $\frac{\pi}{2}$

B. $\frac{\pi}{3}$

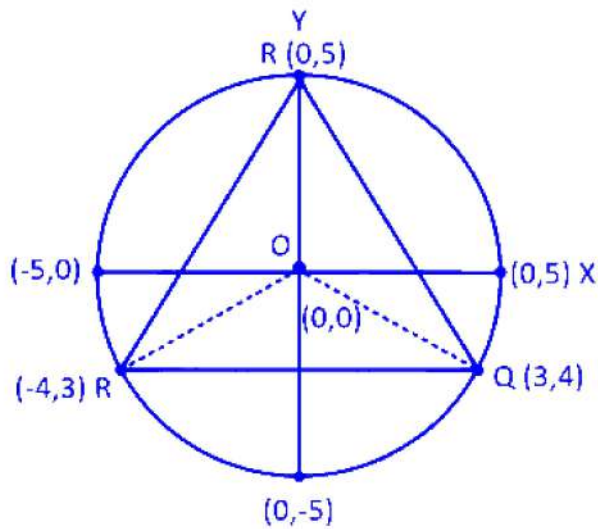
C. $\frac{\pi}{4}$

D. $\frac{\pi}{6}$

Answer: C

Solution:

$$x^2 + y^2 = 25$$



$$RQ = \sqrt{(3+4)^2 + (4-3)^2}$$

$$= \sqrt{49+1} = 5\sqrt{2}$$

$$OQ = OR = 5$$

$$\cos \angle ROQ = \frac{OR^2 + OQ^2 - RQ^2}{2OR \cdot OQ}$$

$$= \frac{25 + 25 - 50}{2 \times 5 \times 5} = 0$$

$$\Rightarrow \angle ROQ = 90^\circ$$

Now,

$$\angle RPQ = \frac{1}{2} \angle ROQ = \frac{1}{2} \times 90^\circ = 45^\circ = \frac{\pi}{4}$$

Question58

The locus of the point of intersection of perpendicular tangents drawn to the circle $x^2 + y^2 = 10$ is

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. $x^2 + y^2 = 5$

B. $x^2 + y^2 = 20$

C. $x^2 + y^2 = 25$

D. $x^2 + y^2 = 100$

Answer: B

Solution:

To determine the locus of the point of intersection of perpendicular tangents to the circle given by $x^2 + y^2 = 10$, we need to consider the concept of a director circle.

The director circle is a concentric circle whose radius is $\sqrt{2}$ times the radius of the original circle.

For the circle $x^2 + y^2 = 10$, the radius is $\sqrt{10}$. Thus, the radius of the director circle is:

$$\sqrt{2} \times \sqrt{10} = \sqrt{20}$$

Thus, the equation of the director circle is:

$$x^2 + y^2 = 20$$

Question59

The normal drawn at $(1, 1)$ to the circle $x^2 + y^2 - 4x + 6y - 4 = 0$ is

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. $4x + 3y = 7$

B. $4x + y = 5$

C. $x + y = 2$

D. $4x - y = 3$

Answer: B

Solution:

To find the equation of the normal to the circle $x^2 + y^2 - 4x + 6y - 4 = 0$ at the point $(1, 1)$, we first determine the circle's center.

The given equation can be rewritten as:

$$(x^2 - 4x) + (y^2 + 6y) = 4$$

Completing the square for both x and y :

$$(x - 2)^2 - 4 + (y + 3)^2 - 9 = 4$$

$$(x - 2)^2 + (y + 3)^2 = 17$$

Thus, the center of the circle is $(2, -3)$.

The normal line from $(1, 1)$ passes through the center $(2, -3)$. The slope of this normal line is calculated as follows:

$$\text{slope} = \frac{-3-1}{2-1} = -4$$

Using the point-slope form of a line with point $(1, 1)$:

$$y - 1 = -4(x - 1)$$

Simplifying this equation:

$$y - 1 = -4x + 4$$

$$y = -4x + 5$$

In standard form, this can be written as:

$$4x + y - 5 = 0$$

Thus, the equation of the normal is $4x + y - 5 = 0$.

Question60

Parametric equations of the circle $2x^2 + 2y^2 = 9$ are

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. $x = \frac{3}{2}\cos \theta, \quad y = \frac{3}{2}\sin \theta$

B. $x = \frac{3}{\sqrt{2}} \cos \theta, \quad y = 3 \sin \theta$

C. $x = \frac{3}{\sqrt{2}} \sin \theta, \quad y = \frac{3}{\sqrt{2}} \cos \theta$

D. $x = 3 \sin \theta, \quad y = \frac{3}{2} \cos \theta$

Answer: C

Solution:

Given equation of circle is

$$\begin{aligned} 2x^2 + 2y^2 &= 9 \\ \Rightarrow 2(x^2 + y^2) &= 9 \Rightarrow x^2 + y^2 = 9/2 \\ \Rightarrow x^2 + y^2 &= \left(\frac{3}{\sqrt{2}}\right)^2 \end{aligned}$$

Now, the parametric equation of circle is

$$x = r \cos \theta, y = r \sin \theta$$

$$\therefore x = \frac{3}{\sqrt{2}} \cos \theta \quad \left[\begin{array}{l} \because r^2 = x^2 + y^2 \\ r = \frac{\sqrt{3}}{2} \end{array} \right]$$

$$\text{and } y = \frac{3}{\sqrt{2}} \sin \theta$$

Question61

Angle between the circles $x^2 + y^2 - 4x - 6y - 3 = 0$ and $x^2 + y^2 + 8x - 4y + 11 = 0$ is

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. $\frac{\pi}{3}$

B. $\frac{\pi}{6}$

C. $\frac{\pi}{2}$

D. $\frac{\pi}{4}$

Answer: A

Solution:

To find the angle between the circles c_1 and c_2 , where:

$$c_1 : x^2 + y^2 - 4x - 6y - 3 = 0$$

$$c_2 : x^2 + y^2 + 8x - 4y + 11 = 0,$$

we can use the formula:

$$\cos \theta = \frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2}$$

where r_1 and r_2 are the radii of the circles, and d is the distance between their centers.

First, calculate the radii:

$$r_1 = \sqrt{2^2 + 3^2 + 3} = \sqrt{16} = 4$$

$$r_2 = \sqrt{4^2 + 2^2 - 11} = \sqrt{9} = 3$$

Next, determine the centers of the circles:

The center of c_1 is $(2, 3)$.

The center of c_2 is $(-4, 2)$.

Now, calculate the distance d between the centers:

$$d = \sqrt{(-4 - 2)^2 + (2 - 3)^2} = \sqrt{36 + 1} = \sqrt{37}$$

Substitute all values into the formula for the angle:

$$\cos \theta = \left| \frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2} \right| = \left| \frac{16 + 9 - 37}{2 \times 4 \times 3} \right| = \frac{1}{2}$$

Thus, the angle θ is $\frac{\pi}{3}$.

Question62

From a point $(1, 0)$ on the circle $x^2 + y^2 - 2x + 2y + 1 = 0$ if chords are drawn to this circle, then locus of the poles of these chords with respect the circle $x^2 + y^2 = 4$ is

AP EAPCET 2024 - 22th May Morning Shift

Options:

A. $x = 4$

B. $x + 2y = 5$

C. $x^2 + y^2 - x - y = 0$

D. $2y^2 = (x + 1)$

Answer: A

Solution:

$$x^2 + y^2 - 2x + 2y + 1 = 0$$

is a circle with centre $(1, -1)$ and radius 1. When chords are drawn from the point $(1, 0)$ on the circle, the locus will be $x^2 + y^2 = 4$.

For a given point (x_1, y_1) on the circle $x^2 + y^2 = 4$, the equation of the chord joining the point of tangency to the circle is $xx_1 + yy_1 = 4$ given original point

$(1, 0)$.

$$\Rightarrow x \cdot 1 + y \cdot 0 = 4 \Rightarrow x = 4$$

Thus, the locus of poles of these chords is a vertical line.

Question 63

If A and B are the centres of similitude with respect to the circles $x^2 + y^2 - 14x + 6y + 33 = 0$ and $x^2 + y^2 + 30x - 2y + 1 = 0$, then the mid-point of AB is

AP EAPCET 2024 - 22th May Morning Shift

Options:

A. $\left(\frac{7}{3}, \frac{4}{5}\right)$

B. $\left(\frac{3}{2}, \frac{1}{5}\right)$

C. $\left(\frac{39}{2}, \frac{-7}{4}\right)$

D. $\left(\frac{39}{4}, \frac{-7}{2}\right)$

Answer: D

Solution:

$$x^2 + y^2 - 14x + 6y + 33 = 0 \quad \dots (i)$$

$$x^2 + y^2 + 30x - 2y + 1 = 0 \quad \dots (ii)$$

using Eq. (i), we can write it as

$$\begin{aligned}(x-7)^2 - 49 + (y+3)^2 - 9 + 33 &= 0 \\ (x-7)^2 + (y+3)^2 - 25 &= 0\end{aligned}$$

So, centre $(7, -3)$ and radius $= 5$ using Eq. (ii) we can write it as

$$\begin{aligned}(x+15)^2 - 225 + (y-1)^2 - 1 + 1 &= 0 \\ (x+15)^2 + (y-1)^2 &= 225\end{aligned}$$

So, centre $(-15, 1)$ and radius $= 15$

Now, A(External centre similitude)

$$= \frac{R_2 C_1 - R_1 C_2}{R_2 - R_1}$$

And B (Internal centre similitude) $= \frac{R_2 C_1 + R_1 C_2}{R_2 + R_1}$, where C_1 and C_2 are

centers and R_1 and R_2 are their radii.

$$\begin{aligned}A &= \frac{15(7, -3) - 5(-15, 1)}{15 - 5} \\ &= \frac{(105, -45) + (75, -5)}{10} \\ &= \frac{(180, -50)}{10} = (18, -5) \\ B &= \frac{15(7, -3) + 5(-15, 1)}{15 + 5} \\ &= \frac{(105, -45) + (-75, 5)}{20} \\ &= \frac{(30, -40)}{20} = (1.5, -2)\end{aligned}$$

Finally mid-point M of line segment

$$\begin{aligned}M &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left(\frac{18 + 1.5}{2}, \frac{-5 + (-2)}{2} \right) \\ M &= \left(\frac{19.5}{2}, \frac{-7}{2} \right) = \left(\frac{39}{4}, \frac{-7}{2} \right)\end{aligned}$$

Question 64

C_1 is the circle with centre at $O(0, 0)$ and radius 4, C_2 is a variable circle with centre at (α, β) and radius 5. If the common chord of C_1 and C_2 has slope $\frac{3}{4}$ and of maximum length, then one of the possible values of $\alpha + \beta$ is

AP EAPCET 2024 - 22th May Morning Shift

Options:

A. $\frac{21}{5}$

B. $\frac{3}{5}$

C. $\frac{1}{5}$

D. $\frac{19}{5}$

Answer: B

Solution:

Given that C_1 is a circle centered at $O(0, 0)$ with a radius of 4, and C_2 is a variable circle with a center at (α, β) and a radius of 5.

The equations of the circles are:

$$C_1 : x^2 + y^2 = 16$$

$$C_2 : (x - \alpha)^2 + (y - \beta)^2 = 25$$

For the common chord to exist and have maximum length, the distance between the centers $(\sqrt{\alpha^2 + \beta^2})$ is:

$$\sqrt{5^2 - 4^2} = 3$$

The slope of the common chord is given as $\frac{3}{4}$. Therefore, the equation of the chord can be written as:

$$3x - 4y = 0 \quad \dots (i)$$

Since the distance between the points $(0, 0)$ and (α, β) should be 3, we find:

$$\left| \frac{3\alpha - 4\beta}{\sqrt{3^2 + 4^2}} \right| = 3$$

Simplifying further, we have:

$$3\alpha - 4\beta \pm 15 = 0 \quad \dots (ii)$$

Given that the product of the slopes of perpendicular lines is -1, and the line connecting the centers O and (α, β) is perpendicular to the chord, we have:

$$\frac{3}{4} \times \frac{\beta}{\alpha} = -1$$

Which simplifies to:

$$3\beta + 4\alpha = 0 \quad \dots (iii)$$

By solving equations (ii) and (iii), we find:

$$\alpha = -\frac{9}{5}$$

$$\beta = \frac{12}{5}$$

Therefore, $\alpha + \beta = \frac{3}{5}$.

Question65

If the pair of tangents drawn to the circle $x^2 + y^2 = a^2$ from the point $(10, 4)$ are perpendicular. then $a =$

AP EAPCET 2024 - 22th May Morning Shift

Options:

A. $\sqrt{58}$

B. 58

C. $2\sqrt{63}$

D. $2\sqrt{45}$

Answer: A

Solution:

To determine the radius a for which the tangents from the point $(10, 4)$ to the circle $x^2 + y^2 = a^2$ are perpendicular, we use the property that tangents from a given point (x_1, y_1) to a circle are perpendicular if:

$$x_1^2 + y_1^2 = 2r^2$$

Given that the point is $(10, 4)$ and the circle is defined by $x^2 + y^2 = a^2$, we set $x_1 = 10$, $y_1 = 4$, and $r = a$. Plugging these into the formula gives:

$$10^2 + 4^2 = 2a^2$$

Calculating further:

$$100 + 16 = 2a^2$$

$$116 = 2a^2$$

Dividing by 2:

$$a^2 = \frac{116}{2} = 58$$

Thus, the radius a is:

$$a = \sqrt{58}$$

Question66

If $x - 4 = 0$ is the radical axis of two orthogonal circles out of which one is $x^2 + y^2 = 36$, then the centre of the other circle is

AP EAPCET 2024 - 22th May Morning Shift

Options:

A. $(8, 0)$

B. $(9, 0)$

C. $(6, 0)$

D. $(12, 0)$

Answer: B

Solution:

Given the circle $x^2 + y^2 = 36$, we have a circle S_2 centered at $(0, 0)$ with a radius of 6. The radical axis of two orthogonal circles is given by $x - 4 = 0$.

For two circles to be orthogonal, the condition is:

$$2(g_1g_2 + f_1f_2) = c_1 + c_2$$

where (g_1, f_1) and (g_2, f_2) are the centers of the circles, and c_1 and c_2 are the constants in their respective equation forms $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$.

Let's denote the other circle as S_1 . According to the given problem, the radical axis is:

$$x - 4 = 0$$

The equation of the other circle S_1 can be expressed as:

$$S_1 - S_2 = k(x - 4)$$

So:

$$S_1 = S_2 + k(x - 4)$$

This translates to:

$$S_1 = x^2 + y^2 - 36 + kx - 4k = 0$$

Substituting into the orthogonality condition:

$$c_1 + c_2 = 0$$

Hence:

$$-4k - 36 - 36 = 0$$

Solving for k :

$$-4k - 72 = 0 \Rightarrow k = -18$$

Therefore, the equation of the other circle S_1 becomes:

$$x^2 + y^2 - 18x + 36 = 0$$

The center of this circle is $(9, 0)$.

Question67

The perimeter of the locus of the point P which divides the line segment QA internally in the ratio $1 : 2$, where $A = (4, 4)$ and Q lies on the circle $x^2 + y^2 = 9$, is

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. 8π

B. 4π

C. π

D. 9π

Answer: B

Solution:

Let $Q = (x_1, y_1)$ and $A = (4, 4)$.

The circle is defined by the equation $x^2 + y^2 = 9$.

The coordinates of the point P , which divides the line segment QA in the ratio $m : n = 1 : 2$, are determined by the formula:

$$P \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Substituting the given ratio and coordinates:

$$P\left(\frac{1 \cdot 4 + 2 \cdot x_1}{1+2}, \frac{1 \cdot 4 + 2 \cdot y_1}{1+2}\right) \Rightarrow P\left(\frac{4+2x_1}{3}, \frac{4+2y_1}{3}\right)$$

Since Q is on the circle $x^2 + y^2 = 9$, we have:

$$x_1^2 + y_1^2 = 9$$

Using parametric equations for the circle, let $x_1 = 3 \cos \theta$ and $y_1 = 3 \sin \theta$. Substituting into the expression for P from Eq. (i):

$$P\left(\frac{4+6 \cos \theta}{3}, \frac{4+6 \sin \theta}{3}\right) \Rightarrow P\left(\frac{4}{3} + 2 \cos \theta, \frac{4}{3} + 2 \sin \theta\right)$$

Let:

$$x = \frac{4}{3} + 2 \cos \theta \quad \Rightarrow \quad \cos \theta = \frac{x - \frac{4}{3}}{2}$$

$$y = \frac{4}{3} + 2 \sin \theta \quad \Rightarrow \quad \sin \theta = \frac{y - \frac{4}{3}}{2}$$

Substitute into the trigonometric identity $\sin^2 \theta + \cos^2 \theta = 1$:

$$\left(\frac{x - \frac{4}{3}}{2}\right)^2 + \left(\frac{y - \frac{4}{3}}{2}\right)^2 = 1$$

Simplifying further:

$$\left(x - \frac{4}{3}\right)^2 + \left(y - \frac{4}{3}\right)^2 = 4$$

This equation represents a circle with center $\left(\frac{4}{3}, \frac{4}{3}\right)$ and radius 2. Thus, the perimeter of this circle is:

$$2\pi \times \text{radius} = 4\pi$$

Question68

If the equation of the circle whose radius is 3 units and which touches internally the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ at the point $(-1, -1)$ is $x^2 + y^2 + px + qy + r = 0$, then $p + q - r =$

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. 2

B. $\frac{5}{2}$

C. $\frac{26}{5}$

D. 3

Answer: A

Solution:

Radius of circle is 3 units and which touches internally the circle $\Rightarrow C \equiv x^2 + y^2 - 4x - 6y - 12 = 0$ at points $(-1, 1)$

Centre of circle C is $(2, 3)$ and radius is 5 units

If $C_2(h, K)$ is the centre of circle of radius 3. Which touches circle C internally at $(1, -1)$, then

$$\Rightarrow C_2 = 3 \text{ and } \Rightarrow C_1, C_2 = R - r = 5 - 3 = 2$$

Thus, $C_2(h, K)$ divides C in the ratio 2 : 3 internally therefore,

$$\begin{aligned}\Rightarrow h &= \frac{2(-1) + 3 \times 2}{2 + 3} = \frac{4}{5} \\ \Rightarrow k &= \frac{2(-1) + 3 \times 3}{2 + 3} = \frac{7}{5}\end{aligned}$$

Hence, equations of circle C_2 is

$$\begin{aligned}\Rightarrow \left(x - \frac{4}{5}\right)^2 + \left(y - \frac{7}{5}\right)^2 &= (3)^2 \\ \Rightarrow x^2 + \frac{16}{25} - \frac{8x}{5} + y^2 + \frac{49}{25} - \frac{14y}{5} &= 9 \\ \Rightarrow x^2 + y^2 - \frac{8x}{5} - \frac{14y}{5} - \frac{32}{5} &= 0\end{aligned}$$

On comparing with given equation of circle

$$x^2 + y^2 + px + qy + r = 0$$

$$\text{We get, } p = \frac{-8}{5}, q = \frac{-14}{5}, r = \frac{-32}{5}$$

$$\begin{aligned}\Rightarrow p + q - r &= \frac{-8}{5} - \frac{14}{5} + \frac{32}{5} = 2 \\ \Rightarrow p + q - r &= 2\end{aligned}$$

Question69

The equation of the circle touching the circle

$x^2 + y^2 - 6x + 6y + 17 = 0$ externally and to which the lines
 $x^2 - 3xy - 3x + 9y = 0$ are normal is

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. $x^2 + y^2 - 3x + 2y - 2 = 0$

B. $x^2 + y^2 - 6x - 2y + 1 = 0$

C. $x^2 + y^2 + 6x - 2y - 1 = 0$

D. $x^2 + y^2 - 9x - 3y + 2 = 0$

Answer: B

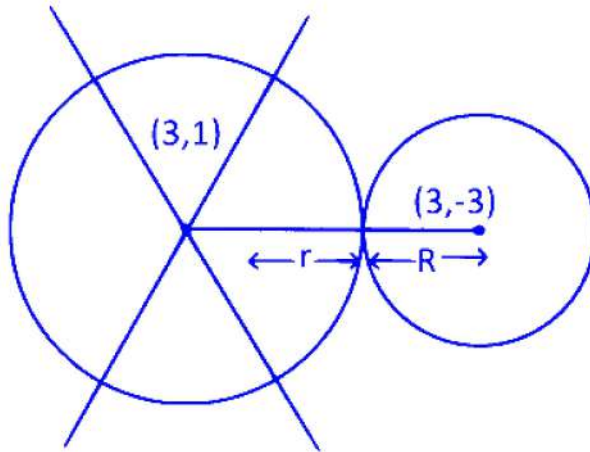
Solution:

Equation of circle

$$x^2 + y^2 - 6x - 6y + 17 = 0 \quad \dots (i)$$

and equation of line

$$x^2 - 3xy - 3x + 9y = 0 \quad \dots (ii)$$



From Eq. (ii), we get

$$\Rightarrow x^2 - 3xy - 3x + 9y = 0$$

$$\Rightarrow x^2 - 3x - 3xy + 9y = 0$$

$$\Rightarrow x(x - 3) - 3y(x - 3) = 0$$

$$\Rightarrow (x - 3)(x - 3y) = 0$$

$$\Rightarrow x = 3, x = 3y$$

$$\Rightarrow x = 3, y = 1$$

$$R = \sqrt{9 + 9 - 17}$$

$$R = \sqrt{1} = 1$$

According to the given condition,

$$\Rightarrow C_1 C_2 = R + r$$

$$\Rightarrow \sqrt{(3 - 3)^2 + (1 + 3)^2} = 1 + r$$

$$\Rightarrow 4 = 1 + r$$

$$\Rightarrow r = 3$$

Equation of circle having coordinates $(3, 1)$ and radius 3 is

$$\Rightarrow (x - 3)^2 + (y - 1)^2 = 9$$

$$\Rightarrow x^2 + 9 - 6x + y^2 + 1 - 2y = 9$$

$$\Rightarrow x^2 + y^2 - 6x - 2y + 1 = 0$$

Above equation is equation of circle which touches the circle.

Question70

The pole of the straight line $9x + y - 28 = 0$ with respect to the circle $2x^2 + 2y^2 - 3x + 5y - 7 = 0$ is

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. $(-1, 3)$

B. $(2, -3)$

C. $(3, -1)$

D. $(3, -3)$

Answer: C

Solution:

We have, equation of straight line

$$9x + y - 28 = 0 \quad \dots (i)$$

and equation of circle

$$C \equiv 2x^2 + 2y^2 - 3x + 5y - 7 = 0 \quad \dots (ii)$$

Let (x_1, y_1) be the required point, line (i) must coincide with the polar of (x_1, y_1) whose equation is

$$\Rightarrow 2xx_1 + 2yy_1 - \frac{3}{2}(x + x_1) + \frac{5}{2}(y + y_1) - 7$$

$$= 0$$

$$\Rightarrow x(4x_1 - 3) + y(4y_1 + 5) - 3x_1 + 5y_1 - 14 = 0 \quad \dots (iii)$$

Since, Eqs. (i) and (iii) are same, we have

$$\frac{4x_1 - 3}{9} = \frac{y_1 + 5}{1} = \frac{-3x_1 + 5y_1 - 14}{-28}$$

$$\Rightarrow x_1 = 9y_1 + 12$$

$$\Rightarrow 3x_1 - 117y_1 = 126$$

On solving, we get

$$x = 3 \Rightarrow y = -1$$

So, the required point is $(3, -1)$.

Question 71

The equation of a circle which touches the straight lines $x + y = 2$, $x - y = 2$ and also touches the circle $x^2 + y^2 = 1$ is

AP EAPCET 2024 - 21th May Evening Shift

Options:

- A. $(x + \sqrt{2})^2 + y^2 = 3 - \sqrt{2}$
- B. $(x + \sqrt{2})^2 + y^2 = 1 - 2\sqrt{2}$
- C. $(x - \sqrt{2})^2 + y^2 = 2(1 - \sqrt{2})$
- D. $(x - \sqrt{2})^2 + y^2 = 3 - 2\sqrt{2}$

Answer: D

Solution:

We have, equations of lines

$$\Rightarrow L_1 \equiv x + y = 2 \quad \dots (i)$$

$$\Rightarrow L_2 \equiv x - y = 2 \quad \dots (ii)$$

Let the coordinate of centre of circle be (g, f) which touches the line i.e. it will satisfy the equations of lines L_1 and L_2 .

$$\Rightarrow g + f = 2 \quad \dots (iii)$$

$$\Rightarrow g - f = 2 \quad \dots (iv)$$

On solving Eqs. (iii) and (iv), we get $(g, f) = (2, 0)$

The center of the required circle will lie on bisector of acute angle between these tangents i.e. on.

$$\frac{x+y-2}{\sqrt{2}} = \frac{x-y-2}{\sqrt{2}} \text{ i.e.}$$

Let it be $(h, 0)$ where h is (positive). If r be the radius, then it touches the circle $x^2 + y^2 = 1$ externally

According to the condition,

$$C_1 C_2 = r_1 + r_2$$

$$h = 1 + r \quad \dots (v)$$

$$\therefore h = 2 \pm \sqrt{2r} = 1 + r$$

On solving, we get, $r = \sqrt{2} - 1$

Hence, the equation of circle with center $(\sqrt{2}, 0)$ and radius $(\sqrt{2} - 1)$ is

$$\Rightarrow (x - \sqrt{2})^2 + y^2 = (\sqrt{2} - 1)^2$$

$$\Rightarrow (x - \sqrt{2})^2 + y^2 = 3 - 2\sqrt{2}$$

Question 72

The radical axis of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ and $2x^2 + 2y^2 + 3x + 8y + 2c = 0$ touches the circle $x^2 + y^2 + 2x + 2y + 1 = 0$. Then,

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. $g = \frac{3}{8}$ or $f = 1$

B. $g = \frac{2}{3}$ or $t = 3$

C. $g = \frac{1}{2}$ or $f = 1$

D. $g = \frac{3}{4}$ or $f = 2$

Answer: D

Solution:

We have two given circles:

$$S_1 \equiv x^2 + y^2 + 2gx + 2fy + c = 0$$

$$S_2 \equiv 2x^2 + 2y^2 + 3x + 8y + 2c = 0$$

The equation of the second circle can be simplified by dividing by 2:

$$S_2 \equiv x^2 + y^2 + \frac{3}{2}x + 4y + c = 0$$

The radical axis of the circles $S_1 = 0$ and $S_2 = 0$ is determined by subtracting one equation from the other:

$$S_1 - S_2 = 0$$

$$\Rightarrow (x^2 + y^2 + 2gx + 2fy + c) - \left(x^2 + y^2 + \frac{3}{2}x + 4y + c\right) = 0$$

$$\Rightarrow 2gx + 2fy - \frac{3}{2}x - 4y = 0$$

$$\Rightarrow x\left(g - \frac{3}{4}\right) + y(f - 2) = 0$$

This radical axis touches the circle:

$$S_3 \equiv x^2 + y^2 + 2x + 2y + 1 = 0$$

The center of this circle is $(-1, -1)$ and it has a radius of 1. For tangency, the distance from the center to the radical axis must equal the radius, i.e., $p = r$:

$$\frac{-1(g - \frac{3}{4}) - (f - 2)}{\sqrt{(g - \frac{3}{4})^2 + (f - 2)^2}} = 1$$

Squaring both sides, we arrive at:

$$(g - \frac{3}{4})^2 + (f - 2)^2 + 2(g - \frac{3}{4})(f - 2)$$

$$= (g - \frac{3}{4})^2 + (f - 2)^2$$

$$\therefore 2(g - \frac{3}{4})(f - 2) = 0$$

This implies either $g - \frac{3}{4} = 0$ or $f - 2 = 0$:

$$g = \frac{3}{4} \quad \text{or} \quad f = 2$$

Question 73

$2x - 3y + 1 = 0$ and $4x - 5y - 1 = 0$ are the equations of two diameters of the circle $S \equiv x^2 + y^2 + 2gx + 2fy - 11 = 0$. Q and R are the points of contact of the tangents drawn from the point $P(-2, -2)$ to this circle. If C is the centre of the circle $S = 0$, then the area (in square units) of the quadrilateral $PQCR$ is

AP EAPCET 2024 - 21th May Morning Shift

Options:

A. 25

B. 30

C. 24

D. 36

Answer: B

Solution:

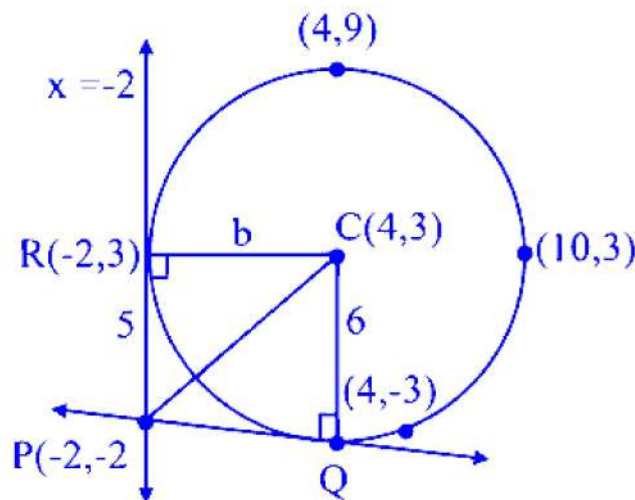
The point of intersection of the given linear equation is $(4, 3)$.

So, center of the circle will be $(4, 3)$.

$$\begin{aligned}\therefore S &\equiv x^2 + y^2 - 8x - 6y - 11 = 0 \\ \text{Radius} &= \sqrt{(-4)^2 + (-3)^2 - (-11)} \\ &= \sqrt{36} = 6\end{aligned}$$

The most rightward point of circle will be $(-2, 3)$. The x -coordinate of this point is same as the x -coordinate of the point $P(-2, -2)$.

So, the line $x = -2$ will be a tangent from P and it will touch the circle at $(-2; 3)$.



Clearly, $\triangle CPR \cong \triangle CPQ$ and $\triangle CPR$ is a right-angled triangle.

$\therefore$ Area of quadrilateral $PQCR = 2 \times$ Area of $\triangle CPR$

$$\begin{aligned}&= 2 \times \left(\frac{1}{2} \times 5 \times 6 \right) \\ &= 30\end{aligned}$$

Question74

If the inverse point of the point $(-1, 1)$ with respect to the circle $x^2 + y^2 - 2x + 2y - 1 = 0$ is (p, q) , then $p^2 + q^2 =$

AP EAPCET 2024 - 21th May Morning Shift

Options:

A. $\frac{1}{16}$

B. $\frac{1}{8}$

C. $\frac{1}{4}$

D. $\frac{1}{2}$

Answer: B

Solution:

Given, if the inverse point of the point $(-1, +1)$ w.r.t. the

Circle $x^2 + y^2 - 2x + 2y - 1 = 0$ is (p, q) then, $p^2 + q^2 = ?$

Now, $x^2 - 2x + y^2 + 2y = 1$

1 Add or subtract

$$(x^2 - 2x + 1 - 1) + (y^2 + 2y + 1 - 1) = 1$$

$$(x^2 - 2x + 1) - 1 + (y^2 + 2y + 1) - 1 = 1$$

$$(x - 1)^2 - 1 + (y + 1)^2 - 1 = 1$$

$$(x - 1)^2 + (y + 1)^2 = 3$$

So, center of circle $(1, -1)$ and Radius $= \sqrt{3}$

Now, find the distance from the point $(-1, 1)$ to the center of the circle.

$$\sqrt{(-1 - 1)^2 + (1 - (-1))^2}$$

$$= \sqrt{(-2)^2 + (2)^2} = 2\sqrt{2}$$

Now, determine the inverse point (p, q) of $(-1, 1)$ with respect to the circle lies on the same line passing through the point.

Here, $(x_1, y_1) = (-1, 1)$

$$(a, b) = (1, -1),$$

$$r^2 = 3$$

Formula for inverse point

$$p = \left(a + \left\{ r^2 + (x_1 - a) / \left((x_1 - a)^2 + (y_1 - b)^2 \right) \right\} \right)$$

$$q = \left(b + \left\{ r^2 (y_1 - b) / \left((x_1 - a)^2 + (y_1 - b)^2 \right) \right\} \right)$$

$$p = 1 + \frac{3x - 2}{8} = \frac{8 - 6}{8} = \frac{2}{8} = \frac{1}{4}$$

$$q = -1 + \frac{3(1 + 1)}{4 + 4}$$

$$= -1 + \frac{6}{8} = -\frac{1}{4}$$

$$p^2 + q^2 = \left(\frac{1}{4} \right)^2 + \left(-\frac{1}{4} \right)^2 = \frac{1}{8}$$

Question75

If (a, b) is the mid-point of the chord $2x - y + 3 = 0$ of the circle $x^2 + y^2 + 6x - 4y + 4 = 0$, then $2a + 3b =$

AP EAPCET 2024 - 21th May Morning Shift

Options:

A. -1

B. 0

C. 1

D. 3

Answer: C

Solution:

Mid-point (a, b) of the chord. $(2x - y + 3 = 0)$ of the circle. $x^2 + y^2 + 6x - 4y + 4 = 0$

$$(x^2 + 6x) + (y^2 - 4y) + 4 = 0$$

$$(x^2 + 9 - 9 + 6x) + (y^2 + 4 - 4 - 4y) + 4 = 0$$

$$(x^2 + 9 + 9 + 6x) - 9 + (y^2 + 4 - 4y) = 0$$

$$(x + 3)^2 + (y - 2)^2 = 9$$

Center $(-3, 2)$ and Radius $= 3$

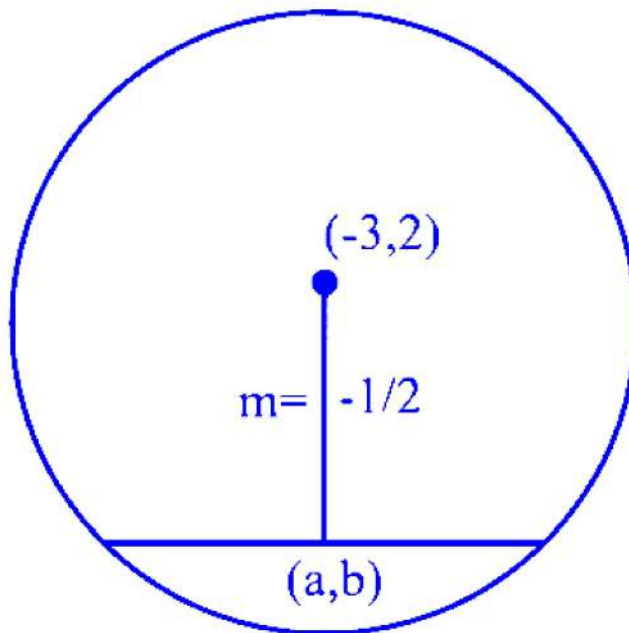
The chord equation, $2x - y + 3 = 0$

The mid-point (a, b) is perpendicular bisector of the chord which passes through center of the circle $(-3, 2)$. The equation of the perpendicular bisector of the chord can be found using the given chord equation and the center of the circle rewriting chord equation.

$$y = 2x + 3$$

Slope of this line is 2 . Therefore, the slope of perpendicular bisector is $-\frac{1}{2}$

Now, center $(-3, 2)m = \frac{-1}{2}$



$$y - 2 = -\frac{1}{2}(x + 3)$$

$$y = -\frac{1}{2}x + \frac{1}{2}$$

The mid-point (a, b) of the chord is the intersection. Point of the original chord $2x - y \neq 3 = 0$ and perpendicular bisectr $y = \frac{-1}{2}x + \frac{1}{2}$.

On substitute the value,

$$2x - \left(-\frac{1}{2}x + \frac{1}{2}\right) + 3 = 0$$

On solving $x = 11$ and $y = 1$

So, the mid-point (a, b) is $(-1, 1)$.

Calculate $2a + 3b$

$$\Rightarrow 2(-1) + 3(1) \Rightarrow 1$$

Therefore, $2a + 3b = 1$.

Question76

If a direct common tangent drawn to the circle

$x^2 + y^2 - 6x + 4y + 9 = 0$ and $x^2 + y^2 + 2x - 2y + 1 = 0$ touches the circles at A and B , then $AB =$

AP EAPCET 2024 - 21th May Morning Shift

Options:

A. 9

B. 16

C. $4\sqrt{6}$

D. $2\sqrt{6}$

Answer: D

Solution:

Given, first circle,

$$x^2 + y^2 - 6x + 4y + 9 = 0$$

By complete the square,

$$(x - 3)^2 + (y + 2)^2 = 4$$

Center $C_1(3, -2)$, Radius $(R_1) = 2$

Similarly, for second circle.

$$x^2 + y^2 + 2x - 2y + 1 = 0$$

By complete the square,

$$(x + 1)^2 + (y - 1)^2 = 1$$

Center $C_2(-1, 1)$, Radius $R_2 = 1$

Distance between C_1 and C_2

$$d = \sqrt{(3 - (-1))^2 + (-2 - 1)^2} = \sqrt{4^2 + (-3)^2} = 5$$

$$\begin{aligned}\text{Length of tangent} &= \sqrt{d^2 - (R_1 - R_2)^2} \\ &= \sqrt{5^2 - (2 - 1)^2} \\ &= \sqrt{25 - 1} \\ &= \sqrt{24} \\ &= 2\sqrt{6}\end{aligned}$$

Therefore, the length of the common tangent is $2\sqrt{6}$ unit.

Question 77

The radius of the circle which cuts the circles

$x^2 + y^2 - 4x - 4y + 7 = 0$, $x^2 + y^2 + 4x - 4y + 6 = 0$ and $x^2 + y^2 + 4x + 4y + 5 = 0$ orthogonally is

AP EAPCET 2024 - 21th May Morning Shift

Options:

A. $\frac{\sqrt{193}}{4\sqrt{2}}$

B. $\frac{\sqrt{193}}{8}$

C. $\frac{\sqrt{193}}{4}$

D. $\frac{\sqrt{193}}{2\sqrt{2}}$

Answer: A

Solution:

Given, 1st circle

$$= x^2 + y^2 - 4x - 4y + 7 = 0$$

$$\text{2nd circle} = x^2 + y^2 + 4x - 4y + 6 = 0$$

$$\text{3rd circle} = x^2 + y^2 + 4x + 4y + 5 = 0$$

Now, find the center and radii of the given circles.

$$\text{1st circle } x^2 + y^2 - 4x - 4y + 7 = 0$$

By complete the square

$$(x - 2)^2 + (y - 2)^2 = 2^2 + 2^2 - 7$$

$$(x - 2)^2 + (y - 2)^2 = 1$$

$$\text{Center } C_1(2, 2) \text{ Radius } R_1 = 1$$

$$\text{Similarly, 2nd Circle } (x + 2)^2 + (y - 2)^2 = 2$$

$$\text{Center } C_2(-2, 2) \text{ and Radius } R_2 = \sqrt{2}$$

$$\text{3rd circle } (x + 2)^2 + (y + 2)^2 = 3$$

$$C_3(-2, -2), R_3 = \sqrt{3}$$

Now, let the Radius R and center (a, b) be the circle which cut each of given circles orthogonally, then the below condition must hold.

$$(a - 2)^2 + (b - 2)^2 = R^2 + 1$$

$$(a + 2)^2 + (b - 2)^2 = R^2 + 2$$

$$(a + 2)^2 + (b + 2)^2 = R^2 + 3$$

On subtract Eq. (i) from Eq. (ii), we get

$$a = \frac{1}{8}$$

On subtracting Eq. (i) from Eq. (iii), we get

$$b = \frac{1}{8}$$

On substitute a and b in Eq. (i)

$$\left(\frac{1}{8} - 2\right)^2 + \left(\frac{1}{8} - 2\right)^2 = R^2 + 1$$

$$\left(\frac{-15}{8}\right)^2 + \left(\frac{-15}{8}\right)^2 = R^2 + 1$$

$$\frac{450}{64} = R^2 + 1$$

$$R^2 = \frac{193}{32}$$

$$R = \sqrt{\frac{193}{32}}$$

$$R = \frac{\sqrt{193}}{4\sqrt{2}}$$

Therefore, the radius of the circle that cuts all the given circle orthogonally is $\frac{\sqrt{193}}{4\sqrt{2}}$ unit.

Question78

$A(2, 3), B(-1, 1)$ are two points. If P is a variable point such that $\angle APB = 90^\circ$, then locus of P is

AP EAPCET 2024 - 20th May Evening Shift

Options:

A. $x^2 + y^2 - x - 4y + 1 = 0$

B. $x^2 + y^2 + x + 4y - 1 = 0$

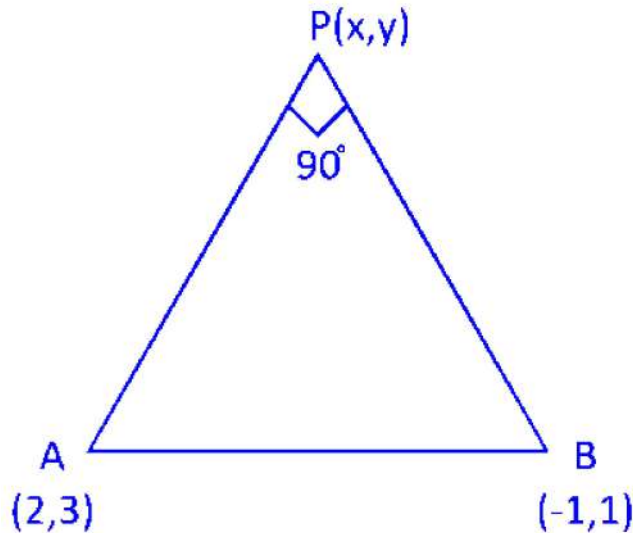
C. $x^2 + y^2 - x + 4y - 1 = 0$

D. $x^2 + y^2 + x - 4y + 1 = 0$

Answer: A

Solution:

In $\triangle APB$, we have



$$m_1 = \text{Slope of } AP = \frac{y-3}{x-2}$$

$$m_2 = \text{Slope of } BP = \frac{y-1}{x+1}$$

$$\text{Now, } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\begin{aligned} \tan 90^\circ &= \left| \frac{\frac{(y-3)}{(x-2)} - \frac{(y-1)}{(x+1)}}{1 + \frac{(y-3)}{(x-2)} \frac{(y-1)}{(x+1)}} \right| \\ &= \left| \frac{(y-3)(x+1) - (y-1)(x-2)}{(x-2)(x+1) + (y-3)(y-1)} \right| \\ \frac{1}{\infty} = 0 &= \frac{(x-2)(x+1) + (y-3)(y-1)}{(y-3)(x+1) - (y-1)(x-2)} \\ \Rightarrow x^2 - x - 2 + y^2 - 4y + 3 &= 0 \\ \Rightarrow x^2 - x + y^2 - 4y + 1 &= 0 \end{aligned}$$

Question 79

The largest among the distances from the point $P(15, 9)$ to the points on the circle $x^2 + y^2 - 6x - 8y - 11 = 0$ is

AP EAPCET 2024 - 20th May Evening Shift

Options:

A. 12

B. 13

C. 19

D. 7

Answer: C

Solution:

Given the equation of the circle:

$$x^2 + y^2 - 6x - 8y - 11 = 0$$

we can determine the center of the circle, C , which is $(3, 4)$, and the radius, r , as follows:

$$r = \sqrt{9 + 16 + 11} = \sqrt{36} = 6$$

Now, let's consider the point $P(15, 9)$.

The distance between the point P and the center C is calculated as:

$$CP = \sqrt{(15 - 3)^2 + (9 - 4)^2} = \sqrt{144 + 25} = \sqrt{169} = 13$$

To find the largest possible distance from the point P to any point on the circle, we add the radius to CP :

$$\text{Greatest distance from } P \text{ to the circle} = CP + r = 13 + 6 = 19$$

Question80

The circle $x^2 + y^2 - 8x - 12y + \alpha = 0$ lies in the first quadrant without touching the coordinate axes. If $(6, 6)$ is an interior point to the circle, then

AP EAPCET 2024 - 20th May Evening Shift

Options:

- A. $4 < \alpha < 6$
- B. $6 < \alpha < 16$
- C. $16 < \alpha < 48$
- D. $36 < \alpha < 48$

Answer: D

Solution:

Given the circle equation:

$$x^2 + y^2 - 8x - 12y + \alpha = 0$$

And the point $P(6, 6)$ is an interior point of the circle. To determine the range of α , we perform the following steps:

Point $P(6, 6)$ lies inside the circle, thus:

$$6^2 + 6^2 - 8 \cdot 6 - 12 \cdot 6 + \alpha < 0$$

$$36 + 36 - 48 - 72 + \alpha < 0$$

Simplifying:

$$72 - 48 - 72 + \alpha < 0$$

$$\alpha < 48$$

To ensure the circle does not touch the coordinate axes and remains in the first quadrant, the radius must be less than both the x-coordinate and y-coordinate of the center. The center of the circle (h, k) is found by completing the square:

$$h = \frac{8}{2} = 4, \quad k = \frac{12}{2} = 6$$

Thus, the center is at $(4, 6)$ and the radius $\sqrt{4^2 + 6^2 - \alpha} < 4$ and also < 6 :

$$\sqrt{52 - \alpha} < 4 \quad \text{and} \quad \sqrt{52 - \alpha} < 6$$

Solving these gives:

$$52 - \alpha < 16 \quad \text{and} \quad 52 - \alpha < 36$$

Which simplifies to:

$$\alpha > 36$$

Combining the inequalities from steps 1 and 2, we have:

$$36 < \alpha < 48$$

Question81

The equation of the circle whose diameter is the common chord of the circles $x^2 + y^2 - 6x - 7 = 0$ and $x^2 + y^2 - 10x + 16 = 0$ is

AP EAPCET 2024 - 20th May Evening Shift

Options:

A. $8x^2 + 8y^2 - 92x + 197 = 0$

B. $x^2 + y^2 - 23x + 197 = 0$

C. $x^2 + y^2 - \frac{23}{2}x + \frac{197}{4} = 0$

D. $4x^2 + 4y^2 - 46x + 197 = 0$

Answer: A

Solution:

To find the equation of the circle whose diameter is the common chord of the circles given by:

$$x^2 + y^2 - 6x - 7 = 0 \quad (i)$$

$$x^2 + y^2 - 10x + 16 = 0 \quad (ii)$$

First, we find the common chord by subtracting equation (i) from equation (ii):

$$\begin{aligned} x^2 + y^2 - 10x + 16 - (x^2 + y^2 - 6x - 7) &= 0, \\ \Rightarrow -4x + 23 &= 0, \\ \Rightarrow x &= \frac{23}{4}. \end{aligned}$$

Substitute $x = \frac{23}{4}$ back into either equation (let's use equation (ii)) to find the y -coordinates of the intersection points:

$$\begin{aligned} x^2 + y^2 - 10x &= -16, \\ \left(\frac{23}{4}\right)^2 + y^2 - 10 \cdot \frac{23}{4} &= -16, \\ y &= \frac{\pm 3\sqrt{15}}{4}. \end{aligned}$$

The points of intersection are $\left(\frac{23}{4}, \frac{-3\sqrt{15}}{4}\right)$ and $\left(\frac{23}{4}, \frac{3\sqrt{15}}{4}\right)$.

The midpoint of the chord (diameter) is:

$$\left(\frac{23}{4}, 0\right)$$

The radius is the distance from this midpoint to either endpoint:

$$d = \sqrt{\left(\frac{23}{4} - \frac{23}{4}\right)^2 + \left(0 - \frac{3\sqrt{15}}{4}\right)^2}$$

$$= \frac{3\sqrt{15}}{4}$$

Thus, the equation of the circle is:

$$\left(x - \frac{23}{4}\right)^2 + y^2 = \left(\frac{3\sqrt{15}}{4}\right)^2,$$

$$\Rightarrow \left(x - \frac{23}{4}\right)^2 + y^2 = \frac{135}{16},$$

$$\Rightarrow 16\left(x^2 - \frac{23}{2}x\right) + 16y^2 = 135,$$

$$\Rightarrow 8x^2 + 8y^2 - 92x + 197 = 0.$$

This is the equation of the circle whose diameter is the common chord of the given circles.

Question82

If the locus of the mid-point of the chords of the circle $x^2 + y^2 = 25$, which subtend a right angle at the origin is given by $\frac{x^2}{\alpha^2} + \frac{y^2}{\alpha^2} = 1$, then $|\alpha| =$

AP EAPCET 2024 - 20th May Evening Shift

Options:

- A. $\frac{2}{5}$
- B. $\frac{5}{\sqrt{2}}$
- C. $\frac{2}{25}$
- D. $5\sqrt{2}$

Answer: B

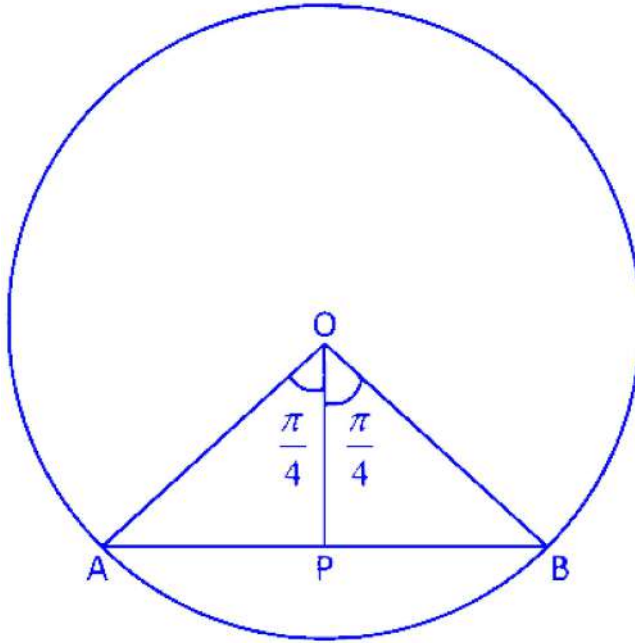
Solution:

Let mid-point be $P(h, k)$

and origin $O(0, 0)$ is also the centre of the circle.

Since, chord making right angle at origin and P is mid-point, OP will bisect the right angle.

$OP = r \cos 45^\circ$, where r is radius of given circle.



$$\Rightarrow \sqrt{(h-0)^2 + (k-0)^2} = 5 \times \frac{1}{\sqrt{2}}$$

$$\Rightarrow h^2 + k^2 = \left(\frac{5}{\sqrt{2}}\right)^2 = \frac{25}{2}$$

$$\therefore OP = \frac{5}{\sqrt{2}}$$

Let the coordinates of P be (x_1, y_1)

$$OP = \sqrt{x_1^2 + y_1^2} = \frac{5}{\sqrt{2}}$$

Question83

The radical centre of the circles $x^2 + y^2 + 2x + 3y + 1 = 0$,
 $x^2 + y^2 + x - y + 3 = 0$, $x^2 + y^2 - 3x + 2y + 5 = 0$

AP EAPCET 2024 - 20th May Evening Shift

Options:

A. $\left(-\frac{7}{38}, \frac{6}{19}\right)$

B. $\left(\frac{6}{19}, \frac{14}{19}\right)$

C. $\left(\frac{14}{19}, \frac{6}{19}\right)$

D. $(\frac{2}{19}, \frac{3}{19})$

Answer: C

Solution:

Let $S_1 : x^2 + y^2 + 2x + 3y + 1 = 0$

$S_2 : x^2 + y^2 + x - y + 3 = 0$

$S_3 : x^2 + y^2 - 3x + 2y + 5 = 0$

The radical axis of the circles will be

$$S_1 - S_2 = 0$$

$$\Rightarrow x + 4y - 2 = 0 \quad \dots (i)$$

$$\text{and } S_2 - S_3 = 0$$

$$4x - 3y - 2 = 0 \quad \dots (ii)$$

$$\text{and } S_3 - S_1 = 0$$

$$\Rightarrow -5x - y + 4 = 0 \quad \dots (iii)$$

On multiply Eqs. (i) by 4 and then subtract from (ii), we get

$$y = \frac{6}{19}$$

On substitute this value in Eq. (i), we get

$$x = \frac{14}{19}$$

Since, x and y values also satisfies Eqs. (iii).

Thus, the radical centre of three circles is $(\frac{14}{19}, \frac{6}{19})$.

Question84

If a circle is inscribed in an equilateral triangle of side a , then the area of any square (in sq units) inscribed in this circle is

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. $\frac{2a^2}{3}$

B. $\sqrt{3} \frac{a^2}{2}$

C. $\frac{a^2}{2\sqrt{3}}$

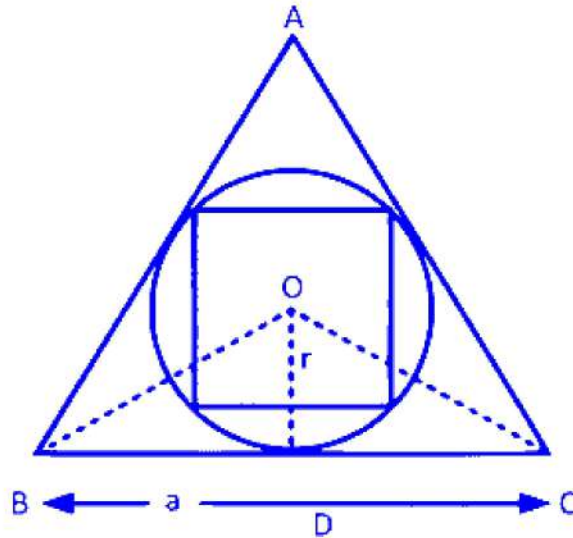
D. $\frac{a^2}{6}$

Answer: D

Solution:

In figure BA and BC are tangents to the circle. So, OB is the angle bisector of $\angle B$.

$$\begin{aligned} \angle B &= 60^\circ \\ \text{Then, } \angle OBD &= \frac{60}{2} = 30^\circ \end{aligned}$$



In $\triangle OBD$, $\angle ODB = 90^\circ$

$$\begin{aligned} \tan 30^\circ &= \frac{OD}{BD} \\ \frac{1}{\sqrt{3}} &= \frac{r}{\frac{a}{2}} = \frac{2r}{a} \\ r &= \frac{a}{2\sqrt{3}} \end{aligned}$$

Now, diagonal of square = diameter of circle

$$= 2r = 2 \times \frac{a}{2\sqrt{3}} = \frac{a}{\sqrt{3}}$$

Now, area of square = $\frac{1}{2}(2r)^2$

$$= \frac{1}{2} \times \left(\frac{a}{\sqrt{3}} \right)^2 = \frac{a^2}{6}$$

Question85

If the line segment joining the points $(1, 0)$ and $(0, 1)$ subtends an angle of 45° at a variable point P , then the equation of the locus of P

is

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. $(x^2 + y^2 - 1)(x^2 + y^2 - 2x - 2y + 1) = 0, x \neq 0, 1$

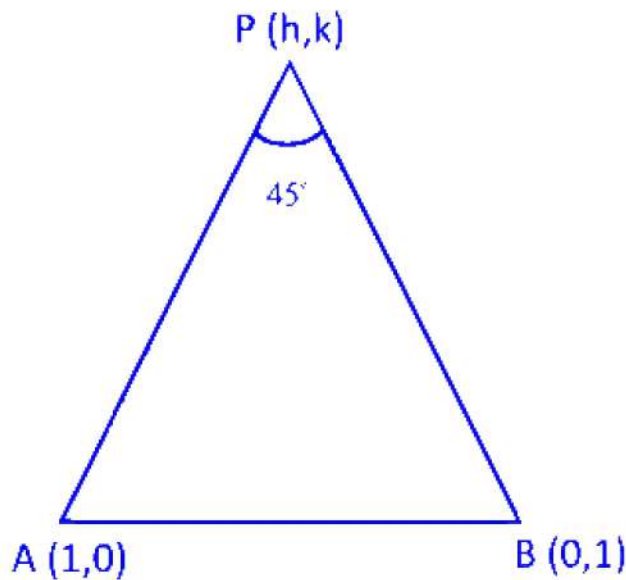
B. $(x^2 + y^2 - 1)(x^2 + y^2 + 2x + 2y + 1) = 0, x \neq 0, 1$

C. $x^2 + y^2 + 2x + 2y + 1 = 0$

D. $x^2 + y^2 = 4$

Answer: A

Solution:



$$\text{Slope of } AP = \frac{k}{h-1}$$

$$\text{Slope of } BP = \frac{k-1}{h}$$

$$\tan 45^\circ = \left| \frac{\frac{k}{h-1} - \frac{k-1}{h}}{1 + \frac{k(k-1)}{h(h-1)}} \right|$$

$$\pm 1 = \frac{kh - (k-1)(h-1)}{h(h-1) + k(k-1)}$$

$$\pm 1 = \frac{kh - kh + h + k - 1}{h^2 - h + k^2 - k}$$

$$\Rightarrow \pm (h^2 - h + k^2 - k) = h + k - 1$$

$$\text{If, } h + k - 1 = -(h^2 - h + k^2 - k),$$

$$\Rightarrow h + k - 1 = -h^2 + h - k^2 + k \quad h^2 + k^2 - 1$$

$$\text{If } h + k - 1 = h^2 - h + k^2 - 12$$

$$\Rightarrow h^2 + k^2 - 2h - 2k + 1 = 0$$

$$\text{Equation of locus of } P \text{ is } (x^2 + y^2 - 1) = 0$$

$$\text{or } x^2 + y^2 - 2x - 2y + 1 = 0$$

Hence, equation of locus of p is

$$(x^2 + y^2 - 1)(x^2 + y^2 - 2x - 2y + 1) = 0$$

$$x \neq 0, 1$$

Question 86

Equation of the circle having its centre on the line $2x + y + 3 = 0$ and having the lines $3x + 4y - 18 = 0$, $3x + 4y + 2 = 0$ as tangents is

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. $x^2 + y^2 + 6x + 8y + 4 = 0$

B. $x^2 + y^2 - 6x - 8y + 18 = 0$

C. $x^2 + y^2 - 8x + 10y + 37 = 0$

D. $x^2 + y^2 + 8x - 10y + 37 = 0$

Answer: D

Solution:

Let centre of circle be (h, k) .

Centre of circle lies on $2x + y + 3 = 0$

$$2h + k + 3 = 0$$

The two tangents $3x + 4y - 18 = 0$ and $3x + 4y + 2 = 0$ are parallel as their slopes are equal

$$2r = \frac{|2+18|}{\sqrt{3^2+4^2}} = \frac{20}{5} = 4$$

$$\therefore r = 2 \text{ units}$$

$$\text{Now, } r = \frac{|3h+4k-18|}{\sqrt{3^2+4^2}} = 2$$

$$\Rightarrow 3h + 4k - 18 = \pm 10$$

$$3h + 4k - 18 + 10 = 0$$

$$\text{or } 3h + 4k - 18 - 10 = 0$$

$$\Rightarrow 3h + 4k - 8 = 0$$

$$\text{Or } 3h + 4k - 28 = 0$$

From Eqs. (i) and (ii), we get $h = -4, k = 5$

From Eqs. (i) and (iii), we get $h = -8, k = 13$

Neglecting $h = -8$ and $k = 13$

because distance from centre to tangent $\neq 2$

$\therefore$ Centre of circle is $(-4, 5)$

$\therefore$ Equation of circle is

$$(x + 4)^2 + (y - 5)^2 = 2^2$$

$$x^2 + y^2 + 8x - 10y + 16 + 25 = 4$$

$$x^2 + y^2 + 8x - 10y + 37 = 0$$

Question87

If power of a point $(4, 2)$ with respect to the circle

$x^2 + y^2 - 2\alpha x + 6y + \alpha^2 - 16 = 0$ is 9 , then the sum of the lengths of all possible intercepts made by such circles on the coordinate axes is

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. $16 + 4\sqrt{6}$

B. $16 + 4\sqrt{6} - 6\sqrt{2}$

C. $16 + 4\sqrt{6} + 6\sqrt{2}$

D. $16 + 6\sqrt{2}$

Answer: A

Solution:

The power of the point $(4, 2)$ with respect to the given circle is 9.

To find the circle's equation, use the power of the point:

$$16 + 4 - 8\alpha + 12 + \alpha^2 - 16 = 9$$

Solving this equation gives:

$$\alpha^2 - 8\alpha + 7 = 0$$

which results in:

$$\alpha = 1 \quad \text{or} \quad \alpha = 7$$

Thus, the possible equations of the circles are:

$$x^2 + y^2 - 2x + 6y - 15 = 0$$

$$x^2 + y^2 - 14x + 6y + 33 = 0$$

For the equation $x^2 + y^2 - 2x + 6y - 15 = 0$:

Length of intercepts on the X-axis:

$$2\sqrt{g^2 - c} = 2\sqrt{1 + 15} = 2\sqrt{16} = 8$$

Length of intercepts on the Y-axis:

$$2\sqrt{f^2 - c} = 2\sqrt{9 + 15} = 2\sqrt{24} = 4\sqrt{6}$$

For the equation $x^2 + y^2 - 14x + 6y + 33 = 0$:

Length of intercept on the X-axis:

$$2\sqrt{g^2 - c} = 2\sqrt{49 - 33} = 2\sqrt{16} = 8$$

Length of intercept on the Y-axis is not possible (as the term under the square root is negative).

Therefore, the sum of the lengths of all possible intercepts is:

$$8 + 8 + 4\sqrt{6} = 16 + 4\sqrt{6}$$

Question88

Let α be an integer multiple of 8 . If S is the set of all possible values of α such that the line $6x + 8y + \alpha = 0$ intersects the circle $x^2 + y^2 - 4x - 6y + 9 = 0$ at two distinct points, then the number of elements in S is

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. 4

B. 6

C. 2

D. 1

Answer: A

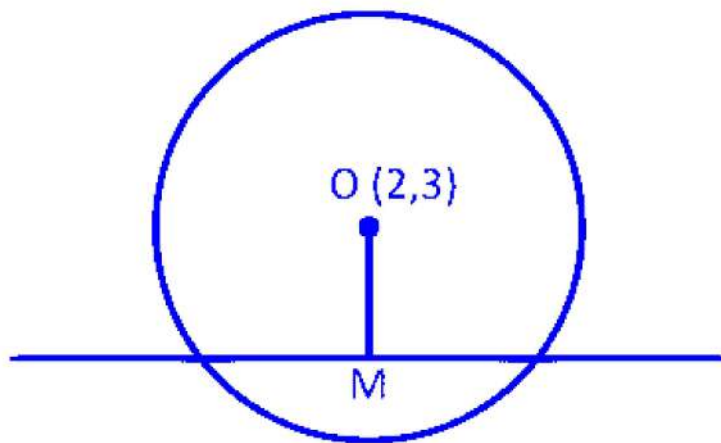
Solution:

We have,

Equation of circle is

$$x^2 + y^2 - 4x - 6y + 9 = 0$$

Centre of circle = $(2, 3)$, $r = 2$



As the line intersects the circle at 2 points, perpendicular distance from centre to line is less than radius

$$0 < \frac{|12 + 24 + \alpha|}{\sqrt{6^2 + 8^2}} < 2$$

$$0 < |36 + \alpha| < 20$$

$$\Rightarrow \text{Either } 0 < 36 + \alpha < -16$$

$$\text{or } 0 < -(36 + \alpha) < 20$$

$$\Rightarrow \text{Either } -36 < \alpha < -16, \text{ the possible values of } \alpha \text{ are } -32 - 24 [\because \alpha \text{ is a multiple of } 8]$$

If $-36 > \alpha > -56$, then possible values of α are $-40, -48$.

Hence, the number of elements in set $s = 4$.

Question89

If the circle $x^2 + y^2 - 8x - 8y + 28 = 0$ and $x^2 + y^2 - 8x - 6y + 25 - \alpha^2 = 0$ have only one common tangent, then $\alpha =$

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. $\alpha = 4$

B. $\alpha = 2$

C. $\alpha = 1$

D. $\alpha = 5$

Answer: C

Solution:

C_1 : Centre of circle

$$x^2 + y^2 - 8x - 8y + 28 = 0 \text{ is } (4, 4)$$

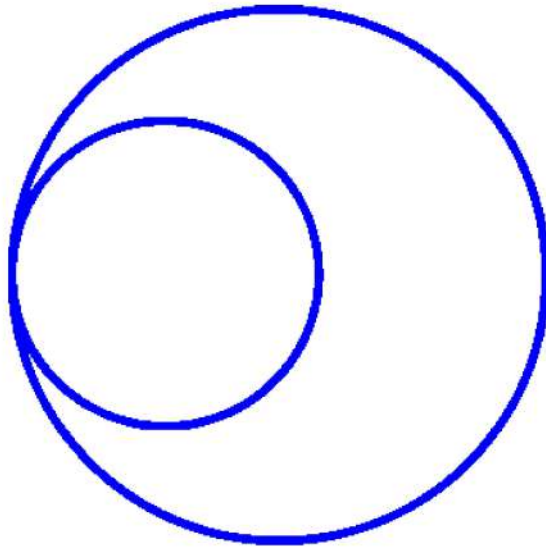
$$r_1 = \sqrt{16 + 16 - 28} = 2$$

C_2 : Centre of circle

$$x^2 + y^2 - 8x - 6y + 25 - \alpha^2 = 0 \text{ is } (4, 3)$$

$$r_2 = \sqrt{16 + 9 - 25 + \alpha^2} = \alpha$$

As the two circles have one common tangent, then the two circle coincides internally at point



$$C_1 C_2 = |r_1 - r_2| \sqrt{(4 - 4)^2 + (4 - 3)^2} = |2 - \alpha|$$

$$1 = |2 - \alpha| \Rightarrow 2 - \alpha = \pm 1$$

$$\Rightarrow 2 - \alpha = 1 \text{ or } 2 - \alpha = -1$$

$$\Rightarrow \alpha = 1 \text{ or } \alpha = 3$$

Question90

If the equation of the circle passing through the points of intersection of the circles $x^2 - 2x + y^2 - 4y - 4 = 0$, $x^2 + 2x + y^2 + 4y - 4 = 0$ and the point $(3, 3)$ is given by $x^2 + y^2 + \alpha x + \beta y + \gamma = 0$, then $3(\alpha + \beta + \gamma) =$

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. 32

B. -32

C. -26

D. 26

Answer: C

Solution:

The equation of the circle passing through the intersection of two given circles can be described by:

$$x^2 - 2x + y^2 - 4y - 4 + \lambda(x^2 + 2x + y^2 + 4y - 4) = 0$$

This simplifies to:

$$x^2(1 + \lambda) + y^2(1 + \lambda) + x(-2 + 2\lambda) + y(-4 + 4\lambda) - 4 - 4\lambda = 0$$

The circle in equation (i) must pass through the point (3, 3), so we substitute these coordinates into the equation:

$$9(1 + \lambda) + 9(1 + \lambda) + 3(-2 + 2\lambda) + 3(-4 + 4\lambda) - 4 - 4\lambda = 0$$

Simplifying the above expression yields:

$$18 + 18\lambda - 6 + 6\lambda - 12 + 12\lambda - 4 - 4\lambda = 0$$

This further simplifies to:

$$32\lambda - 4 = 0 \implies \lambda = \frac{1}{8}$$

Substituting $\lambda = \frac{1}{8}$ back into equation (i), the equation of the circle becomes:

$$\begin{aligned} \frac{9}{8}x^2 + \frac{9}{8}y^2 - \frac{14}{8}x - \frac{28}{8}y - \frac{36}{8} &= 0 \\ \implies x^2 + y^2 - \frac{14}{9}x - \frac{28}{9}y - 4 &= 0 \end{aligned}$$

From this, we identify $\alpha = \frac{-14}{9}$, $\beta = \frac{-28}{9}$, and $\gamma = -4$.

Now, calculating $3(\alpha + \beta + \gamma)$:

$$3\left(-\frac{14}{9} - \frac{28}{9} - 4\right) = -\frac{78}{3} = -26$$

Question91

The angle subtended by the chord $x + y - 1 = 0$ of the circle $x^2 + y^2 - 2x + 4y + 4 = 0$ at the origin is

AP EAPCET 2024 - 19th May Evening Shift

Options:

A. $\cos^{-1}\left(\frac{6}{\sqrt{34}}\right)$

B. $\frac{\pi}{2}$

C. $\cos^{-1}\left(\frac{2}{\sqrt{13}}\right)$

D. $\frac{\pi}{3}$

Answer: A

Solution:

Given

Chord:

$$x + y - 1 = 0$$

Circle:

$$x^2 + y^2 - 2x + 4y + 4 = 0$$

We have to find the angle subtended by this chord at the origin.

Step 1: Find the points of intersection of chord and circle

From the chord:

$$y = 1 - x$$

Substitute in the circle:

$$x^2 + (1 - x)^2 - 2x + 4(1 - x) + 4 = 0$$

$$x^2 + 1 - 2x + x^2 - 2x + 4 - 4x + 4 = 0$$

$$2x^2 - 8x + 9 = 0$$

Let the roots be x_1, x_2 . Then:

$$x_1 + x_2 = \frac{8}{2} = 4, \quad x_1 x_2 = \frac{9}{2}$$

Corresponding points are:

$$A(x_1, 1 - x_1), \quad B(x_2, 1 - x_2)$$

Step 2: Slopes of lines OA and OB

Slope of OA:

$$m_1 = \frac{1 - x_1}{x_1}$$

Slope of OB:

$$m_2 = \frac{1 - x_2}{x_2}$$

Find $m_1 + m_2$

$$\begin{aligned} m_1 + m_2 &= \left(\frac{1}{x_1} - 1 \right) + \left(\frac{1}{x_2} - 1 \right) \\ &= \left(\frac{x_1 + x_2}{x_1 x_2} \right) - 2 \\ &= \frac{4}{9/2} - 2 = \frac{8}{9} - 2 = -\frac{10}{9} \end{aligned}$$

Find $m_1 m_2$

$$\begin{aligned} m_1 m_2 &= \frac{(1 - x_1)(1 - x_2)}{x_1 x_2} \\ &= \frac{1 - (x_1 + x_2) + x_1 x_2}{x_1 x_2} \\ &= \frac{1 - 4 + 9/2}{9/2} = \frac{3/2}{9/2} = \frac{1}{3} \end{aligned}$$

Step 3: Use formula for angle between two lines

For slopes m_1, m_2 :

$$\cos \theta = \frac{|1 + m_1 m_2|}{\sqrt{(1 + m_1^2)(1 + m_2^2)}}$$

First find:

$$\begin{aligned} m_1^2 + m_2^2 &= (m_1 + m_2)^2 - 2m_1 m_2 \\ &= \left(-\frac{10}{9} \right)^2 - 2 \left(\frac{1}{3} \right) = \frac{100}{81} - \frac{2}{3} = \frac{100 - 54}{81} = \frac{46}{81} \end{aligned}$$

Now:

$$\begin{aligned}(1 + m_1^2)(1 + m_2^2) &= 1 + (m_1^2 + m_2^2) + m_1^2 m_2^2 \\ &= 1 + \frac{46}{81} + \frac{1}{9} = \frac{81 + 46 + 9}{81} = \frac{136}{81}\end{aligned}$$

Also:

$$1 + m_1 m_2 = 1 + \frac{1}{3} = \frac{4}{3}$$

Step 4: Substitute in cosine formula

$$\cos \theta = \frac{4/3}{\sqrt{136/81}} = \frac{4/3}{\sqrt{136}/9} = \frac{4}{3} \times \frac{9}{\sqrt{136}} = \frac{12}{\sqrt{136}}$$

Since $136 = 4 \times 34$:

$$\cos \theta = \frac{12}{2\sqrt{34}} = \frac{6}{\sqrt{34}}$$

 **Final Answer**

$$\theta = \cos^{-1}\left(\frac{6}{\sqrt{34}}\right)$$

Correct option: (A)

Question92

Let P be any point on the circle $x^2 + y^2 = 25$. Let L be the chord of contact of P with respect to the circle $x^2 + y^2 = 9$. The locus of the poles of the lines L with respect to the circle $x^2 + y^2 = 36$ is

AP EAPCET 2024 - 19th May Evening Shift

Options:

A. $y^2 = 20x$

B. $\frac{x^2}{9} + \frac{y^2}{36} = 1$

C. $x^2 + y^2 = 400$

D. $\frac{x^2}{25} - \frac{y^2}{16} = 1$

Answer: C

Solution:

Given a point P on the circle $x^2 + y^2 = 25$, let the coordinates of P be (x_1, y_1) . Thus, we have:

$$x_1^2 + y_1^2 = 25 \quad (\text{Equation 1})$$

The chord of contact from point P to the circle $x^2 + y^2 = 9$ is given by the equation:

$$xx_1 + yy_1 = 9$$

Let (h, k) be the pole of this chord with respect to the circle $x^2 + y^2 = 36$. The polar of (h, k) is given by:

$$hx + ky = 36$$

Since the line L is the polar of P with respect to the circle $x^2 + y^2 = 36$, the equation of the polar line, in terms of (x_1, y_1) , is simply:

$$xx_1 + yy_1 = 0$$

The ratio given by comparing the polar equations is:

$$\frac{h}{x_1} = \frac{k}{y_1} = \frac{36}{9} = 4$$

This implies:

$$x_1 = \frac{h}{4} \quad \text{and} \quad y_1 = \frac{k}{4}$$

Substituting x_1 and y_1 from above into Equation 1, we have:

$$\left(\frac{h}{4}\right)^2 + \left(\frac{k}{4}\right)^2 = 25$$

Simplifying:

$$\frac{h^2}{16} + \frac{k^2}{16} = 25$$

Multiplying through by 16:

$$h^2 + k^2 = 400$$

Therefore, the locus of the poles (h, k) is given by:

$$x^2 + y^2 = 400$$

Question93

If the circles $S \equiv x^2 + y^2 - 14x + 6y + 33 = 0$ and $S^1 \equiv x^2 + y^2 - a^2 = 0 (a \in N)$ have 4 common tangents, then possible number of values of a is

AP EAPCET 2024 - 19th May Evening Shift

Options:

A. 13

B. 5

C. 14

D. 2

Answer: D

Solution:

We are given the circle equation:

$$S \equiv x^2 + y^2 - 14x + 6y + 33 = 0$$

The center, C_1 , of this circle is $(7, -3)$, and we can determine its radius, r_1 , as follows:

$$r_1 = \sqrt{7^2 + (-3)^2 - 33} = \sqrt{49 + 9 - 33} = \sqrt{25} = 5$$

For the circle $S^1 \equiv x^2 + y^2 = a^2$, the center, C_2 , is $(0, 0)$ and its radius, r_2 , is a .

To have four common tangents between the two circles, the condition is:

$$C_1C_2 > r_1 + r_2$$

Calculating C_1C_2 , we have:

$$C_1C_2 = \sqrt{(7-0)^2 + (-3-0)^2} = \sqrt{49+9} = \sqrt{58}$$

Thus, we require:

$$\sqrt{58} > a + 5$$

Approximating $\sqrt{58} \approx 7.6$, we get:

$$a + 5 < 7.6$$

$$a < 7.6 - 5$$

$$a < 2.6$$

Since a is a natural number ($a \in \mathbb{N}$), the possible values of a are 1 and 2.

Therefore, there are 2 possible values of a .

If the area of the circum-circle of triangle formed by the line $2x + 5y + \alpha = 0$ and the positive coordinate axes is $\frac{29\pi}{4} Sq$, units, then $|\alpha| =$

AP EAPCET 2024 - 19th May Evening Shift

Options:

A. 25

B. 10

C. 20

D. 400

Answer: B

Solution:

The line $2x + 5y + \alpha = 0$ intersects the x -axis and y -axis at points $A\left(-\frac{\alpha}{2}, 0\right)$ and $B\left(0, -\frac{\alpha}{5}\right)$ respectively. This forms a right-angled triangle with the axes.

The circumcenter of a right-angled triangle is at the midpoint of the hypotenuse AB . Thus, the center of the circumcircle is $\left(\frac{-\alpha}{4}, \frac{-\alpha}{10}\right)$.

To find the radius of the circumcircle, we calculate the distance from the center to one of the vertices, say A :

$$\begin{aligned}\text{Radius of the circle} &= \sqrt{\left(\frac{-\alpha}{2} - \frac{-\alpha}{4}\right)^2 + \left(0 - \frac{-\alpha}{10}\right)^2} \\ &= \sqrt{\left(-\frac{\alpha}{4}\right)^2 + \left(\frac{\alpha}{10}\right)^2} \\ &= \sqrt{\frac{\alpha^2}{16} + \frac{\alpha^2}{100}} \\ &= \sqrt{\frac{25\alpha^2 + 4\alpha^2}{400}} \\ &= \frac{\sqrt{29\alpha^2}}{20}.\end{aligned}$$

The area of the circumcircle is given by πr^2 :

$$\begin{aligned}\pi \left(\frac{29}{400}\alpha^2\right) &= \frac{29\pi}{4} \\ \Rightarrow \frac{29\alpha^2}{400} &= \frac{29}{4} \\ \Rightarrow \alpha^2 &= 100 \\ \Rightarrow |\alpha| &= 10.\end{aligned}$$

Question95

The circle $S \equiv x^2 + y^2 - 2x - 4y + 1 = 0$ cuts the Y -axis at A, B ($OA > OB$). If the radical axis of $S \equiv 0$ and $S' \equiv x^2 + y^2 - 4x - 2y + 4 = 0$ cuts the Y -axis at C , then the ratio in which C divides AB is

AP EAPCET 2024 - 19th May Evening Shift

Options:

A. $7 + 2\sqrt{3} : -7 + 2\sqrt{3}$

B. $\sqrt{3} + 2 : \sqrt{3} - 2$

C. $6 - 2\sqrt{3} : 2\sqrt{3} - 6$

D. $-3 : \sqrt{3}$

Answer: A

Solution:

We are given a circle defined by the equation:

$$S \equiv x^2 + y^2 - 2x - 4y + 1 = 0$$

This can be rewritten by completing the square as:

$$(x - 1)^2 + (y - 2)^2 = 4$$

To find where this circle intersects the Y -axis, set $x = 0$:

$$y^2 - 4y + 1 = 0$$

Solving this quadratic equation:

$$y = \frac{4 \pm \sqrt{16 - 4}}{2}$$

$$y = 2 \pm \sqrt{3}$$

Thus, the points of intersection are $A(0, 2 + \sqrt{3})$ and $B(0, 2 - \sqrt{3})$. Since $OA > OB$, we have $OA = 2 + \sqrt{3}$ and $OB = 2 - \sqrt{3}$.

The radical axis of the circles $S \equiv x^2 + y^2 - 2x - 4y + 1 = 0$ and $S' \equiv x^2 + y^2 - 4x - 2y + 4 = 0$ is found by subtracting:

$$S - S' = 0$$

$$(x^2 + y^2 - 2x - 4y + 1) - (x^2 + y^2 - 4x - 2y + 4) = 0$$

$$-2x + 4x - 4y + 2y + 1 - 4 = 0$$

$$2x - 2y - 3 = 0$$

This simplifies to the line equation:

$$2x - 2y = 3$$

Now, solve for y when $x = 0$ (where it intersects the Y-axis):

$$2(0) - 2y = 3$$

$$y = -\frac{3}{2}$$

Thus, the point C where it intersects the Y-axis is $(0, -\frac{3}{2})$.

Assume C divides AB in the ratio $k : 1$. Using section formula for coordinates:

$$-\frac{3}{2} = \frac{k(2-\sqrt{3})+(2+\sqrt{3})}{k+1}$$

Solving for k :

$$-3k - 3 = 4k - 2\sqrt{3}k + 4 + 2\sqrt{3}$$

Simplifying:

$$-7k + 2\sqrt{3}k = 7 + 2\sqrt{3}$$

$$k = \frac{7+2\sqrt{3}}{-7+2\sqrt{3}}$$

Therefore, the ratio in which C divides the segment AB is:

$$(7 + 2\sqrt{3}) : (-7 + 2\sqrt{3})$$

Question96

If the circle $S = 0$ cuts the circles $x^2 + y^2 - 2x + 6y = 0$, $x^2 + y^2 - 4x - 2y + 6 = 0$ and $x^2 + y^2 - 12x + 2y + 3 = 0$ orthogonally, then equation of the tangent at $(0, 3)$ on $S = 0$ is

AP EAPCET 2024 - 19th May Evening Shift

Options:

A. $x + y - 3 = 0$

B. $y = 3$

C. $x = 0$

D. $x - y + 3 = 0$

Answer: B

Solution:

Let the equation of the desired circle be:

$$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$$

The circle S is orthogonal to three given circles. Using the property of orthogonal circles, we find the equations involving g , f , and c .

For the Circle $x^2 + y^2 - 2x + 6y = 0$:

Since S and this circle cut orthogonally, we have:

$$2 \cdot g \cdot (-1) + 2 \cdot f \cdot 3 = c + 0$$

Simplifying:

$$-2g + 6f = c \quad \dots(i)$$

For the Circle $x^2 + y^2 - 4x - 2y + 6 = 0$:

Orthogonality gives us:

$$2g \cdot (-2) + 2f \cdot (-1) = c + 6$$

Simplifying:

$$-4g - 2f = c + 6$$

For the Circle $x^2 + y^2 - 12x + 2y + 3 = 0$:

Orthogonality yields:

$$2g \cdot (-6) + 2f \cdot 1 = c + 3$$

Simplifying:

$$-12g + 2f = c + 3$$

From equation (i), substituting c from the orthogonality conditions, we proceed to simplify:

Solving the System of Equations:

Using equations:

$$-4g - 2f = c + 6$$

$$-12g + 2f = c + 3$$

Rearrange equation (i):

$$c = -2g + 6f \quad (\text{equation i})$$

From $-4g - 2f = -2g + 6f + 6$:

$$-2g - 8f = 6 \Rightarrow g + 4f = -3 \quad \dots(ii)$$

From $-12g + 2f = -2g + 6f + 3$:

$$-10g - 4f = 3 \Rightarrow 10g + 4f = -3$$

Solving equations (ii) and the above, we find:

$$g = 0, \quad f = -\frac{3}{4}$$

Substituting into equation (i) for c :

$$c = 6 \times \left(-\frac{3}{4}\right) = -\frac{9}{2}$$

Equation of the Circle S :

$$x^2 + y^2 - \frac{3}{2}y - \frac{9}{2} = 0$$

Finding the Equation of the Tangent at $(0, 3)$:

The equation of the tangent to this circle at the point (x_1, y_1) is given by:

$$xx_1 + yy_1 - \frac{3}{2}\left(\frac{y+y_1}{2}\right) - \frac{9}{2} = 0$$

Substituting point $(0, 3)$:

$$3y - \frac{3}{2}\left(\frac{y+3}{2}\right) - \frac{9}{2} = 0$$

Simplifying:

$$3y - \frac{3}{4}y - \frac{9}{4} - \frac{9}{2} = 0$$

Convert all terms to common denominator:

$$\frac{12y}{4} - \frac{3y}{4} - \frac{9}{4} - \frac{18}{4} = 0$$

Re-arrange the equation:

$$\frac{9y}{4} = \frac{27}{4}$$

Solve for y :

$$y = 3$$

Thus, the equation of the tangent is:

$$y = 3$$

Question97

If θ is the angle between the tangents drawn from the point $(2, 3)$ to the circle $x^2 + y^2 - 6x + 4y + 12 = 0$ then $\theta =$

AP EAPCET 2024 - 18th May Morning Shift

Options:

A. $\cos^{-1} \left(\frac{5}{13} \right)$

B. $\sin^{-1} \left(\frac{4}{5} \right)$

C. $2 \tan^{-1} \left(\frac{5}{12} \right)$

D. $\tan^{-1} \left(\frac{5}{12} \right)$

Answer: D

Solution:

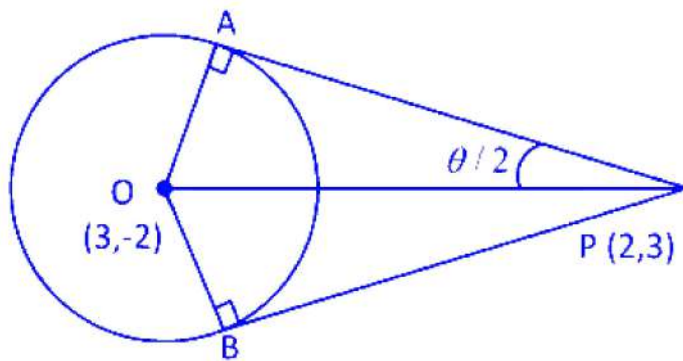
We have,

Equation, of circle is

$$x^2 + y^2 - 6x + 4y + 12 = 0$$

$$\text{Centre of circle} = (3, -2)$$

$$\text{radius} = \sqrt{9 + 4 - 12} = 1$$



$$OP = \sqrt{(3-2)^2 + (-2-3)^2}$$

$$= \sqrt{1 + 25} = \sqrt{26}$$

$$AP = \sqrt{OP^2 - AO^2}$$

$$= \sqrt{26 - 1} = 5$$

$$\text{In } \triangle MOP, \tan \frac{\theta}{2} = \frac{AO}{AP} = \frac{1}{5}$$

$$\tan \theta = \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}} = \frac{2 \times \frac{1}{5}}{1 - \left(\frac{1}{5}\right)^2} = \frac{\frac{2}{5}}{\frac{24}{25}} = \frac{5}{12}$$

$$\text{Hence, } \theta = \tan^{-1} \frac{5}{12}$$

Question98

If $2x - 3y + 3 = 0$ and $x + 2y + k = 0$ are conjugate lines with respect to the circle $S = x^2 + y^2 + 8x - 6y - 24 = 0$, then the length of the tangent drawn from the point $\left(\frac{k}{4}, \frac{k}{3}\right)$ to the circle $S = 0$, is

AP EAPCET 2024 - 18th May Morning Shift

Options:

A. 7

B. 1

C. 12

D. 24

Answer: B

Solution:

Let (a, b) be the point on the line

$$x + 2y + K = 0$$

We have, equation of circle is

$$x^2 + y^2 + 8x - 6y - 24 = 0$$

Equation of polar for the given circle is obtained as

$$ax + by + 4(x + d) - 3(y + b) - 24 = 0$$

$$(a + 4)x + (b - 3)y + 4a - 3 - 24$$

Now, comparing this line with

$$2x - 3y + 3 = 0 \text{ we get}$$

$$\frac{a + 4}{2} = \frac{b - 3}{-3} = \frac{4a - y - 24}{3}$$

$$-3a - 12 = 2b - 6 \Rightarrow 3a + 2b = -6$$

$$\text{and } 3a + 12 = 8a - 6b - 4s$$

$$\Rightarrow 5a - 6b = 60 \dots (ii)$$

On solving Eqs. (i) and (ii), we get

$$a = 3 \text{ and } b = \frac{-15}{2}$$

Now, $\left(3, \frac{15}{2}\right)$ lies on line $x + 2y + k = 0$

$$3 - 15 + K = 0 \Rightarrow K = 12$$

Required length of tangent from a point

$$\begin{aligned} & \left(\frac{12}{4}, \frac{12}{3} \right) \text{ i.e. } (3, 4) \\ & = \sqrt{3^3 + 4^2 + 24 - 24 - 24} \\ & = \sqrt{9 + 16 - 24} = 1 \end{aligned}$$

Question99

If $Q(h, k)$ is the inverse point of the point $P(1, 2)$ with respect to the circle $x^2 + y^2 - 4x + 1 = 0$, then $2h + k =$

AP EAPCET 2024 - 18th May Morning Shift

Options:

- A. 3
- B. 4
- C. 7
- D. 11

Answer: B

Solution:

The equation of pole with respect to the point $\beta, 2$ to the given circle is

$$\begin{aligned} x + 2y - 2(x + 1) + 1 &= 0 \\ \Rightarrow -x + 2y - 1 &= 0 \text{ or } x - 2y + 1 = 0 \end{aligned}$$

Inverse of point $P(1, 2)$ is the foot (h, k) of the perpendicular from the point $(1, 2)$ to the line $x - 2y + 1 = 0$

$$\begin{aligned} \Rightarrow \frac{h-1}{1} &= \frac{k-2}{-2} = - \left[\frac{1-4+1}{1^2+(-2)^2} \right] \\ \Rightarrow h-1 &= \frac{K-2}{-2} = \frac{2}{5} \\ \Rightarrow h-1 &= \frac{2}{5} \text{ and } \frac{K-2}{-2} = \frac{2}{5} \\ \Rightarrow h &= \frac{7}{5} \text{ and } K = \frac{-4}{5} + 2 = \frac{6}{5} \end{aligned}$$

$$\text{Hence, } 2h + K = \frac{14}{5} + \frac{6}{5} = \frac{20}{5} = 4$$

Question100

If (a, b) and (c, d) are the internal and external centres of similitudes of the circles $x^2 + y^2 + 4x - 5 = 0$ and $x^2 + y^2 - 6y + 8 = 0$ respectively, then $(a + d)(b + c) =$

AP EAPCET 2024 - 18th May Morning Shift

Options:

A. 4

B. 9

C. 13

D. 22

Answer: C

Solution:

We have, equation of two circles

$$x^2 + y^2 + 4x - 5 = 0$$

$$\text{and } x^2 + y^2 - 6y + 8 = 0$$

Centre and radius of both circles are

$$c_1 = (-2, 0), r_1 = \sqrt{4 + 5} = 3$$

$$c_2 = (0, 3), r_2 = \sqrt{9 - 8} = 1$$

Let P and Q be the internal and external centre of similitude, it divides C_1C_2 in the ratio 3 : 1.

$$P = \frac{x_2 + C_1}{3 + 1} = \left(\frac{-2}{4}, \frac{9}{4}\right) = \left(-\frac{1}{2}, \frac{9}{4}\right)$$

$$\text{Here, } a = -\frac{1}{2} \text{ and } b = \frac{9}{4}$$

$$Q = \frac{3C_2 - C_1}{C_1 - C_2} = \left(\frac{2}{2}, \frac{9}{2}\right) = \left(1, \frac{9}{2}\right)$$

$$\text{Here, } c = 1 \text{ and } d = \frac{9}{2}$$

$$\begin{aligned} \text{Hence, } (a + c)(b + d) &= \left(-\frac{1}{2} + 1\right) \left(\frac{9}{4} + 1\right) \\ &= \frac{4 \times 13}{4} = 13 \end{aligned}$$

A circle s passes through the points of intersection of the circles $x^2 + y^2 - 2x + 2y - 2 = 0$ and $x^2 + y^2 + 2x - 2y + 1 = 0$. If the centre of this circle S lies on the line $x - y + 6 = 0$, then the radius of the circle S is

AP EAPCET 2024 - 18th May Morning Shift

Options:

A. $\sqrt{5}$

B. 5

C. $\sqrt{41}$

D. $\sqrt{14}$

Answer: D

Solution:

The equation of a circle that passes through the intersection points of the given circles is determined as follows:

First, start with the general form that includes both circles and a parameter λ :

$$x^2 + y^2 - 2x + 2y - 2 + \lambda(x^2 + y^2 + 2x - 2y + 1) = 0$$

Simplifying this equation leads to:

$$(1 + \lambda)x^2 + (1 + \lambda)y^2 + (2\lambda - 2)x + (2 - 2\lambda)y + (-2 + \lambda) = 0$$

To rewrite in standard circle form, factor out the coefficients:

$$x^2 + y^2 + 2x \left(\frac{\lambda-1}{1+\lambda} \right) + 2y \left(\frac{1-\lambda}{1+\lambda} \right) - \frac{2-\lambda}{1+\lambda} = 0$$

The center of the circle (h, k) from this equation is:

$$\left(\frac{1-\lambda}{1+\lambda}, \frac{\lambda-1}{1+\lambda} \right)$$

Since this center lies on the line $x - y + 6 = 0$, we substitute the center into the line's equation:

$$1 - \lambda - (\lambda - 1) + 6(1 + \lambda) = 0$$

Solving:

$$1 - \lambda - \lambda + 1 + 6\lambda + 6 = 0 \Rightarrow 4\lambda + 8 = 0 \Rightarrow \lambda = -2$$

Substitute $\lambda = -2$ back into the center, we get:

$$(h, k) = (-3, 3)$$

Finally, calculate the radius of the circle:

$$\text{Radius} = \sqrt{(-3)^2 + 3^2 - \frac{2-(-2)}{1+(-2)}} = \sqrt{(-3)^2 + 3^2 - \frac{4}{-1}}$$

Solving further:

$$\text{Radius} = \sqrt{9 + 9 + 4} = \sqrt{14}$$

Question102

The locus of mid-points of points of intersection of $x \cos \theta + y \sin \theta = 1$ with the coordinate axes is

AP EAPCET 2022 - 5th July Morning Shift

Options:

A. $x^2 + y^2 = 4$

B. $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{4}$

C. $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{2}$

D. $x^2 + y^2 = 2$

Answer: B

Solution:

Let the line cut the axes in A and B and if (h, k) be the mid-point of AB , then

$$2h = \frac{1}{\cos \theta}, 2k = \frac{1}{\sin \theta}$$

In order to find the locus, eliminate the variable θ by $\cos^2 \theta + \sin^2 \theta = 1$

$$\frac{1}{4h^2} + \frac{1}{4k^2} = 1 \Rightarrow \frac{1}{x^2} + \frac{1}{y^2} = 4$$

Question103

The radius of the circle having. $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ as its tangents is

AP EAPCET 2022 - 5th July Morning Shift

Options:

A. $\frac{3}{2}$

B. 3

C. 6

D. $\frac{3}{4}$

Answer: D

Solution:

Equation of the given tangents are

$$E_1 = 3x - 4y + 4 = 0$$

$$E_2 = 6x - 8y - 7 = 0$$

$$\Rightarrow 3x - 4y - \frac{7}{2} = 0$$

$$\text{Here, } a = 3, b = -4, c_1 = 4, c_2 = \frac{-7}{2}$$

Since, slopes of the given tangents are equal, i.e. $\frac{3}{4}$.

$\therefore$ The given tangents are parallel,

$$d = \left| \frac{c_2 - c_1}{\sqrt{a^2 + b^2}} \right| = \left| \frac{4 - \left(\frac{-7}{2}\right)}{\sqrt{3^2 + (-4)^2}} \right|$$

$$d = \frac{\frac{15}{2}}{\sqrt{9 + 16}} = \frac{15}{2 \times 5} = \frac{3}{2}$$

As we know that distance between two parallel tangents of a circle is equal to the diameter of that circle.

$$\therefore \text{Diameter of given circle} = \frac{3}{2}$$

$$\begin{aligned} \therefore \text{Radius of the given circle} &= \frac{\text{Diameter}}{2} \\ &= \left(\frac{3/2}{2} \right) = \frac{3}{4} \end{aligned}$$

Question 104

A circle is such that $(x - 2) \cos \theta + (y - 2) \sin \theta = 1$ touches it for all values of θ . Then, the circle is

AP EAPCET 2022 - 5th July Morning Shift

Options:

A. $x^2 + y^2 - 4x - 4y + 7 = 0$

B. $x^2 + y^2 + 4x + 4y + 7 = 0$

C. $x^2 + y^2 - 4x - 4y - 7 = 0$

D. $x^2 + y^2 + 4x + 4y - 7 = 0$

Answer: A

Solution:

Since, the line $(x - 2) \cos \theta + (y - 2) \sin \theta = 1$ touches a circle. So it is a tangent equation to a circle.

Equation of tangent to a circle at (x_1, y_1) is $(x - h)x_1 + (y - k)y_1 = a^2$ to a circle

$$(x - h)^2 + (y - k)^2 = a^2$$

Then, after comparing

$$x - h = x - 2, y - k = y - 2 \text{ and } a^2 = 1$$

$$x_1 = \cos \theta, y_1 = \sin \theta$$

$\therefore$ Required equation of circle

$$\begin{aligned} (x - 2)^2 + (y - 2)^2 &= 1 \\ \Rightarrow x^2 + y^2 - 4x - 4y + 7 &= 0 \end{aligned}$$

Question105

The least distance of the point $(10, 7)$ from the circle $x^2 + y^2 - 4x - 2y - 20 = 0$ is

AP EAPCET 2022 - 5th July Morning Shift

Options:

A. 6

B. 7

C. 4

D. 5

Answer: D

Solution:

Given equation $(x^2 + y^2 - 4x - 2y - 20 = 0)$

So, $C = (2, 1)$ and radius $= \sqrt{(g)^2 + (f)^2 - c}$

$$= \sqrt{(2)^2 + (1)^2 + 20}$$

Radius $= 5$

When substituted $x = 10$ and $y = 7$ in equation, then value becomes

$$(10)^2 + (7)^2 - 4(10) - 2(7) - 20 = 75$$

which is greater than zero.

Thus, the point $(10, 7)$ lies outside the circle.

Its distance from the centre of the circle $(2, 1)$ is

$$\sqrt{(10 - 2)^2 + (7 - 1)^2} = 10 \text{ units}$$

So, the minimum distance from the circle therefore becomes $10 - 5 = 5$ units

Question106

Suppose that the x -coordinates of the points A and B satisfy $x^2 + 2x - a^2 = 0$ and their y -coordinates satisfy $y^2 + 4y - b^2 = 0$. Then, the equation of the circle with AB as its diameter is

AP EAPCET 2022 - 5th July Morning Shift

Options:

A. $x^2 + y^2 + 2x + 4y - a^2 - b^2 = 0$

B. $x^2 + y^2 + 2x + 4y + a^2 + b^2 = 0$

C. $x^2 + y^2 - 2x - 4y - a^2 - b^2 = 0$

D. $x^2 + y^2 - 2x - 4y + a^2 + b^2 = 0$

Answer: A

Solution:

Let x_1 and x_2 are roots of equation $x^2 + 2x - a^2 = 0$ and y_1 and y_2 are roots of equation $y^2 + 4y - b^2 = (x - x_1)(x - x_2) = 0$ and $y^2 + 4y - b^2 \rightarrow y_1, y_2 = (y - y_1)(y - y_2)$ (roots of equation)

Let $A(x_1, y_1)$ and $B(x_2, y_2)$.

Now, the equation of circle with diameter AB is given by $\Rightarrow (x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

$$\Rightarrow x^2 + 2x - a^2 + y^2 + 4y - b^2 = 0$$

$$\Rightarrow x^2 + y^2 + 2x + 4y = a^2 + b^2$$

$$\Rightarrow x^2 + y^2 + 2x + 4y - a^2 - b^2 = 0$$

Question107

The radical centre of the three circles

$x^2 + y^2 - 1 = 0$, $x^2 + y^2 - 8x + 15 = 0$ and $x^2 + y^2 + 10y + 24 = 0$ is

AP EAPCET 2022 - 5th July Morning Shift

Options:

A. $(2, \frac{-5}{2})$

B. $(2, \frac{5}{2})$

C. $(-2, \frac{5}{2})$

D. $(-2, \frac{-5}{2})$

Answer: A

Solution:

$$\text{Let } S_1 : x^2 + y^2 - 1 = 0 \quad \dots (i)$$

$$S_2 : x^2 + y^2 - 8x + 15 = 0 \quad \dots (ii)$$

$$\text{and } S_3 : x^2 + y^2 + 10y + 24 = 0 \quad \dots (iii)$$

From Eqs. (i) and (ii),

$$\begin{aligned} 8x - 15 &= 1 \\ \Rightarrow x &= 2 \end{aligned}$$

Now, from Eqs. (i) and (iii),

$$\begin{aligned} -10y - 24 &= 1 \\ \Rightarrow -10y &= 25 \Rightarrow y = \frac{-5}{2} \end{aligned}$$

Hence, the radical centre is $(2, \frac{-5}{2})$.

Question 108

For any real number t , the point $\left(\frac{8t}{1+t^2}, \frac{4(1-t^2)}{1+t^2} \right)$ lies on a / an

AP EAPCET 2022 - 4th July Evening Shift

Options:

- A. circle of radius 2
- B. circle of radius 4
- C. ellipse with 4 as its major axis length
- D. ellipse with 4 as its minor axis length

Answer: B

Solution:

$$\begin{aligned} \text{Let } x &= \frac{8t}{1+t^2} \quad \dots (i) \\ y &= \frac{4(1-t^2)}{1+t^2} \quad \dots (ii) \end{aligned}$$

Squaring and adding Eqs. (i) and (ii), we get

$$\begin{aligned}x^2 + y^2 &= \frac{64t^2}{(1+t^2)^2} + \frac{16(1-2t^2+t^4)}{(1+t^2)^2} \\&= \frac{16(1+2t^2+t^4)}{(1+t^2)^2} = \frac{16(1+t^2)^2}{(1+t^2)^2} = 16 \\ \Rightarrow x^2 + y^2 &= (4)^2\end{aligned}$$

which is circle of radius 4 units.

Question109

The area of the circle passing through the points $(5, \pm 2)$, $(1, 2)$ is

AP EAPCET 2022 - 4th July Evening Shift

Options:

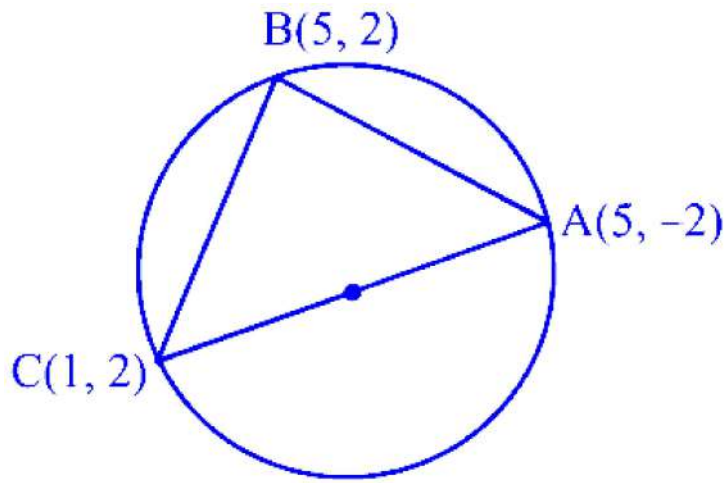
- A. 8π
- B. 4π
- C. 2π
- D. 16π

Answer: A

Solution:

Centre of circle passing through $(1, 2)$, $(5, 2)$ and $(5, -2)$ is mid-point of $C(1, 2)$ and $A(5, -2)$. We know that diameter always form a right angle on a circle.

$$\therefore \text{Centre} = \left(\frac{1+5}{2}, \frac{2-2}{2} \right) = (3, 0)$$



$$\begin{aligned}\text{and Radius} &= \frac{\text{Diameter}}{2} \\ &= \frac{\sqrt{(5-1)^2 + (-2-2)^2}}{2} = 2\sqrt{2}\end{aligned}$$

Thus, area is $\pi(2\sqrt{2})^2 = 8\pi$ (units)²

Question110

The ratio of the largest and shortest distances from the point $(2, -7)$ to the circle $x^2 + y^2 - 14x - 10y - 151 = 0$ is

AP EAPCET 2022 - 4th July Evening Shift

Options:

- A. 15 : 13
- B. 7 : 1
- C. 3 : 2
- D. 14 : 1

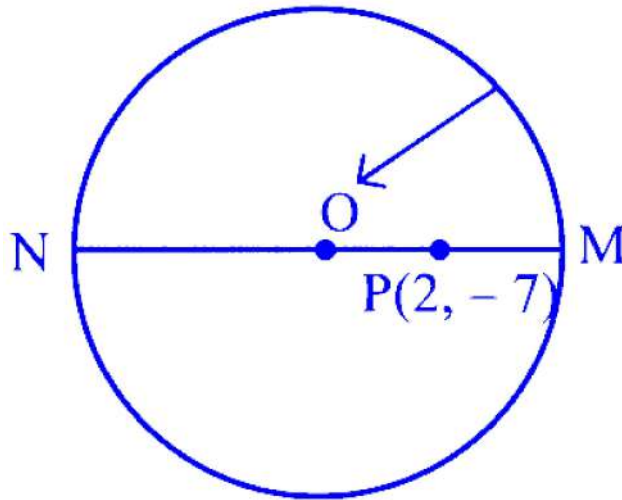
Answer: D

Solution:

To find the shortest and longest distances from the point $(2, -7)$ to the circle

$$x^2 + y^2 - 14x - 10y - 151 = 0$$

So, to find the solution we first draw the diagram,



So, we have to find the coordinates of the centre O and the radius of the circle, i.e $OM = ON = \text{radius}$ Thus, the shortest distance $PM = OM - OP$ and the longest distance $PN = MN - PM$ First we have to determine the point $P(2, -7)$ lies in which side of circle. So, we put the value $x = 2, y = -7$ in the left hand side of the equation, we get $x^2 + y^2 - 14x - 10y - 151 = (2)^2$

$$= (2)^2 + (-7)^2 - 14 \times 2 - 10(-7) - 151 = -56 < 0$$

Thus, the point lies inside of the circle. Thus, we know any equation of the circle is in the form $x^2 + y^2 + 2gx + 2fy + c = 0$.

Thus, $g = -7$, and $f = -5, c = -151$

So, centre is $(-g, -f) = (7, 5)$

$$\begin{aligned} \text{and radius } &= r = OM = ON \\ &= \sqrt{7^2 + 5^2 - (-151)} \\ &= \sqrt{49 + 25 + 151} \\ &= \sqrt{225} = 15 \end{aligned}$$

Now, distance between centre $O(7, 5)$ and $P(2, -7)$ is $OP = \sqrt{(7-2)^2 + (5+7)^2}$

$$= \sqrt{25 + 144} = \sqrt{169} = 13$$

Thus, the shortest distance from the point $P(2, -7)$ to the circle is $PM = OM - OP = 15 - 13 = 2$ units and the longest distance is

$$\begin{aligned} PN &= MN - PM = (OM + ON) - PM \\ &= (15 + 15) - 2 = 28 \text{ units} \end{aligned}$$

Thus, the required ratio is $28 : 2 = 14 : 1$

Question111

A circle has its centre in the first quadrant and passes through $(2, 3)$. If this circle makes intercepts of length 3 and 4 respectively on $x = 2$ and $y = 3$, its equation is

AP EAPCET 2022 - 4th July Evening Shift

Options:

A. $x^2 + y^2 + 3x - 5y + 8 = 0$

B. $x^2 + y^2 - 4x - 6y + 13 = 0$

C. $x^2 + y^2 - 6x - 8y + 23 = 0$

D. $x^2 + y^2 - 8x - 9y + 30 = 0$

Answer: D

Solution:

Let the centre of circle be (h, k) . Equation of circle will be

$$x^2 + y^2 - 2hx - 2ky + c = 0 \quad \dots (i)$$

Circle passes through $(2, 3)$

$$2^2 + 3^2 - 4h - 6k + c = 0$$

$$\Rightarrow 4h + 6k - c = 13$$

$$\text{or } c = 4h + 6k - 13$$

Equation of circle be

$$x^2 + y^2 - 2hx - 2ky + 4h + 6k - 13$$

This circle makes intercept of length 3 on the line $x = 2$

$|y_1 - y_2| = 3$, when $x = 2$ is substituted in the equation of circle, where y_1 and y_2 are the two roots

$$4 + y^2 - 4h - 2ky + 4h + 6k - 13 = 0$$

$$\Rightarrow y^2 - 2ky + 6k - 9 = 0$$

$$y_1 + y_2 = 2k \text{ and } y_1 y_2 = 6k - 9$$

$$|y_1 - y_2| = 3$$

$$\sqrt{(y_1 + y_2)^2 - 4y_1 y_2} = 3$$

$$\Rightarrow 4k^2 - 4(6k - 9) = 9$$

$$\Rightarrow 4k^2 - 24k + 36 - 9 = 0 \Rightarrow 4k^2 - 24k + 27 = 0$$

$$\Rightarrow k = \frac{9}{2}, k \neq \frac{3}{2}$$

[$\because$ when substitute $k = \frac{3}{2}$ in $y_1 y_2$, we get $y_1 y_2$ as 0]

Now, similarly substitute $y = 3$

$$x^2 + 9 - 2hx - 6k + 4h + 6k - 13 = 0$$

$$\Rightarrow x^2 - 2hx + 4h - 4 = 0$$

$$\Rightarrow x_1 + x_2 = 2h, x_1 x_2 = 4h - 4$$

$$\therefore |x_1 - x_2| = 4$$

$$\sqrt{(x_1 + x_2)^2 - 4x_1 x_2} = 4$$

$$\Rightarrow 4h^2 - 16h + 16 = 16 \Rightarrow 4h(h - 4) = 0$$

$$h = 4 \text{ and } h \neq 0$$

$\therefore$ Equation of the required circle is

$$x^2 + y^2 - 8x - 9y + 30 = 0$$

Question 112

The image of the point $(3, 4)$ with respect to the radical axis of the circles $x^2 + y^2 + 8x + 2y + 10 = 0$ and $x^2 + y^2 + 7x + 3y + 10 = 0$ is

AP EAPCET 2022 - 4th July Evening Shift

Options:

A. $(3, 4)$

B. $(-4, -3)$

C. $(4, 3)$

D. $(-3, -4)$

Answer: C

Solution:

The given equation of circles

$$S_1 : x^2 + y^2 + 8x + 2y + 10 = 0$$

$$S_2 : x^2 + y^2 + 7x + 3y + 10 = 0$$

As we know the equation of radical axis of two circles is given by

$$S_1 - S_2 = 0$$

$$\Rightarrow (x^2 + y^2 + 8x + 2y + 10) - (x^2 + y^2 + 7x + 3y + 10) = 0$$

$$\Rightarrow (8x - 7x) + (2y - 3y) = 0$$

$$\Rightarrow x - y = 0$$

Thus, equation of radical axis of the given circle is

$$x - y = 0$$

Hence, the image of the point $(3, 4)$ with respect to $y = x$ line is $(4, 3)$.

Question113

The locus of centers of the circles, possessing the same area and having $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ as their common tangent, is

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. $12x - 16y - 15 = 0$

B. $3x - 4y + \frac{11}{2} = 0$

C. $12x - 16y + 15 = 0$

D. $3x - 4y - \frac{11}{2} = 0$

Answer: A

Solution:

1. Same area $\Rightarrow$ same radius

Circles having the same area must have the same radius.

2. Given common tangents:

$$L_1 : 3x - 4y + 4 = 0, \quad L_2 : 6x - 8y - 7 = 0$$

3. Condition for same radius:

The distance of the center (x, y) from both tangent lines must be equal.

4. Distance formula from a line:

Distance of (x, y) from $ax + by + c = 0$ is

$$\frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$$

5. Apply distance equality:

$$\frac{|3x - 4y + 4|}{\sqrt{3^2 + (-4)^2}} = \frac{|6x - 8y - 7|}{\sqrt{6^2 + (-8)^2}}$$

6. Simplify denominators:

$$\sqrt{25} = 5, \quad \sqrt{100} = 10$$

$$\frac{|3x - 4y + 4|}{5} = \frac{|6x - 8y - 7|}{10}$$

7. Cross-multiply (taking same sign case):

$$2(3x - 4y + 4) = 6x - 8y - 7$$

8. Simplify:

$$6x - 8y + 8 = 6x - 8y - 7$$

$$\Rightarrow 15 = 0 \quad (\text{shift constants})$$

Rearranging gives:

$$12x - 16y - 15 = 0$$

☒ The correct answer is: A $12x - 16y - 15 = 0$

Question 114

For any two non-zero real numbers a and b if this line $\frac{x}{a} + \frac{y}{b} = 1$ is a tangent to the circle $x^2 + y^2 = 1$, then which of the following is true?

AP EAPCET 2022 - 4th July Morning Shift

Options:

- A. $(\frac{1}{a}, \frac{1}{b})$ lies inside the circle
- B. (a, b) lies inside the circle
- C. $(\frac{1}{a}, \frac{1}{b})$ lies on the circle
- D. (a, b) lies on the circle.

Answer: C

Solution:

Curve : $x^2 + y^2 = 1$

Let $(\frac{1}{a}, \frac{1}{b}) = (x_1, y_1)$ touches the circles, then equation of tangent at (x_1, y_1) is

$$\begin{aligned} &xx_1 + yy_1 = 1 \\ \Rightarrow &x \cdot \left(\frac{1}{a}\right) + y \cdot \left(\frac{1}{b}\right) = 1 \Rightarrow \frac{x}{a} + \frac{y}{b} = 1 \end{aligned}$$

Question115

The length of the intercept on the line $4x - 3y - 10 = 0$ by the circle $x^2 + y^2 - 2x + 4y - 20 = 0$ is

AP EAPCET 2022 - 4th July Morning Shift

Options:

- A. 5
- B. 2
- C. 10
- D. 6

Answer: C

Solution:

$$4x - 3y - 10 = 0$$

$$\Rightarrow x = \frac{3y + 10}{4}$$

$$\text{and } x^2 + y^2 - 2x + 4y - 20 = 0$$

$$\Rightarrow \left(\frac{3y + 10}{4} \right)^2 + y^2 - 2 \left(\frac{3y + 10}{4} \right) + 4y - 20 = 0$$

$$\Rightarrow 9y^2 + 100 + 60y + 16y^2 - 24y - 80 + 64y - 320 = 0$$

$$\Rightarrow 25y^2 + 100y - 300 = 0 \Rightarrow y^2 + 4y - 12 = 0$$

$$\Rightarrow y^2 + 6y - 2y - 12 = 0 \Rightarrow y(y + 6) - 2(y + 6) = 0$$

$$\Rightarrow (y + 6)(y - 2) = 0$$

$$y = -6, 2$$

$$y = -6 \Rightarrow x = \frac{-18 + 10}{4} = -2 \Rightarrow (-2, -6)$$

$$y = 2 \Rightarrow x = \frac{6 + 10}{4} = 4 \Rightarrow (4, 2)$$

Required length

$$= \sqrt{(4 + 2)^2 + (2 + 6)^2}$$

$$= \sqrt{36 + 64} = \sqrt{100} = 10$$

Question116

The pole of the line $\frac{x}{a} + \frac{y}{b} = 1$ with respect to the circle $x^2 + y^2 = c^2$ is

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. $\left(\frac{c^2}{a}, \frac{c^2}{b} \right)$

B. $\left(\frac{c^2}{b}, \frac{c^2}{a} \right)$

C. $\left(\frac{c}{a}, \frac{c}{b} \right)$

D. $\left(\frac{c}{b}, \frac{c}{a} \right)$

Answer: A

Solution:

Let the pole be (x_1, y_1) , then the equation of the polar with respect to the circle $x^2 + y^2 = c^2$ is $xx_1 + yy_1 = c^2$

$$\Rightarrow \frac{xx_1}{c^2} + \frac{yy_1}{c^2} = 1 \quad \dots (i)$$

Eq. (i) and $\frac{x}{a} + \frac{y}{b} = 1$ represents the same line.

$$\therefore \frac{x_1}{c^2} = \frac{1}{a} \Rightarrow x_1 = \frac{c^2}{a}$$
$$\frac{y_1}{c^2} = \frac{1}{b} \Rightarrow y_1 = \frac{c^2}{b}$$

$$\text{Pole : } (x_1, y_1) = \left(\frac{c^2}{a}, \frac{c^2}{b} \right)$$

Question117

If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the Y -axis, then the length of PQ is

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. 5

B. 6

C. 4

D. 3

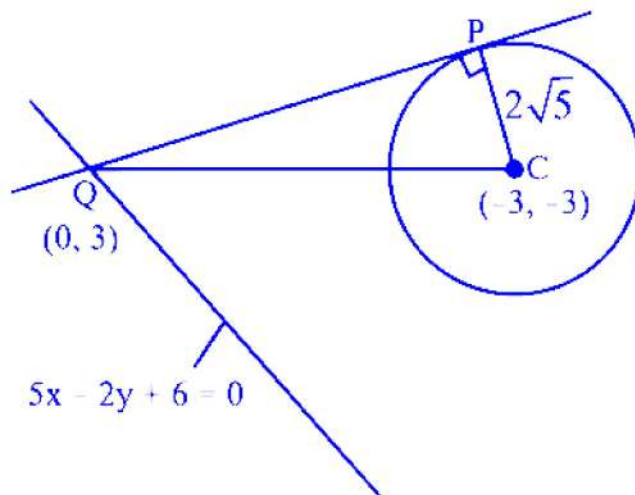
Answer: A

Solution:

Line: $5x - 2y + 6 = 0$

On Y -axis, $x = 0 \Rightarrow y = \frac{6}{2} = 3$

i.e. $Q \equiv (0, 3)$



Circle : $x^2 + y^2 + 6x + 6y - 2 = 0$

Centre $(-g, -f) \equiv (-3, -3)$

Radius $= \sqrt{g^2 + f^2 - c}$

$= \sqrt{9 + 9 + 2} = \sqrt{20}$

$= 2\sqrt{5}$

$CQ = \sqrt{(0+3)^2 + (3+3)^2} = \sqrt{9+36} = \sqrt{45}$

In $\triangle PQC$,

$(CQ)^2 = (CP)^2 + (QP)^2$

$\Rightarrow 45 = 20 + (QP)^2$

$\Rightarrow 25 = (QP)^2 \Rightarrow QP = 5$

Question118

The equation of the pair of straight lines parallel to x -axis and touching the circle $x^2 + y^2 - 6x - 4y - 12 = 0$ is

AP EAPCET 2021 - 20th August Evening Shift

Options:

A. $y^2 - 4y - 21 = 0$

B. $y^2 + 4y - 21 = 0$

C. $y^2 - 4y + 21 = 0$

D. $y^2 + 4y + 21 = 0$

Answer: A

Solution:

Equation parallel to X -axis is given as

$$y = a$$

$$\text{or } y - a = 0 \dots\dots (i)$$

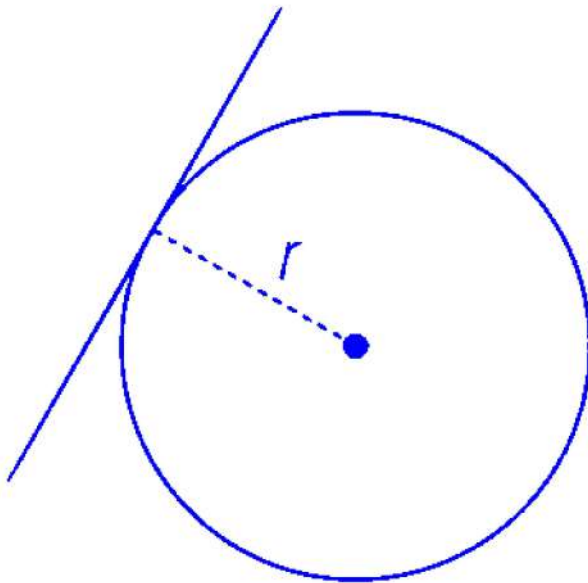
If Eq. (i) touches the circle

$$x^2 + y^2 - 6x - 4y - 12 = 0 \dots\dots (ii)$$

Then, Eq. (i) becomes tangent to circle.

Centre of Eq. (ii) is $(3, 2)$

Radius of Eq. (ii) is $\sqrt{9 + 4 + 12} = 5$



Then, perpendicular distance from centre to line (i) is radius i.e.

Perpendicular distance between $(3, 2)$ and $y - a = 0$ is equal to 5

$$\text{i.e. } \left| \frac{2 - a}{1} \right| = 5$$

$$\Rightarrow 2 - a = \pm 5$$

$$\Rightarrow a = 7, -3$$

$$\therefore \text{Equation of pair of lines} = (y + 3)(y - 7) = 0$$

$$\text{i.e. } (y + 3)(y - 7) = 0$$

$$\Rightarrow y^2 - 4y - 21 = 0$$

Question119

The points where the circle $x^2 + y^2 - 3x - 4y + 2 = 0$ cuts the X -axis are

AP EAPCET 2021 - 20th August Evening Shift

Options:

A. (1, 2) and (2, 0)

B. (2, 0) and (3, 0)

C. (0, 2) and (0, 1)

D. (1, 0) and (2, 0)

Answer: D

Solution:

Given circle,

$$x^2 + y^2 - 3x - 4y + 2 = 0 \dots (i)$$

It cut the X -axis, when $y = 0$

Put $y = 0$ in Eq. (i), we obtain

$$\begin{aligned} x^2 - 3x + 2 &= 0 \\ \Rightarrow (x - 2)(x - 1) &= 0 \\ \therefore x &= 2, 1 \end{aligned}$$

Thus, circle (i) touches X -axis at (2, 0), (1, 0).

Question120

The center and radius of the circle $x^2 + y^2 + 8x + 10y - 8 = 0$ respectively are and units

AP EAPCET 2021 - 20th August Evening Shift

Options:

A. $(-4, -5), 7$

B. $(4, 5), 49$

C. $(-8, -10), 8$

D. $(-4, 5), 7$

Answer: A

Solution:

$$\begin{aligned}x^2 + y^2 + 8x + 10y - 8 &= 0 \\ \Rightarrow (x + 4)^2 + (y + 5)^2 - 16 - 25 - 8 &= 0 \\ \Rightarrow (x + 4)^2 + (y + 5)^2 &= 49 \\ \Rightarrow (x + 4)^2 + (y + 5)^2 &= (7)^2\end{aligned}$$

which is standard form of circle.

Radius = 7 and Centre = $(-4, -5)$

Question121

The poles of the tangents to the circle $x^2 + y^2 = 4$ with respect to the circle $(x + 2)^2 + y^2 = 8$, lie on

AP EAPCET 2021 - 20th August Evening Shift

Options:

A. $y^2 + 8x = 0$

B. $x^2 + 8y = 0$

C. $y^2 - 8x = 0$

D. $x^2 - 8y = 0$

Answer: A

Solution:

We know an important result:

If you draw tangents to the circle $x^2 + y^2 = a^2$, and then find their poles with respect to the circle $(x + a)^2 + y^2 = 2a^2$, all those poles will be on the curve $y^2 + 4ax = 0$.

In our question, $a = 2$. So, we are looking at the circles $x^2 + y^2 = 4$ and $(x + 2)^2 + y^2 = 8$.

Using the result, the poles of the tangents to $x^2 + y^2 = 4$ with respect to $(x + 2)^2 + y^2 = 8$ will be on:

$$y^2 + 4 \times 2x = 0$$

which simplifies to:

$$y^2 + 8x = 0$$

Question122

If the power of the point $(1, 6)$ with respect to the circle $x^2 + y^2 + 4x - 6y - a = 0$ is -16 then a equals

AP EAPCET 2021 - 20th August Evening Shift

Options:

A. 5

B. 11

C. 21

D. 6

Answer: C

Solution:

Given equation of circle is

$$S : x^2 + y^2 + 4x - 6y - a = 0 \dots (i)$$

Power of point w.r.t. Eq. (i) is -16 , i.e.

$$S(1, 6) = -16$$

$$\text{or } (1)^2 + (6)^2 + 4 - 36 - a = -16$$

$$\Rightarrow 5 - a = -16$$

$$\Rightarrow a = 21$$

Question 123

The equation of radical axis of the circles $x^2 + y^2 + 4x + 6y + 7 = 0$ and $4x^2 + 4y^2 + 8x + 12y - 9 = 0$ is

AP EAPCET 2021 - 20th August Evening Shift

Options:

A. $x + y + 1 = 0$

B. $8x + 12y = 0$

C. $8x + 12y + 37 = 0$

D. $2x + 3y + 7 = 0$

Answer: C

Solution:

$$S_1 : x^2 + y^2 + 4x + 6y + 7 = 0$$

$$S_2 : x^2 + y^2 + 2x + 3y - \frac{9}{4} = 0$$

Equation of radical axis

$$S_1 - S_2 = 0$$

$$(x^2 + y^2 + 4x + 6y + 7) - (x^2 + y^2 + 2x + 3y - \frac{9}{4}) = 0$$

$$\Rightarrow 2x + 3y + \frac{37}{4} = 0$$

$$\text{or } 8x + 12y + 37 = 0$$

Question124

The radical axis of the circles $S_1 : x^2 + y^2 - 4x + 6y - 10 = 0$ and $S_2 : x^2 + y^2 + 2x - 6y + 2 = 0$, cut the circle S_1 in

AP EAPCET 2021 - 20th August Evening Shift

Options:

- A. two real and distinct points
- B. one real point
- C. imaginary points
- D. can't be determined

Answer: A

Solution:

$$S_1 : x^2 + y^2 - 4x + 6y - 10 = 0 \dots\dots (i)$$

$$S_2 : x^2 + y^2 + 2x - 6y + 2 = 0 \dots\dots (ii)$$

Equation of radical axis

$$S_1 - S_2 = 0$$

$$\Rightarrow \begin{aligned} &(x^2 + y^2 - 4x + 6y - 10) \\ &- (x^2 + y^2 + 2x - 6y + 2) = 0 \end{aligned}$$

$$\Rightarrow 6x - 12y + 12 = 0$$

$$\Rightarrow x - 2y + 2 = 0 \dots\dots(iii)$$

Given that Eq. (iii) cut the circle (i),

Using Eq. (iii) in Eq. (i) as, $x = 2y - 2$

$$(2y - 2)^2 + y^2 - 4(2y - 2) + 6y - 10 = 0$$

$$\Rightarrow 4y^2 + 4 - 8y + y^2 - 8y + 8 + 6y - 10 = 0$$

$$\Rightarrow 5y^2 - 10y + 2 = 0$$

$$\Rightarrow y = \frac{10 \pm \sqrt{100 - 40}}{10} = \frac{5 \pm \sqrt{15}}{5}$$

$$\Rightarrow x = 2 \left(\frac{5 \pm \sqrt{15}}{5} \right)$$

Thus, Eq. (iii) cuts S_1 at two real and distinct points.

Question125

The locus of a point, which is at a distance of 4 units from $(3, -2)$ in xy -plane is

AP EAPCET 2021 - 20th August Morning Shift

Options:

A. $x^2 + y^2 + 6x - 4y + 16 = 0$

B. $x^2 + y^2 - 6x - 4y + 3 = 0$

C. $x^2 + y^2 - 6x + 4y - 16 = 0$

D. $x^2 + y^2 - 6x + 4y - 3 = 0$

Answer: D

Solution:

Let the point be (h, k) .

So, we have

$$\begin{aligned}(h - 3)^2 + (k + 2)^2 &= 4^2 \\ \Rightarrow h^2 + 9 - 6h + k^2 + 4 + 4k &= 16 \\ \Rightarrow h^2 + k^2 - 6h + 4k - 3 &= 0\end{aligned}$$

Replace $h \rightarrow x$ and $k \rightarrow y$

$$x^2 + y^2 - 6x + 4y - 3 = 0$$

Question126

Find the equation of the circle which passes through origin and cuts off the intercepts -2 and 3 over the X and Y -axes respectively.

AP EAPCET 2021 - 20th August Morning Shift

Options:

A. $x^2 + y^2 - 2x + 8y = 0$

B. $2(x^2 + y^2) + 2x - 3y = 0$

C. $x^2 + y^2 - 2x - 8y = 0$

D. $x^2 + y^2 + 2x - 3y = 0$

Answer: D

Solution:

$$2\sqrt{g^2 - c} = 2;$$

$$2\sqrt{f^2 - c} = 3 \text{ and } c = 0$$

$$g = \pm 1, f = \pm \frac{3}{2}$$

$$x^2 + y^2 \pm 2x \pm 3y = 0$$

It passes through $(-2, 0)$ and $(0, 3)$.

So, centre must be in second quadrant or fourth quadrant.

The x -coordinate of centre is negative.

$$\therefore x^2 + y^2 + 2x - 3y = 0$$

Question127

The angle between the pair of tangents drawn from $(1, 1)$ to the circle $x^2 + y^2 + 4x + 4y - 1 = 0$ is

AP EAPCET 2021 - 20th August Morning Shift

Options:

A. $\frac{\pi}{2}$

B. $\frac{\pi}{4}$

C. $\frac{\pi}{3}$

D. $\frac{\pi}{6}$

Answer: A

Solution:

1. Write the circle in standard form

$$x^2 + y^2 + 4x + 4y - 1 = 0$$

$$(x + 2)^2 + (y + 2)^2 = 9$$

So,

Center $C(-2, -2)$, Radius $r = 3$.

2. Distance of point $P(1, 1)$ from the center

$$CP = \sqrt{(1 + 2)^2 + (1 + 2)^2} = \sqrt{9 + 9} = 3\sqrt{2}$$

3. Use formula for angle between tangents

If tangents are drawn from an external point at distance d from the center,

$$\sin \frac{\theta}{2} = \frac{r}{d}$$

4. Substitute values

$$\sin \frac{\theta}{2} = \frac{3}{3\sqrt{2}} = \frac{1}{\sqrt{2}}$$

5. Find angle

$$\frac{\theta}{2} = \frac{\pi}{4} \Rightarrow \theta = \frac{\pi}{2}$$

 The correct answer is: A — $\frac{\pi}{2}$

Question128

If the circle $x^2 + y^2 - 4x - 8y - 5 = 0$ intersects the line $3x - 4y - m = 0$ in two distinct points, then the number of integral values of ' m ' is

AP EAPCET 2021 - 20th August Morning Shift

Options:

A. 52

B. 51

C. 50

D. 49

Answer: D

Solution:

$$x^2 + y^2 - 4x - 8y - 5 = 0$$

$$\text{Centre} = (2, 4)$$

$$\text{Radius} = 2\sqrt{4 + 16 + 5} = 10$$

$$\text{Line : } 3x - 4y - m = 0$$

Distance of L from Centre $<$ Radius

$$\frac{|3 \cdot 2 - 4 \cdot 4 - m|}{\sqrt{3^2 + 4^2}} < 10$$

$$\Rightarrow \frac{|6 - 16 - m|}{5} < 10$$

$$\Rightarrow |-10 - m| < 50$$

$$\Rightarrow -50 < -10 - m < 50$$

$$\Rightarrow -40 < -m < 60$$

$$\Rightarrow -60 < m < 40$$

Only (d) satisfies the above inequality.

Question 129

Let C be the circle center $(0, 0)$ and radius 3 units. The equation of the locus of the mid-points of the chords of the circle c that subtends an angle of $\frac{2\pi}{3}$ at its centre is

Options:

A. $x^2 + y^2 = \frac{1}{4}$

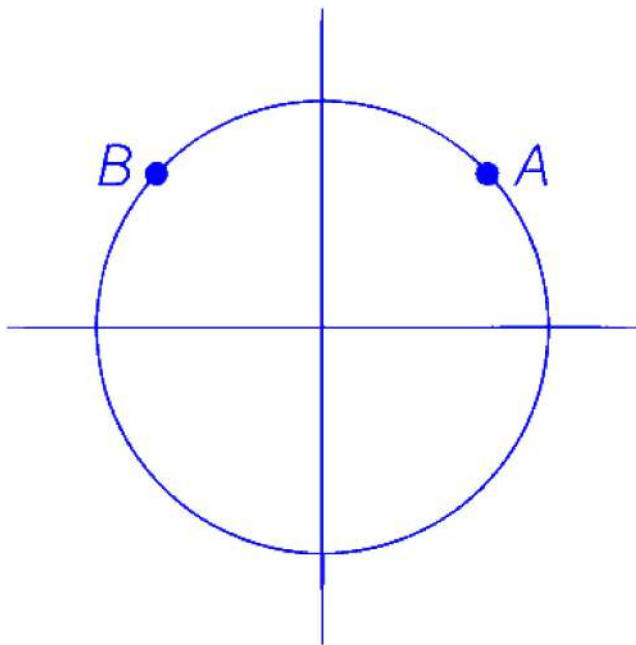
B. $x^2 + y^2 = \frac{27}{4}$

C. $x^2 + y^2 = \frac{9}{4}$

D. $x^2 + y^2 = \frac{5}{4}$

Answer: C

Solution:



Let $A = (3 \cos \theta, 3 \sin \theta)$

Then, $B = (3 \cos (\theta + \frac{2\pi}{3}), 3 \sin (\theta + \frac{2\pi}{3}))$

Let (h, k) be the required point

$$2h = 3 \cos \theta + 3 \cos (\theta + \frac{2\pi}{3})$$

$$= 3 \left[\cos \theta + \cos \theta \left(-\frac{1}{2}\right) - \sin \theta \left(\frac{\sqrt{3}}{2}\right) \right]$$

$$= \frac{3}{2} (\cos \theta - \sqrt{3} \sin \theta)$$

$$2k = 3 \sin \theta + 3 \sin (\theta + \frac{2\pi}{3})$$

$$= 3 \left[\sin \theta + \sin \theta \left(\frac{-1}{2}\right) + \cos \theta \left(\frac{\sqrt{3}}{2}\right) \right]$$

$$= \frac{3}{2}(\sin \theta + \sqrt{3} \cos \theta)$$

$$\cos \theta - \sqrt{3} \sin \theta = \frac{4h}{3} \Rightarrow \sqrt{3} \cos \theta + \sin \theta = \frac{4k}{3}$$

$$\sqrt{3} \cos \theta - 3 \sin \theta = \frac{4h}{\sqrt{3}}$$

$$4 \sin \theta = \frac{4}{3}(k - \sqrt{3}h)$$

$$\sin \theta = \left(\frac{k - \sqrt{3}h}{3} \right)$$

$$\cos \theta = \left(\frac{3k + \sqrt{3}h}{3\sqrt{3}} \right) = \left(\frac{\sqrt{3}k + h}{3} \right)$$

$$\Rightarrow \left(\frac{k - \sqrt{3}h}{3} \right)^2 + \left(\frac{\sqrt{3}k + h}{3} \right)^2 = 1$$

$$\Rightarrow k^2 + 3h^2 - 2\sqrt{3}hk + 3k^2 + h^2 + 2\sqrt{3}kh = 9$$

$$\Rightarrow 4(k^2 + h^2) = 9 \Rightarrow x^2 + y^2 = \frac{9}{4}$$

Question130

The length of the common chord of the circles

$x^2 + y^2 + 3x + 5y + 4 = 0$ and $x^2 + y^2 + 5x + 3y + 4 = 0$ is _____ units.

AP EAPCET 2021 - 20th August Morning Shift

Options:

A. 3

B. 2

C. 6

D. 4

Answer: D

Solution:

$$\begin{aligned}
& x^2 + y^2 + 3x + 5y + 4 = 0 \\
& x^2 + y^2 + 5x + 3y + 4 = 0 \quad P - 2x + 2y = 0 \\
& \Rightarrow x = y \Rightarrow 2x^2 + 8x + 4 = 0 \\
& \Rightarrow x^2 + 4x + 2 = 0 \Rightarrow (x + 2)^2 = 2 \\
& \Rightarrow x + 2 = \pm\sqrt{2} \Rightarrow x = -2 \pm \sqrt{2} \\
& \text{Length} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{2}(x_1 - x_2) \\
& = \sqrt{2}(-2 + \sqrt{2}) - (-2 - \sqrt{2}) = 4
\end{aligned}$$

Question 131

Find the equation of the circle which passes through the point $(1, 2)$ and the points of intersection of the circles

$$x^2 + y^2 - 8x - 6y + 21 = 0 \text{ and } x^2 + y^2 - 2x - 15 = 0$$

AP EAPCET 2021 - 20th August Morning Shift

Options:

- A. $x^2 + y^2 - 18x - 12y + 27 = 0$
- B. $2(x^2 + y^2) - 18x - 12y + 27 = 0$
- C. $3(x^2 + y^2) - 18x - 12y + 27 = 0$
- D. $4(x^2 + y^2) - 18x - 12y + 27 = 0$

Answer: C

Solution:

$$\begin{aligned}
& (x^2 + y^2 - 8x - 6y + 21) + \lambda(x^2 + y^2 - 2x - 15) = 0 \\
& \Rightarrow (1 + 4 - 8 - 12 + 21) + \lambda(1 + 4 - 2 - 15) = 0 \\
& \Rightarrow 6 + (-12\lambda) = 0 \Rightarrow \lambda = 1/2 \\
& \Rightarrow 3(x^2 + y^2) - 18x - 12y + 27 = 0
\end{aligned}$$

Question132

Given, two fixed points $A(-2, 1)$ and $B(3, 0)$. Find the locus of a point P which moves such that the angle $\angle APB$ is always a right angle.

AP EAPCET 2021 - 19th August Evening Shift

Options:

A. $x^2 + y^2 + x + y + 6 = 0$

B. $x^2 + y^2 - x - y - 6 = 0$

C. $x + y + 6 = 0$

D. $2x^2 + 2y^2 - 2x - 2y + 1 = 0$

Answer: B

Solution:

$$\angle APB = 90^\circ$$

$\Rightarrow$ Locus of P is a circle with A and B as extremities of diameter.

$\therefore$ Locus of P is given by

$$(x + 2)(x - 3) + (y - 1)(y - 0) = 0$$

$$x^2 - x - 6 + y^2 - y = 0$$

$$\text{or } x^2 + y^2 - x - y - 6 = 0$$

Question133

The equations of the tangents to the circle $x^2 + y^2 = 4$ drawn from the point $(4, 0)$ are

AP EAPCET 2021 - 19th August Evening Shift

Options:

A. $y = \pm \frac{1}{\sqrt{3}}(x - 4)$

B. $y = \pm \frac{2}{\sqrt{3}}(x - 4)$

C. $x = \pm \frac{1}{\sqrt{3}}(y - 4)$

D. $x = \pm \frac{2}{\sqrt{3}}(y - 4)$

Answer: A

Solution:

Let $S = x^2 + y^2 - 4 = 0$, then $T = xx_1 + yy_1 - 4$

Equation of tangent from $(4, 0)$ is $x_1 = 4, y_1 = 0$ and $S_1 = x_1^2 + y_1^2 - 4$

$$SS_1 = T^2$$

$$\begin{aligned} \Rightarrow (x^2 + y^2 - 4)(16 - 4) &= (4x - 4)^2 \\ \Rightarrow 3(x^2 + y^2 - 4) &= 4(x - 1)^2 \\ \Rightarrow 3x^2 + 3y^2 - 12 &= 4(x^2 - 2x + 1) \\ \Rightarrow 3y^2 &= (x - 4)^2 \\ \Rightarrow y &= \pm \frac{1}{\sqrt{3}}(x - 4) \end{aligned}$$

Question134

If $P(-9, -1)$ is a point on the circle $x^2 + y^2 + 4x + 8y - 38 = 0$, then find equation of the tangent drawn at the other end of the diameter drawn through P

AP EAPCET 2021 - 19th August Evening Shift

Options:

A. $7x - 3y = 60$

B. $7x - 3y = 56$

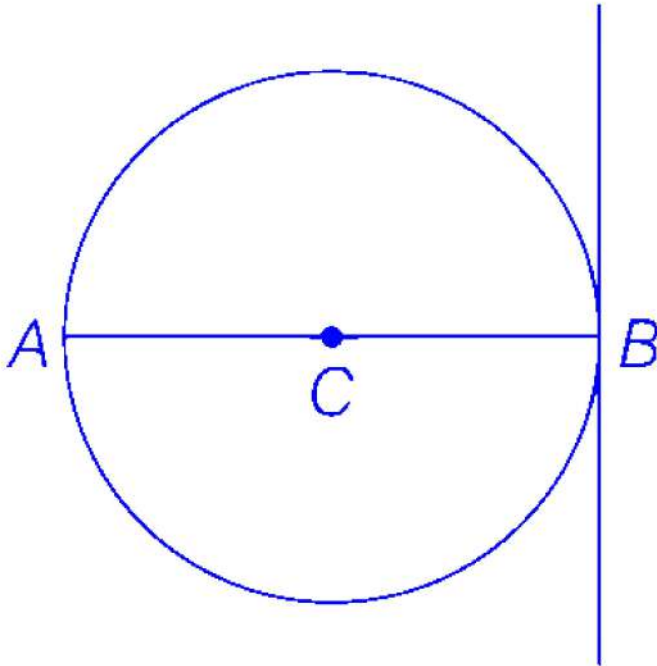
C. $7x + 3y = 56$

D. $7x + 3y = 60$

Answer: B

Solution:

Let $P(-9, -1)$



On the circle

$$x^2 + y^2 + 4x + 8y - 38 = 0$$

$$\therefore 2g = 4 \Rightarrow g = 2$$

$$2f = 8 \Rightarrow f = 4$$

$$\text{Centre} = (-g, -f)$$

Whose centre is $C(-2, -4)$.

Let PB be a diameter. Then, C is mid-point of PB and Let $B(x, y)$.

$$\text{Then, } (-2, -4) = \left(\frac{-9+x}{2}, \frac{-1+y}{2} \right)$$

$$\Rightarrow \frac{-9+x}{2} = -2 \text{ and } \frac{-1+y}{2} = -4$$

$$\Rightarrow x = 5 \text{ and } y = -7$$

$$\therefore B(5, -7)$$

Tangent at $(5, -7)$

$$5x - 7y + 2(x + 5) + 4(y - 7) - 38 = 0$$

$$5x - 7y + 2x + 10 + 4y - 28 - 38 = 0$$

$$\Rightarrow 7x - 3y = 56$$

Question135

Find the equation of a circle whose radius is 5 units and passes through two points on the X -axis, which are at a distance of 4 units from the origin

AP EAPCET 2021 - 19th August Evening Shift

Options:

A. $x^2 + y^2 - 6x - 25 = 0$

B. $x^2 + y^2 - 6y - 25 = 0$

C. $x^2 + y^2 + 6y - 16 = 0$

D. $x^2 + y^2 + 6x - 16 = 0$

Answer: C

Solution:

Given, radius = 5

and points on X-axis at 4 units distance from origin be $(-4, 0)$ and $(4, 0)$.

Let equation of circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \dots (i)$$

Equation (i) passes through $(-4, 0)$ and $(4, 0)$

$$\therefore 16 - 8g + c = 0$$

$$\text{and } 16 + 8g + c = 0$$

On solving, we have

$$c = -16, g = 0$$

$$\therefore \text{radius} = 5$$

$$\Rightarrow \sqrt{g^2 + f^2 - c} = 5$$

$$\text{or } f^2 + 16 = 25$$

$$\text{or } f = \pm 3$$

$\therefore$ Required equation of circle

$$x^2 + y^2 \pm 6y - 16 = 0$$

Question136

If a foot of the normal from the point $(4, 3)$ to a circle is $(2, 1)$ and $2x - y - 2 = 0$, is a diameter of the circle, then the equation of circle is

AP EAPCET 2021 - 19th August Evening Shift

Options:

A. $x^2 + y^2 + 2x + 1 = 0$

B. $x^2 + y^2 + 2x - 1 = 0$

C. $x^2 + y^2 - 2x - 1 = 0$

D. $2(x^2 + y^2) - 2x - 1 = 0$

Answer: C

Solution:

Equation of normal passing through

$(4, 3)$ and $(2, 1)$

$$y - 3 = \frac{1-3}{2-4}(x - 4)$$

$$y - 3 = x - 4$$

$$x - y = 1 \dots (i)$$

Equation of diameter

$$2x - y = 2 \dots (ii)$$

Solving Eqs. (i) and (ii), we have coordinate of centre of circle

$$\therefore x = 1, y = 0$$

Centre of Circle $(1, 0)$

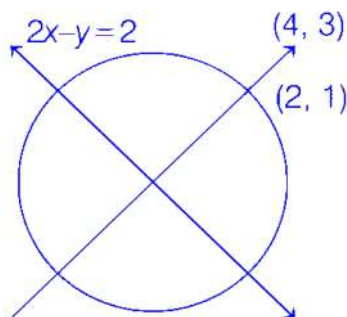
Radius of circle = distance between $(1, 0)$ and $(2, 1)$

$$= \sqrt{(2-1)^2 + (1-0)^2} = \sqrt{2}$$

$$\text{Radius} = \sqrt{2}$$

$\therefore$ Equation of circle

$$\begin{aligned} (x-1)^2 + (y-0)^2 &= (\sqrt{2})^2 \\ \Rightarrow x^2 + y^2 - 2x - 1 &= 0 \end{aligned}$$



Question137

The length of the tangent from any point on the circle

$(x - 3)^2 + (y + 2)^2 = 5r^2$ to the circle $(x - 3)^2 + (y + 2)^2 = r^2$ is 16 units, then the area between the two circles in square units is

AP EAPCET 2021 - 19th August Evening Shift

Options:

- A. 32π
- B. 4π
- C. 8π
- D. 256π

Answer: D

Solution:

Let $P(x, y)$ be any point on the circle, therefore it will satisfy the circle

$$(x_1 - 3)^2 + (y_1 + 2)^2 = 5r^2 \dots (i)$$

The length of tangent drawn from point $P(x_1, y_1)$ to the circle $(x - 3)^2 + (y + 2)^2 = r^2$ is

$$\begin{aligned}\sqrt{(x_1 - 3)^2 + (y_1 + 2)^2 - r^2} &= \sqrt{5r^2 - r^2} \text{ [from Eq. (i)]} \\ &= \sqrt{4r^2} = 2r\end{aligned}$$

According to the question,

$$\begin{aligned}16 &= 2r \\ \Rightarrow r &= 8\end{aligned}$$

$$\begin{aligned}\therefore \text{Area between two circles} &= 5\pi r^2 - \pi r^2 \\ &= 4\pi r^2 = 4\pi \cdot 8^2 \\ &= 256\pi \text{ squnits}\end{aligned}$$

Question138

The equation of the circle, which cuts orthogonally each of the three circles

$$x^2 + y^2 - 2x + 3y - 7 = 0,$$

$$x^2 + y^2 + 5x - 5y + 9 = 0 \text{ and}$$

$$x^2 + y^2 + 7x - 9y + 29 = 0$$

AP EAPCET 2021 - 19th August Evening Shift

Options:

A. $x^2 + y^2 - 16x - 18y - 4 = 0$

B. $x^2 + y^2 = a^2$

C. $x^2 + y^2 - 16x = 0$

D. $y^2 - x^2 + 2x = 0$

Answer: A

Solution:

Let equation of circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \dots (i)$$

This circle is orthogonal to

$$x^2 + y^2 - 2x + 3y - 7 = 0 \dots (ii)$$

$$x^2 + y^2 + 5x - 5y + 9 = 0 \dots (iii)$$

$$x^2 + y^2 + 7x - 9y + 29 = 0 \dots (iv)$$

By condition of orthogonality, we have

$$-2g + 3f = c - 7 \dots (vi)$$

$$5g - 5f = c + 9 \dots (vii)$$

$$7g - 9f = c + 29 \dots (viii)$$

from Eq. (vii) $c = 5g - 5f - 9$

From Eqs. (vi) and (vii), we get

$$-7g + 8f = -16$$

and $g - 2f = 10$ {on subtracting Eq. (vii) from Eq. (viii)}

On solving, we have $g = -8, f = -9, c = -4$

Required equation of circle,

$$\therefore x^2 + y^2 - 16x - 18y - 4 = 0$$

Question139

Find the equations of the tangents drawn to the circle $x^2 + y^2 = 50$ at the points where the line $x + 7 = 0$ meets it.

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. $7x + y + 50 = 0$ and $7x - y + 50 = 0$

B. $x + y = 0$ and $x - y = 0$

C. $x + 7y + 5 = 0$ and $y - 7x + 5 = 0$

D. $x + 7y + 50 = 0$ and $x - 7y + 50 = 0$

Answer: A

Solution:

Given circle $x^2 + y^2 = 50$ (i)

and a line $x + 7 = 0$ (ii)

Point of intersection of Eqs. (i) and (ii) is $x = -7, y = \pm 1$

$\therefore (-7, 1)$ and $(-7, -1)$ are points of contact of Eqs. (i) and (ii),

for circle $x^2 + y^2 = r^2$

Equation of tangent at (x_1, y_1) is

$$xx_1 + yy_1 = r^2$$

$\therefore$ Equation of tangent of Eq. (i) at $(-7, 1)$ and $(-7, -1)$ are

$$-7x + y = 50 \text{ or } 7x - y + 50 = 0$$

$$\text{and } -7x - y = 50 \text{ or } 7x + y + 50 = 0$$

Question140

If the chord of contact of tangents from a point on the circle $x^2 + y^2 = r_1^2$ to the circle $x^2 + y^2 = r_2^2$ touches the circle $x^2 + y^2 = r_3^2$, then r_1, r_2 and r_3 are in

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. AP

B. HP

C. GP

D. AGP

Answer: C

Solution:

(c) Let (x_1, y_1) be any point on the circle

$$x^2 + y^2 = r_1^2$$

Then, $x_1^2 + y_1^2 = r_1^2$ (i)

Equation of chord of contact of tangents from (x_1, y_1) to the circle $x^2 + y^2 = r_2^2$ is

$$xx_1 + yy_1 = r_2^2 \text{ (ii)}$$

$\therefore$ Eq. (ii) touches the circle

$$x^2 + y^2 = r_3^2$$

The length of perpendicular from centre $(0, 0)$ on Eq. (ii) is radius $= r_3$

$$\Rightarrow \frac{r_2^2}{\sqrt{x_1^2 + y_1^2}} = r_3 \Rightarrow \frac{r_2^2}{\sqrt{r_1^2}} = r_3$$

$$\Rightarrow r_2^2 = r_1 r_3$$

$$\Rightarrow r_1, r_2, r_3 \text{ are in GP.}$$

Question141

Find the equation of the circle passing through $(1, -2)$ and touching the X -axis at $(3, 0)$.

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. $x^2 + y^2 + 6x - 4y - 9 = 0$

B. $x^2 + y^2 - 6x - 4y + 9 = 0$

C. $x^2 + y^2 - 6x - 4y - 9 = 0$

D. $x^2 + y^2 - 6x + 4y + 9 = 0$

Answer: D

Solution:

Let (h, k) be centre and r be the radius

Then, $h = 3$

Equation of circle

$$(x - 3)^2 + (y - k)^2 = r^2$$

It passes through $(3, 0)$ and $(1, -2)$

$$\Rightarrow k^2 = r^2$$

$$\text{and } 4 + k^2 + 4k + 4 = r^2$$

$$\Rightarrow 8 + 4k = 0$$

$$\Rightarrow k = -2$$

$\therefore$ Required equation of circle

$$(x - 3)^2 + (y + 2)^2 = 2^2$$

$$\text{or } x^2 + y^2 - 6x + 4y + 9 = 0$$

Question142

Let L_1 be a straight line passing through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle

$x^2 + y^2 - x + 3y = 0$ on L_1 and L_2 are equal, then which of the following equations represent L_1

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. $x + y = 0$ and $x + 7y = 0$

B. $x - y = 0$ and $x + 7y = 0$

C. $x - 7y = 0$ and $x + y = 0$

D. $x - 7y = 0$ and $x - y = 0$

Answer: B

Solution:

Let $L_1 : y = mx$ (equation of line passes through origin)

Intercept made by L_1 and L_2 are equal $\Rightarrow L_1$ and L_2 are at the same distance from centre of circle $x^2 + y^2 - x + 3y = 0$

Now, centre $\left(\frac{1}{2}, -\frac{3}{2}\right)$

[$\because$ centre $= (-g, -f)$]

$$\text{Distance of } L_1 \text{ from } \left(\frac{1}{2}, -\frac{3}{2}\right) = \frac{\left|m \cdot \frac{1}{2} + \frac{3}{2}\right|}{\sqrt{m^2+1}}$$

$$\text{Distance of } L_2 \text{ from } \left(\frac{1}{2}, -\frac{3}{2}\right) = \frac{\left|\frac{1}{2} - \frac{3}{2} - 1\right|}{\sqrt{1+1}}$$

$$\therefore L_2 : x + y - 1 = 0$$

$$[\because y = mx \Rightarrow mx - y = 0]$$

$$\Rightarrow \frac{\left|\frac{m+3}{2}\right|}{\sqrt{m^2+1}} = \frac{2}{\sqrt{2}}$$

Squaring both sides, we get

$$\frac{(m+3)^2}{4(m^2+1)} = \frac{4}{2}$$

$$\begin{aligned} \Rightarrow m^2 + 6m + 9 &= 8(m^2 + 1) \\ \Rightarrow 7m^2 - 6m - 1 &= 0 \\ \Rightarrow m &= \frac{6 \pm \sqrt{36 + 28}}{14} = \frac{6 \pm 8}{14} \\ &= 1, \frac{-1}{7} \\ \therefore L_1 \text{ is } y &= x \\ \text{and} \\ y &= \frac{-1}{7}x \\ \text{or} \\ x - y &= 0 \\ \text{or} \\ x + 7y &= 0 \end{aligned}$$

Question143

The radius of the circle whose center lies at $(1, 2)$ while cutting the circle $x^2 + y^2 + 4x + 16y - 30 = 0$ orthogonally, is units.

AP EAPCET 2021 - 19th August Morning Shift

Options:

- A. $\sqrt{41}$
- B. $\sqrt{31}$
- C. $\sqrt{21}$
- D. $\sqrt{11}$

Answer: D

Solution:

Given, equation of circle

$$x^2 + y^2 + 4x + 16y - 30 = 0 \dots (i)$$

$$\text{Comparing with } x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$$

$$\text{We have, } g_1 = 2, f_1 = 8, c_1 = -30$$

Let equation of circle whose centre at $(1, 2)$ is

$$x^2 + y^2 - 2x - 4y + c = 0 \dots (ii)$$

$$\text{Here, } g_2 = -1, f_2 = -2, c_2 = c$$

Circles Eqs. (i) and (ii) are orthogonal, then

$$\begin{aligned} \Rightarrow 2(g_1g_2 + f_1f_2) &= c_1 + c_2 \\ \text{or } 2(-2 - 16) &= -30 + c \\ c &= -6 \end{aligned}$$

From Eqs. (ii), we get

$$\begin{aligned} x^2 + y^2 - 2x - 4y - 6 &= 0 \\ \text{Radius} &= \sqrt{(-1)^2 + (-2)^2 - (-6)} = \sqrt{11} \end{aligned}$$

Question144

The point which has the same power with respect to each of the circles $x^2 + y^2 - 8x + 40 = 0$, $x^2 + y^2 - 5x + 16 = 0$ and $x^2 + y^2 - 8x + 16y + 160 = 0$ is

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. $(-8, \frac{-15}{2})$

B. $(8, \frac{-15}{2})$

C. $(8, \frac{15}{2})$

D. $(-8, \frac{15}{2})$

Answer: B

Solution:

$$\text{Let } S_1 : x^2 + y^2 - 8x + 40 = 0$$

$$S_2 : x^2 + y^2 - 5x + 16 = 0$$

$$S_3 : x^2 + y^2 - 8x + 16y + 160 = 0$$

The point which has same power is obtained by solving

$$S_2 - S_1 = 0$$

$$\text{and } S_3 - S_1 = 0$$

$$\Rightarrow 3x - 24 = 0$$

$$\text{and } 16y + 120 = 0$$

$$\Rightarrow x = 8 \text{ and } y = \frac{-15}{2} \Rightarrow 2at = 2$$

$\therefore$ Required point is $(8, \frac{-15}{2})$.
